Filtri

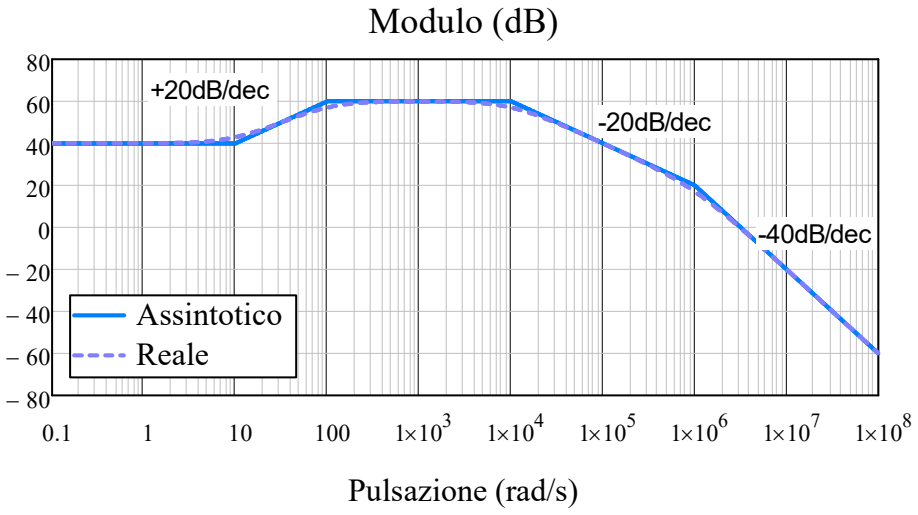
Esercizio 1A

DATI: $A = 100$, $\omega_Z = 10 \cdot rad \cdot s^{-1}$, $\omega_{P_1} = 100 rad \cdot s^{-1}$,

$\omega_{P_2} = 10^4 \cdot rad \cdot s^{-1}$, $\omega_{P_3} = 10^6 \cdot rad \cdot s^{-1}$

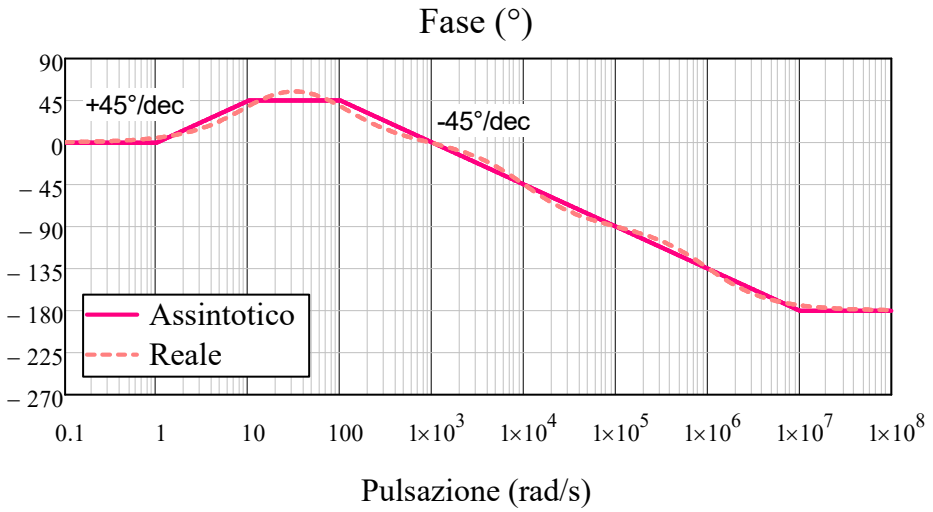
$$W(s) = A \cdot \frac{1 + \frac{s}{\omega_Z}}{\left(1 + \frac{s}{\omega_{P_1}}\right) \cdot \left(1 + \frac{s}{\omega_{P_2}}\right) \cdot \left(1 + \frac{s}{\omega_{P_3}}\right)}$$

Nessun polo e nessuno zero nell'origine. Guadagno positivo. Il diagramma di bode del modulo parte da 40dB.
Per il modulo la pendenza aumenta di 20dB/dec ad ogni zero e diminuisce di 20dB/dec ad ogni polo



Il diagramma di bode della fase parte da 0 poichè non ci sono poli nè zeri nell'origine e A>0

- in corrispondenza dello zero lo sfasamento inizia una decade prima e finisce una decade dopo. Complessivamente aumenta di 90° (45° per ogni decade) tra $\omega_Z/10$ fino a $10\omega_Z$.
- in corrispondenza di ciascun polo lo sfasamento inizia una decade prima e finisce una decade dopo. Complessivamente diminuisce di 90° (45° per ciascuna decade) tra $\omega_P/10$ fino a $10\omega_P$.



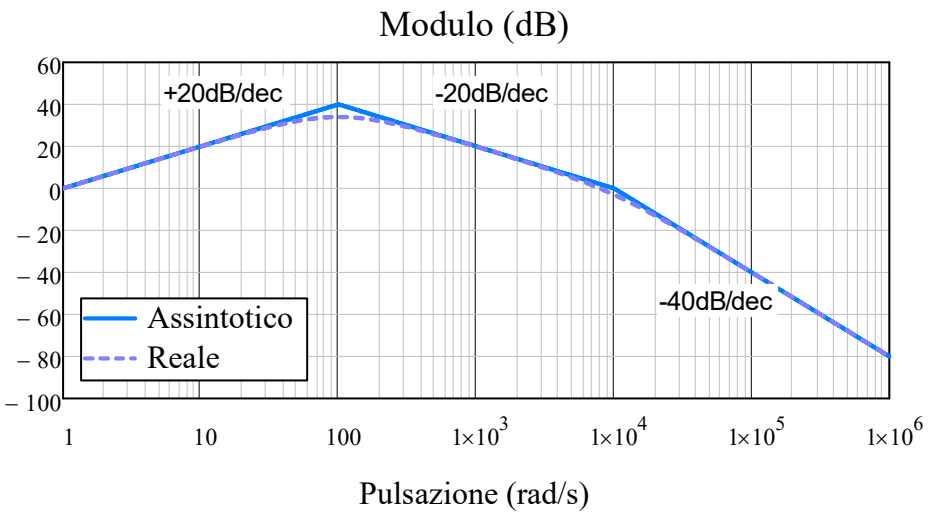
Sfasamento iniziale (0°)							
ω_{Z1}	+45	+45					
ω_{P1}		-45	-45				
ω_{P2}				-45	-45		
ω_{P3}						-45	-45
pendenza totale:	+45	0	-45	-45	-45	-45	-45

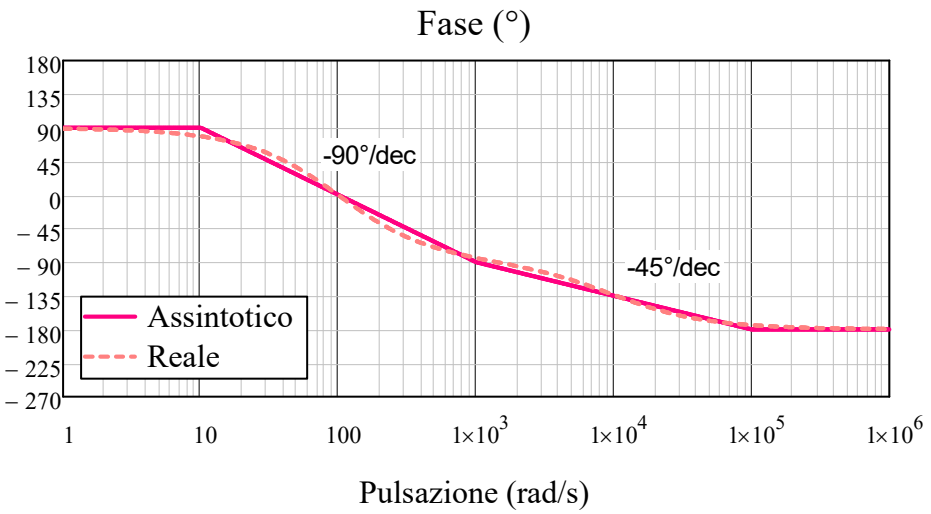
Esercizio 1B

DATI: $A = 1,\ \omega_O = 1s^{-1}\ \omega_{P_1} = 100 rad \cdot s^{-1},\ \omega_{P_2} = 10^4 \cdot rad \cdot s^{-1}$

$$W(s) = A \cdot \frac{\frac{s}{\omega_O}}{\left(1 + \frac{s}{\omega_{P_1}}\right)^2 \cdot \left(1 + \frac{s}{\omega_{P_2}}\right)}$$

Il diagramma di bode del modulo parte con una pendenza di +20dB/dec poichè c'è uno zero nell'origine.
Il diagramma di bode della fase parte da +90° poichè il guadagno è positivo e c'è un polo nell'origine.





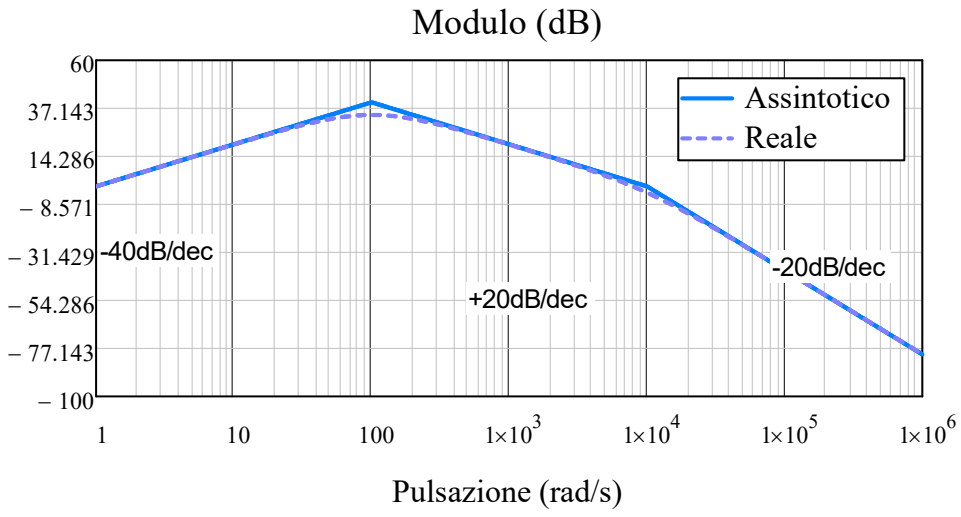
Sfasamento (°/dec)				
ω_{P1} (polo doppio)	-90°/dec	-90°/dec		
ω_{P2}			-45°/dec	-45°/dec
pendenza totale:	-90°/dec	-90°/dec	-45°/dec	-45°/dec

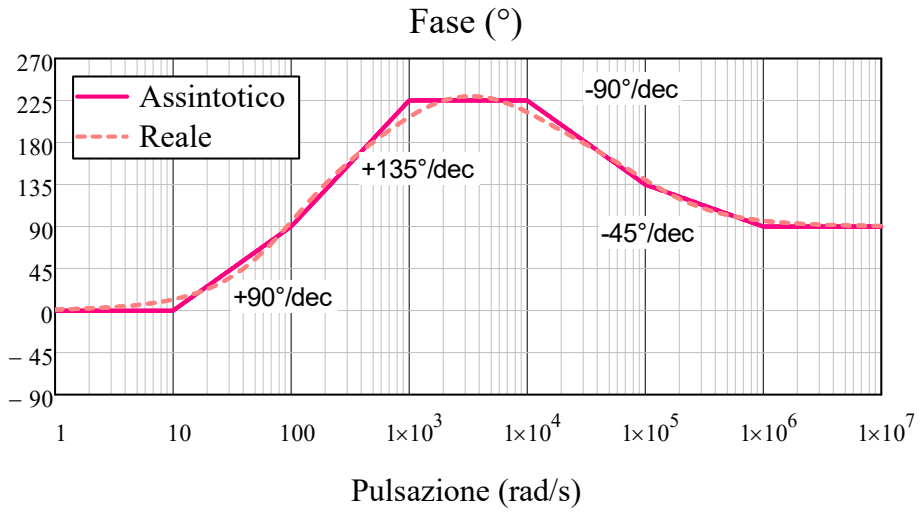
Esercizio 1C

DATI: $A = -10$, $\omega_0 = 1 rad \cdot s^{-1}$, $\omega_{Z_1} = 100 rad \cdot s^{-1}$, $\omega_{Z_2} = 10^3 rad \cdot s^{-1}$, $\omega_{P_1} = 10^4 rad \cdot s^{-1}$, $\omega_{P_2} = 10^5 \cdot rad \cdot s^{-1}$

$$W(s) = A \cdot \frac{\left(1 + \frac{s}{\omega_{Z_1}}\right)^2 \cdot \left(1 + \frac{s}{\omega_{Z_2}}\right)}{\left(\frac{s}{\omega_0}\right)^2 \cdot \left(1 + \frac{s}{\omega_{P_1}}\right) \cdot \left(1 + \frac{s}{\omega_{P_2}}\right)}$$

Il diagramma di bode del modulo parte con una pendenza di -40dB per decade poichè c'è un polo doppio nell'origine

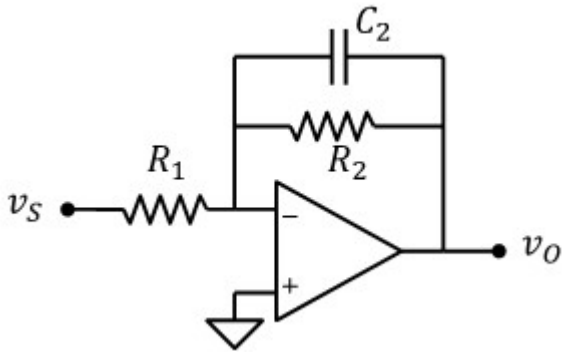




ω_{Z1} (zero doppio)	+90°/dec	+90°/dec			
ω_{Z2}		+45°/dec	+45°/dec		
ω_{P1}			-45°/dec	-45°/dec	
ω_{P2}				-45°/dec	-45°/dec
pendenza totale:	+90°/dec	+135°/dec	0	-90°/dec	-45°/dec

Esercizio 2

DATI: $R_1 = 10k\Omega$, $R_2 = 5k\Omega$, $C_2 = 200nF$



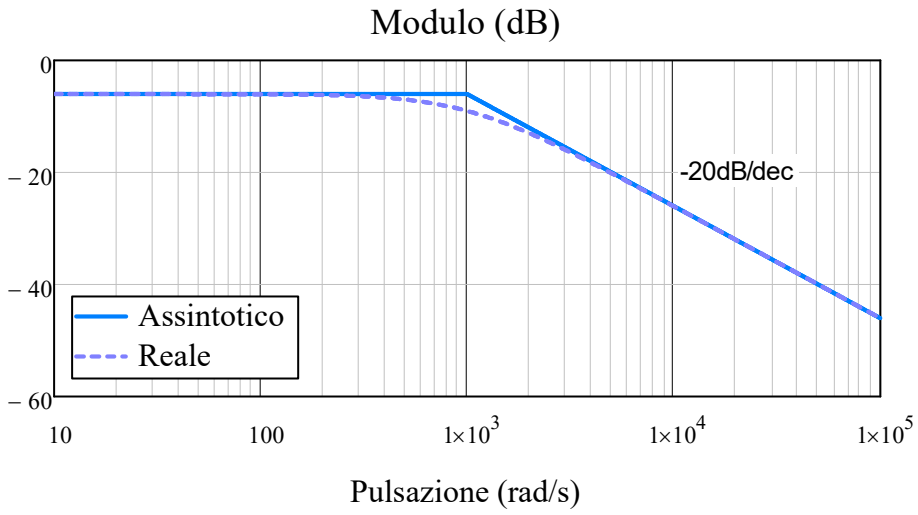
Funzione di trasferimento

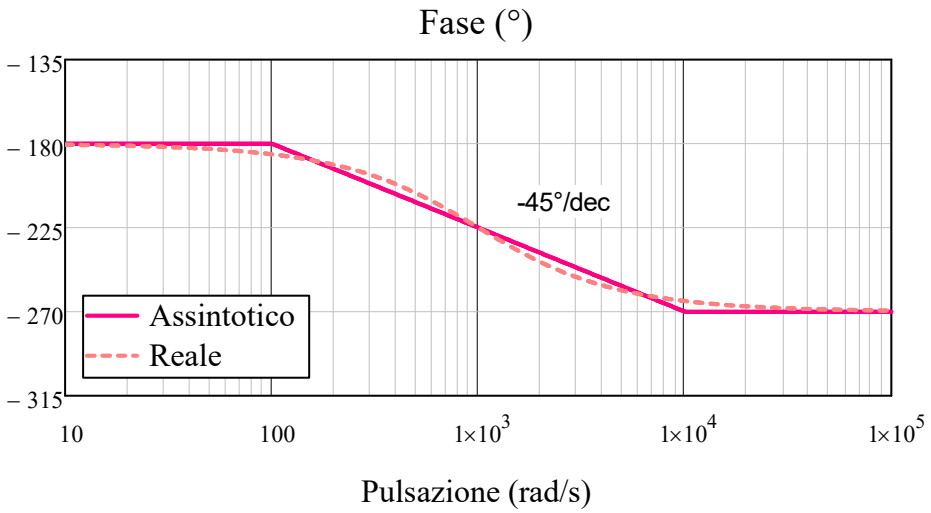
$$Z_2 = \frac{R_2}{1 + j\omega \cdot R_2 \cdot C_2}$$

$$v_O = -v_S \cdot \frac{Z_2}{R_1} = \frac{-R_2}{R_1} \cdot \frac{1}{\left(1 + i\omega \cdot R_2 \cdot C_2\right)} \cdot v_S$$

Definiamo: $A = \frac{-R_2}{R_1} = -0.5$ $\quad \omega_P = \frac{1}{C_2 \cdot R_2} = 1 \times 10^3 \cdot s^{-1}$ $\quad W(s) = A \cdot \frac{1}{\left(1 + \frac{s}{\omega_P}\right)}$

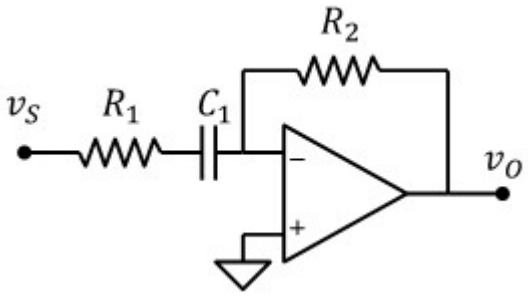
Diagramma di Bode asintotico del modulo e della fase





Esercizio 3

DATI: $R_1 = 10k\Omega$, $R_2 = 50k\Omega$, $C_1 = 10nF$



Funzione di trasferimento

$$Z_1 = \frac{1 + j\omega \cdot R_1 \cdot C_1}{j\omega \cdot C_1}$$

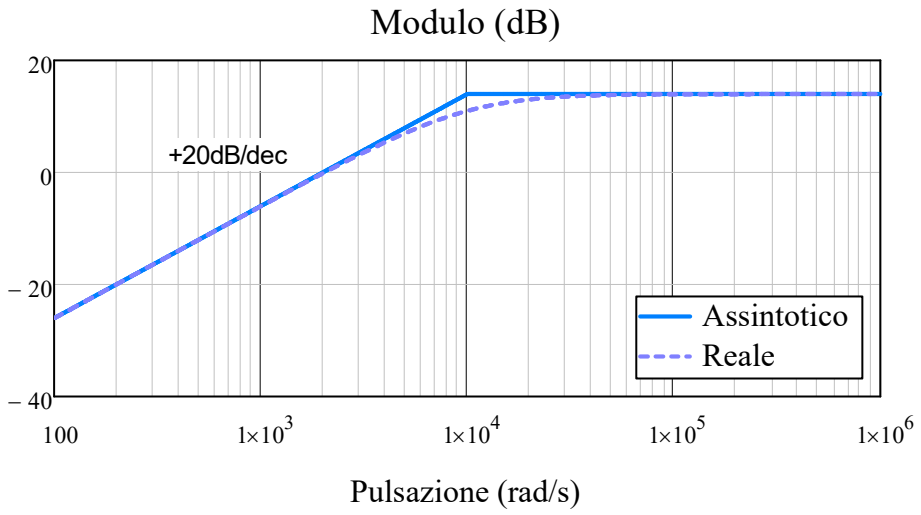
$$v_O = -v_S \cdot \frac{R_2}{Z_1} = \frac{-R_2}{R_1} \cdot \frac{i\omega \cdot C_1 \cdot R_1}{\left(1 + i\omega \cdot R_1 \cdot C_1\right)} \cdot v_S$$

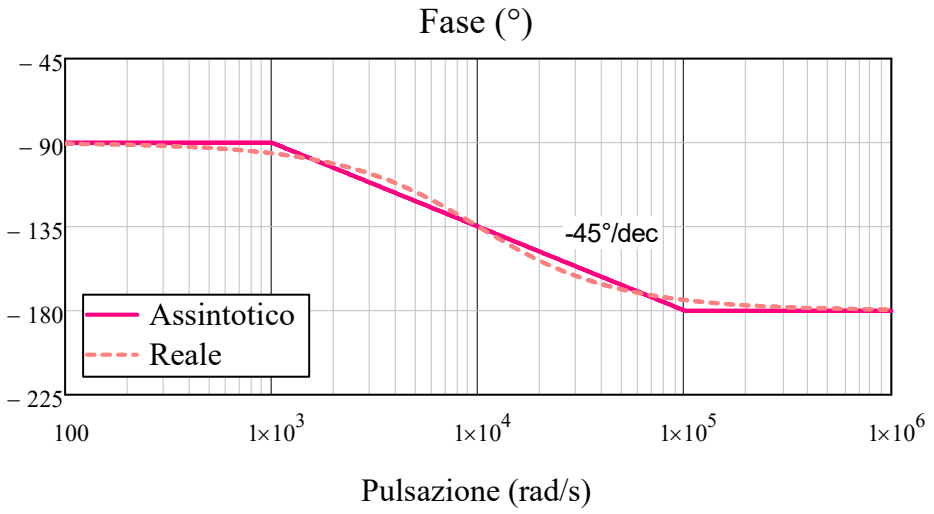
$$\boxed{A = \frac{-R_2}{R_1} = -5}$$

$$\boxed{\omega_P = \frac{1}{C_1 \cdot R_1} = 1 \times 10^4 \cdot s^{-1}}$$

$$\boxed{W(s) = A \cdot \frac{\frac{s}{\omega_P}}{\left(1 + \frac{s}{\omega_P}\right)}}$$

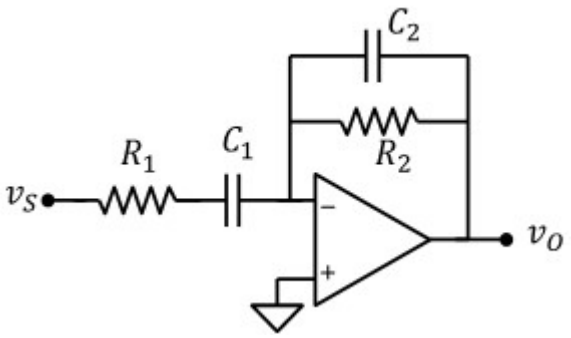
Diagramma di Bode asintotico del modulo e della fase





Esercizio 4

DATI: $R_1 = 10k\Omega$, $R_2 = 5k\Omega$, $C_1 = 10nF$, $C_2 = 2nF$



Funzione di trasferimento

$$Z_1 = \frac{1 + i\omega \cdot R_1 \cdot C_1}{i\omega \cdot C_1} \qquad Z_2 = \frac{R_2}{1 + i\omega \cdot R_2 \cdot C_2}$$

$$v_O = -v_S \cdot \frac{Z_2}{Z_1} = \frac{-R_2}{R_1} \cdot \frac{i\omega \cdot C_1 \cdot R_1}{(1 + i\omega \cdot R_1 \cdot C_1) \cdot (1 + i\omega \cdot R_2 \cdot C_2)} \cdot v_S$$

Definiamo:

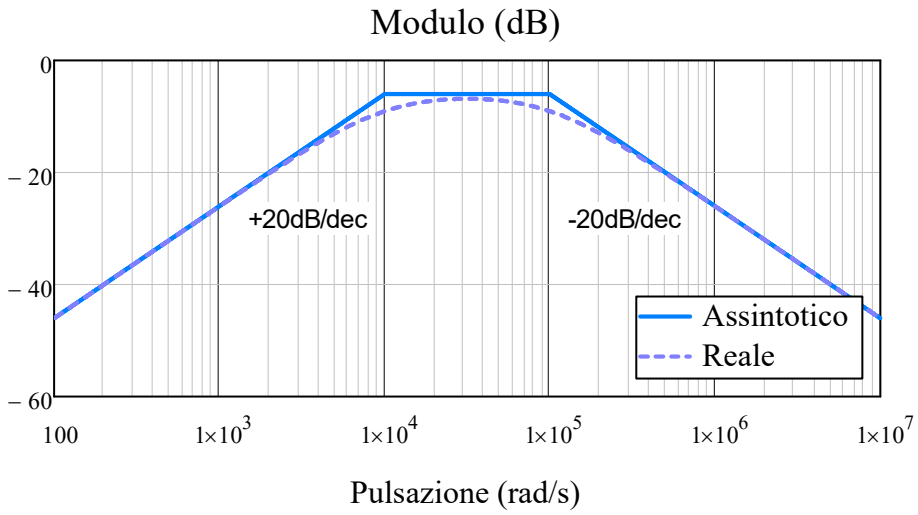
$$A = \frac{-R_2}{R_1} = -0.5$$

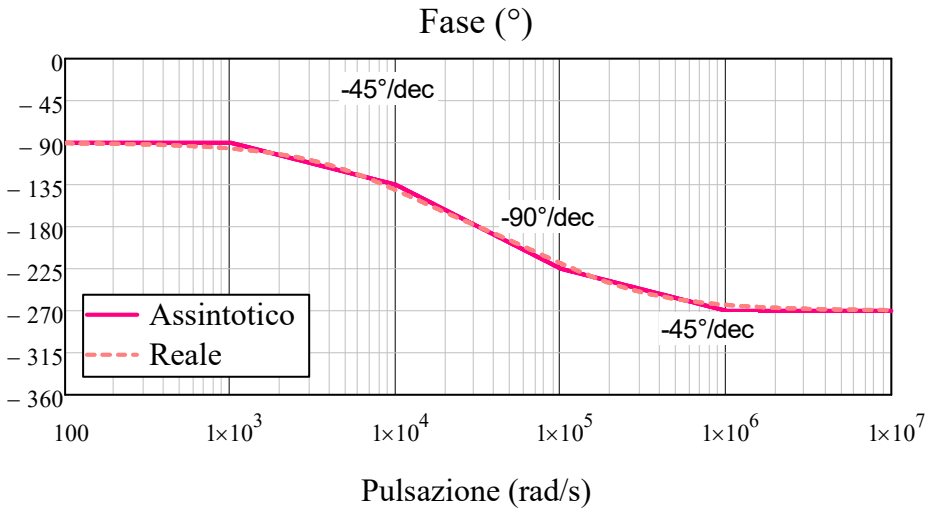
$$\omega_{P_1} = \frac{1}{C_1 \cdot R_1} = 1 \times 10^4 \cdot s^{-1}$$

$$\omega_{P_2} = \frac{1}{C_2 \cdot R_2} = 1 \times 10^5 \cdot s^{-1}$$

$$W(s) = A \cdot \frac{\frac{s}{\omega_{P_1}}}{\left(1 + \frac{s}{\omega_{P_1}}\right) \cdot \left(1 + \frac{s}{\omega_{P_2}}\right)}$$

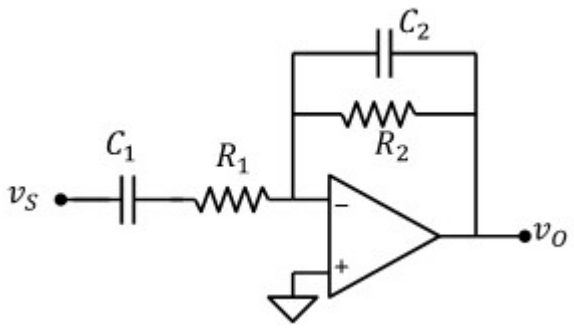
Diagramma di Bode asintotico del modulo e della fase





Esercizio 5

DATI: $\omega_L = 100 s^{-1}$, $\omega_H = 10^5 \cdot s^{-1}$, $A_v = -10$, $R_{IN} = 10 k\Omega$



1) Dimensionamento di resistenze e capacità

Calcolo della funzione di trasferimento:

$$Z_1 = R_1 + \frac{1}{i\omega C_1} = \frac{1 + i\omega \cdot R_1 \cdot C_1}{i\omega \cdot C_1}$$

$$Z_2 = \frac{R_2 \cdot \frac{1}{i\omega \cdot C_2}}{R_2 + \frac{1}{i\omega \cdot C_2}} = \frac{R_2}{1 + i\omega \cdot R_2 \cdot C_2}$$

$$W(\omega) = -\frac{Z_2}{Z_1} = -\frac{i\omega \cdot C_1 \cdot R_2}{(1 + i\omega \cdot R_2 \cdot C_2) \cdot (1 + i\omega \cdot R_1 \cdot C_1)}$$

W(ω) ha uno zero nell'origine e due poli:

I due poli sono: $\omega_{P_1} = \frac{1}{R_1 \cdot C_1}$ $\omega_{P_2} = \frac{1}{R_2 \cdot C_2}$

L'impedenza di ingresso del filtro è $R_{IN} = Z_1$. Ad alta frequenza Z_1 si approssima con R_1, quindi

$$\boxed{R_1 = R_{IN} = 10 \cdot k\Omega}$$

Dobbiamo assumere $\omega_{P1} < \omega_{P2}$. Quindi al centro della banda passante ($\omega_{P1} < \omega < \omega_{P2}$) avremmo:

- $1+j\omega R_1 C_1$ circa uguale a 1
- $1+j\omega R_2 C_1$ circa uguale a 1

il guadagno a centro banda è approssimativamente: $A_v = -\frac{i\omega \cdot C_1 \cdot R_2}{(1) \cdot (i\omega \cdot R_1 \cdot C_1)} = -\frac{R_2}{R_1}$

Bisogna realizzare: $\boxed{R_2 = -A_v \cdot R_1 = 100 \cdot k\Omega}$

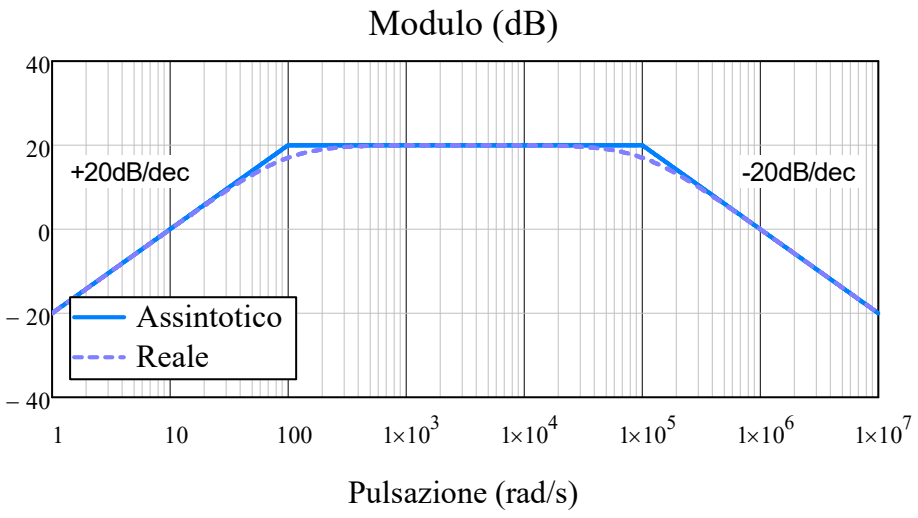
Note le resistenze calcoliamo le capacità:

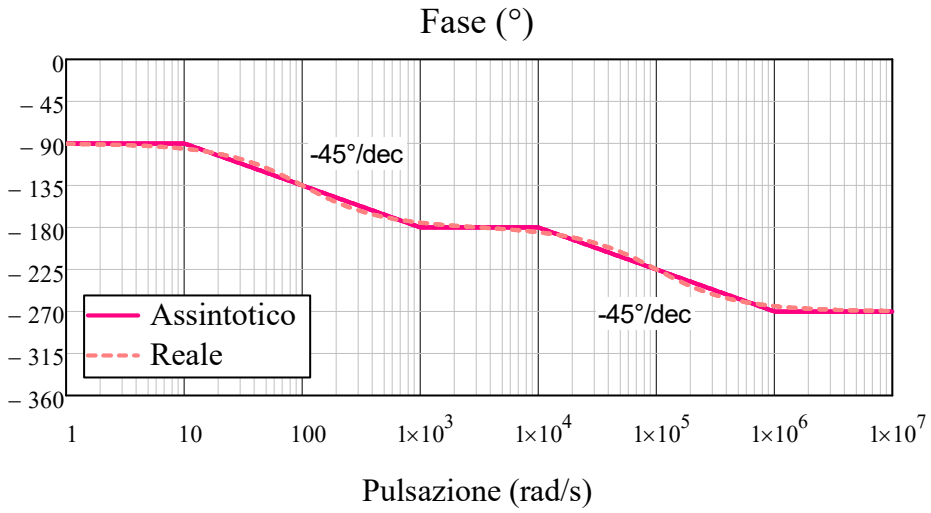
$$\omega_{P_1} = \omega_L = 100 \cdot s^{-1} \qquad \boxed{C_1 = \frac{1}{R_1 \cdot \omega_{P_1}} = 1 \cdot \mu F}$$

$$\omega_{P_2} = \omega_H = 1 \times 10^5 \cdot s^{-1} \qquad \boxed{C_2 = \frac{1}{R_2 \cdot \omega_{P_2}} = 0.1 \cdot nF}$$

$$W(s) = A_v \cdot \frac{\frac{s}{\omega_{P_1}}}{\left(1 + \frac{s}{\omega_{P_1}}\right) \cdot \left(1 + \frac{s}{\omega_{P_2}}\right)}$$

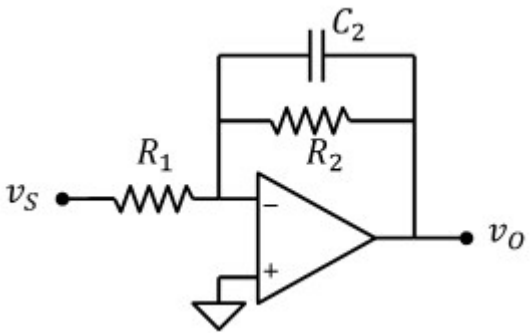
3) Tracciare il diagramma di Bode asintotico del modulo e della fase





Esercizio 6

DATI: $R_2 = 10k\Omega$



1) valore di R_1 affinchè il modulo del guadagno sia $W_0 = 5$

$$A_v = \frac{-R_2}{R_1} \quad \text{(filtro passa basso)} \qquad \boxed{R_1 = \frac{R_2}{W_0} = 2\cdot k\Omega}$$

2) valore di C_2 affinchè la frequenza di taglio sia $\omega_T = 10^3 s^{-1}$

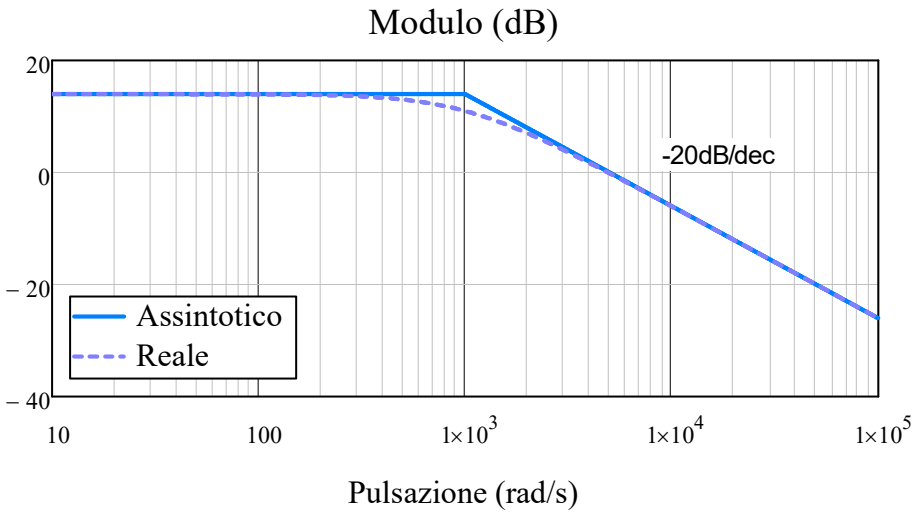
$$\omega_T = (R_2\cdot C_2)^{-1} \qquad \boxed{C_2 = \frac{1}{R_2\cdot\omega_T} = 100\cdot nF}$$

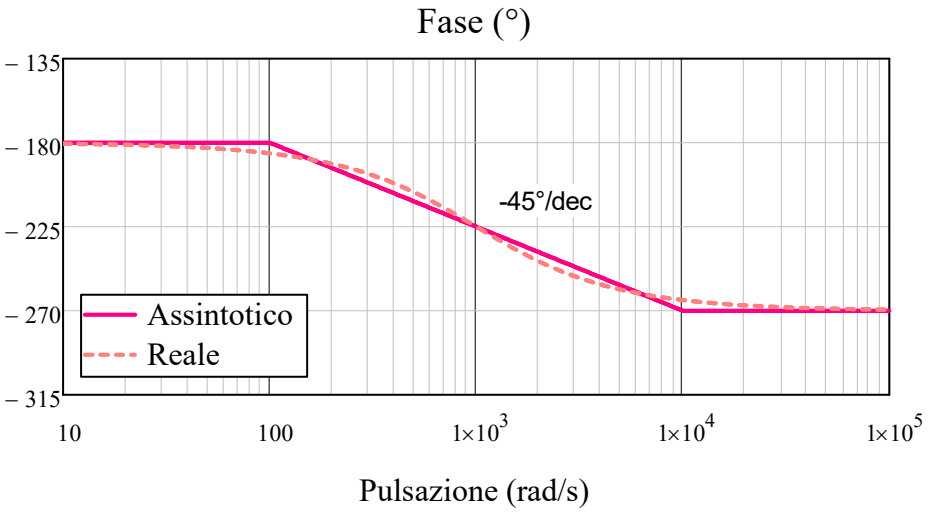
3) Diagramma di Bode asintotico del modulo e della fase

Funzione di trasferimento:

$$Z_2 = \frac{R_2}{1 + i\omega\cdot R_2\cdot C_2} \qquad W(s) = \frac{-Z_2}{R_1} = \frac{-R_2}{R_1}\cdot\frac{1}{1 + s\cdot R_2\cdot C_2}$$

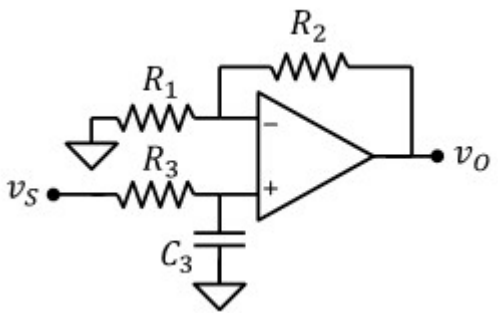
$$A = \frac{-R_2}{R_1} = -5 \qquad \omega_{P_1} = \frac{1}{R_2\cdot C_2} = 1\times 10^3\cdot s^{-1} \qquad W(s) = \frac{A}{1 + \frac{s}{\omega_{P_1}}}$$





Esercizio 7

DATI: $R_1 = 10k\Omega$, $R_3 = 20k\Omega$, $C_2 = 200nF$, $A = 5$, $\omega_P = 1000s^{-1}$



1) Funzione di trasferimento

$$v_P = \frac{1}{1 + i\omega \cdot R_3 \cdot C_3} \qquad v_O = v_P \cdot \left(1 + \frac{R_2}{R_1}\right) = \left(1 + \frac{R_2}{R_1}\right) \cdot \frac{1}{\left(1 + i\omega \cdot R_3 \cdot C_3\right)} \cdot v_S$$

$$\boxed{W(s) = \frac{A}{\left(1 + \frac{s}{\omega_P}\right)}}$$

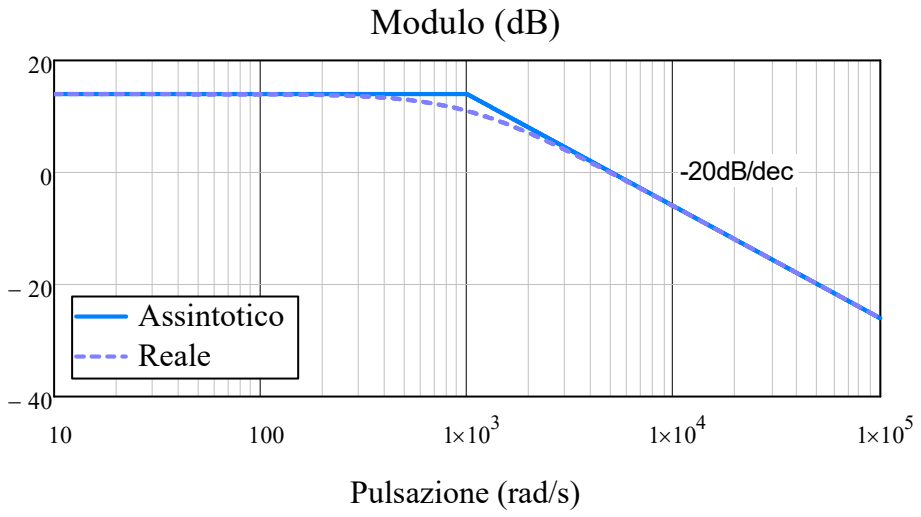
2) Valore di R_2 per avere $A = 5$

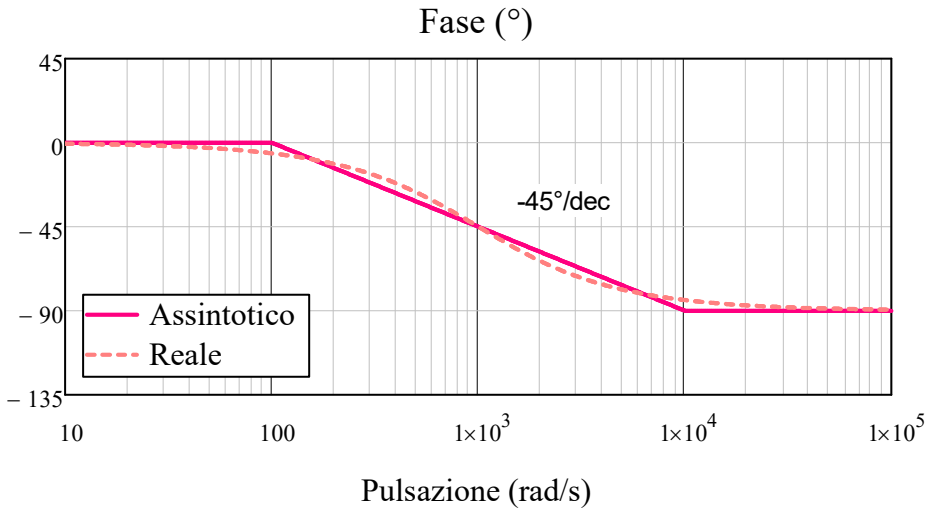
$$A = 1 + \frac{R_2}{R_1} \qquad \boxed{R_2 = R_1 \cdot (A - 1) = 40 \cdot k\Omega}$$

3) Valore di C_3 per avere $\omega_P = 1 \times 10^3 \cdot s^{-1}$

$$\omega_P = \frac{1}{R_3 \cdot C_3} \qquad \boxed{C_3 = \frac{1}{\omega_P \cdot R_3} = 50 \cdot nF}$$

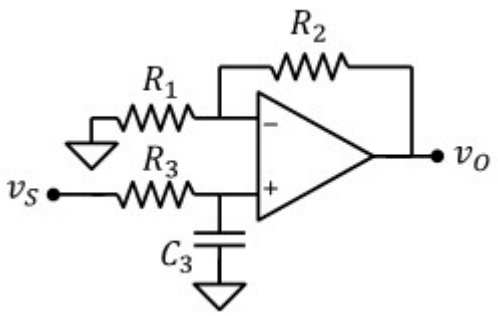
4) Diagramma di Bode asintotico del modulo e della fase





Esercizio 8

DATI: $R_1 = 100k\Omega$, $R_2 = 100k\Omega$, $R_3 = 100\Omega$, $C_3 = 20\mu F$



1) Funzione di trasferimento

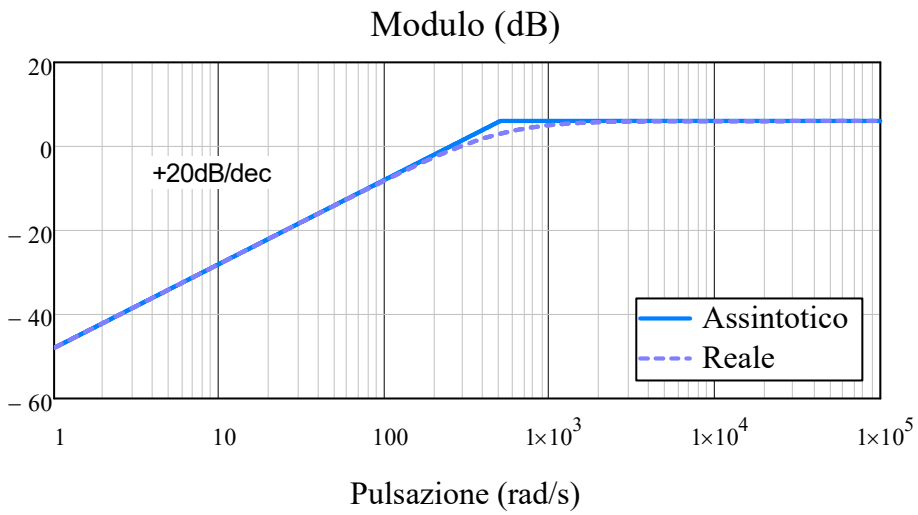
$$v_P = \frac{i\omega \cdot R_3 \cdot C_3}{1 + i\omega \cdot R_3 \cdot C_3} \qquad v_O = v_P \cdot \left(1 + \frac{R_2}{R_1}\right) = \left(1 + \frac{R_2}{R_1}\right) \cdot \frac{i\omega \cdot R_3 \cdot C_3}{\left(1 + i\omega \cdot R_3 \cdot C_3\right)} \cdot v_S$$

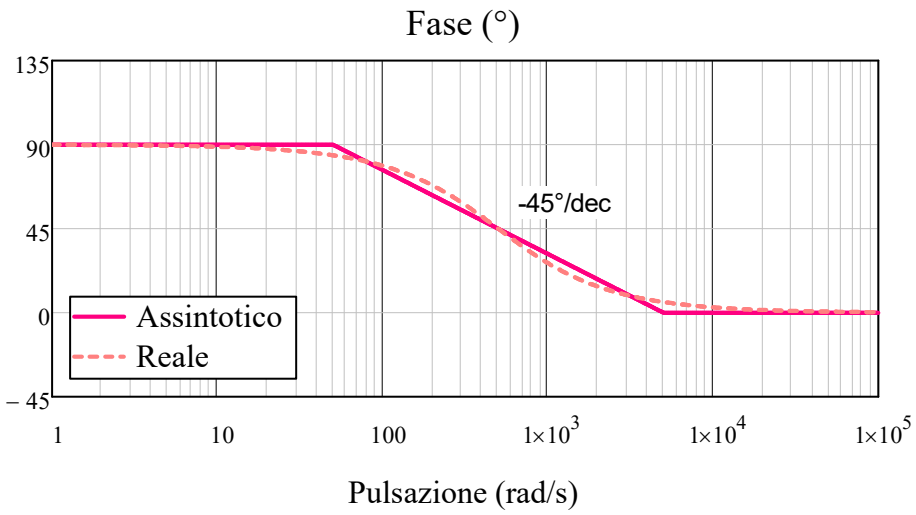
Poniamo:

$$A = 1 + \frac{R_2}{R_1} = 2 \qquad \omega_P = \frac{1}{R_3 \cdot C_3} = 500 \cdot s^{-1}$$

$$\boxed{W(s) = A \cdot \frac{\frac{s}{\omega_P}}{\left(1 + \frac{s}{\omega_P}\right)}}$$

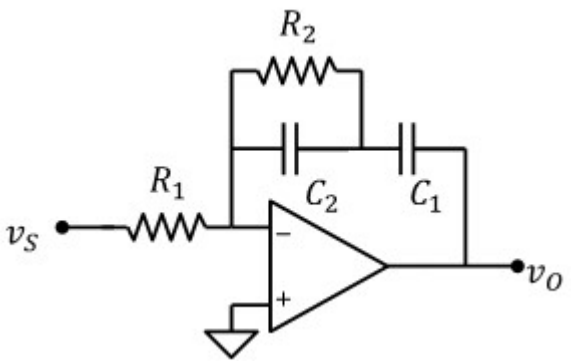
2) Diagramma di Bode asintotico del modulo e della fase





Esercizio 9

DATI: $R_1 = 1k\Omega$, $R_2 = 10k\Omega$, $C_1 = 900nF$, $C_2 = 100nF$



Funzione di trasferimento

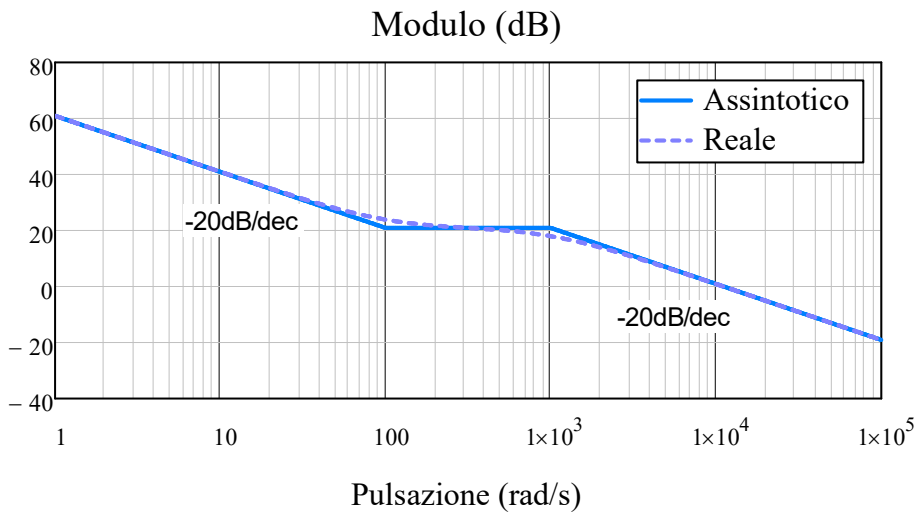
$$Z_2 = \frac{1}{i\omega \cdot C_1} + \frac{R_2}{1 + i\omega \cdot R_2 \cdot C_2} = \frac{1 + i\omega \cdot R_2 \cdot (C_1 + C_2)}{i\omega \cdot C_1 \cdot (1 + i\omega \cdot R_2 \cdot C_2)}$$

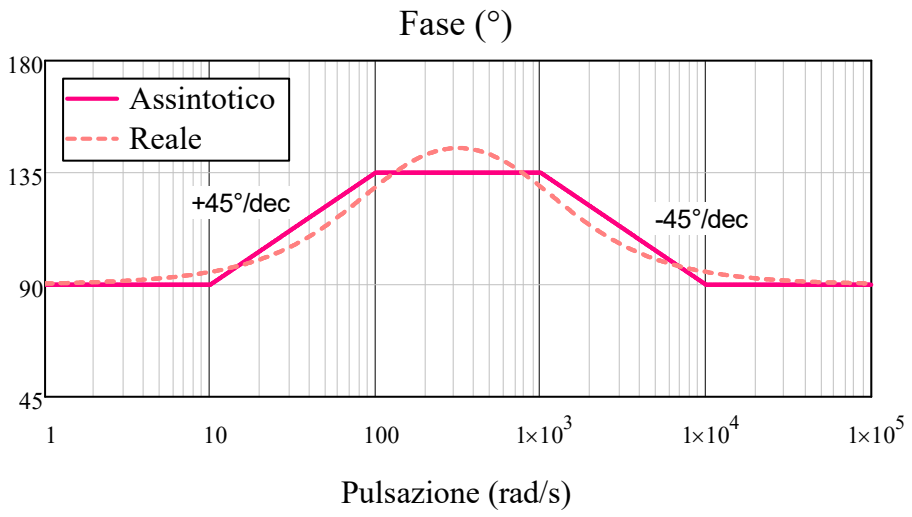
$$v_O = -\frac{Z_2}{R_1} = -\frac{1 + i\omega \cdot R_2 \cdot (C_1 + C_2)}{i\omega \cdot C_1 \cdot R_1 \cdot (1 + i\omega \cdot R_2 \cdot C_2)}$$

Definiamo: $\boxed{\omega_Z = \left[R_2 \cdot (C_1 + C_2)\right]^{-1} = 100 \cdot s^{-1}}$ $\boxed{\omega_P = (R_2 \cdot C_2)^{-1} = 1 \times 10^3 \cdot s^{-1}}$ $\boxed{\omega_O = (C_1 \cdot R_1)^{-1} = 1.11 \times 10^3 \cdot s^{-1}}$

$$\boxed{W(s) = \frac{-\left(1 + \frac{s}{\omega_Z}\right)}{\frac{s}{\omega_O} \cdot \left(1 + \frac{s}{\omega_P}\right)}}$$

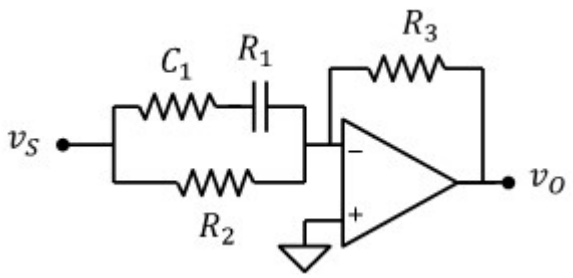
Diagramma di Bode asintotico del modulo e della fase





Esercizio 10

DATI: $R_1 = 1k\Omega$, $R_2 = 9k\Omega$, $R_3 = 45k\Omega$, $C_1 = 1\mu F$



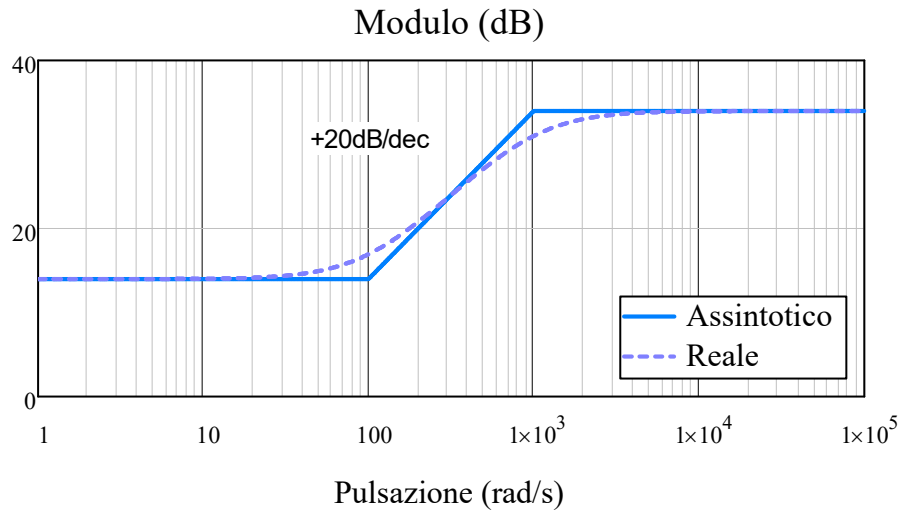
Funzione di trasferimento

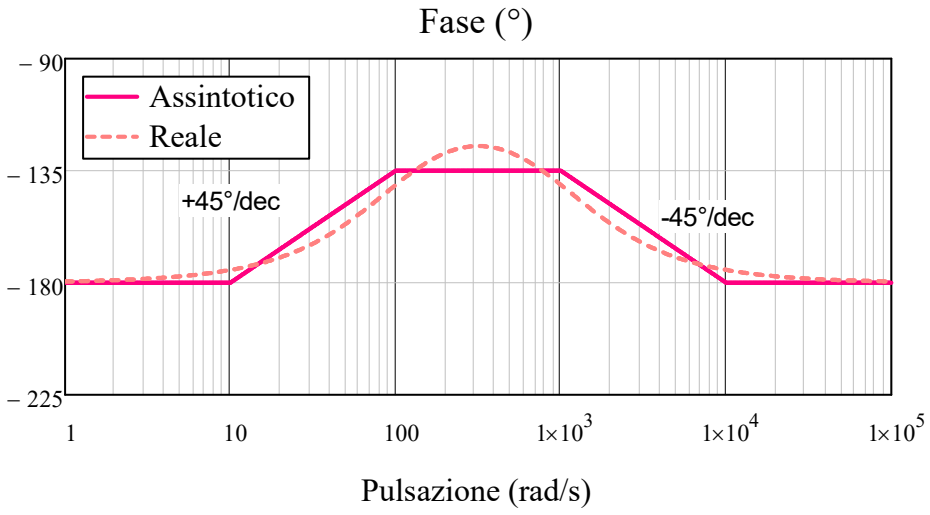
$$v_O = \frac{-R_3}{Z} \qquad Z = \frac{R_2 \cdot Z_1}{R_2 + Z_1} \qquad Z_1 = \frac{1 + i\omega \cdot R_1 \cdot C_1}{i\omega \cdot C_1}$$

$$v_O = \frac{-R_3 \cdot (R_2 + Z_1)}{R_2 \cdot Z_1} = \frac{-R_3}{R_2} \cdot \left(1 + \frac{i\omega \cdot C_1 \cdot R_2}{1 + i\omega \cdot R_1 \cdot C_1}\right) = \frac{-R_3}{R_2} \cdot \left[\frac{1 + i\omega \cdot (R_1 + R_2) \cdot C_1}{1 + i\omega \cdot R_1 \cdot C_1}\right]$$

$$A = \frac{-R_3}{R_2} = -5 \qquad \omega_Z = \frac{1}{(R_1 + R_2) \cdot C_1} = 100 \cdot s^{-1} \qquad \omega_P = \frac{1}{R_1 \cdot C_1} = 1 \times 10^3 \cdot s^{-1} \qquad W(s) = A \cdot \frac{1 + \frac{s}{\omega_Z}}{1 + \frac{s}{\omega_P}}$$

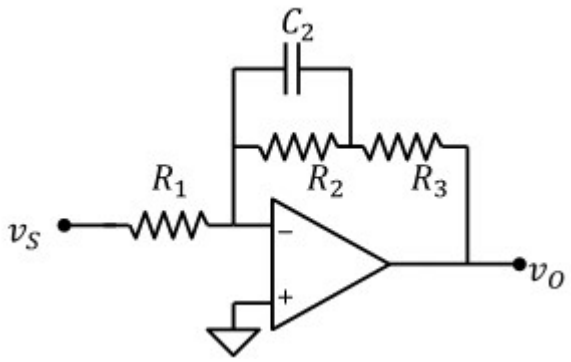
Diagramma di Bode asintotico del modulo e della fase





Esercizio 11

DATI: $R_1 = 2k\Omega$, $R_2 = 198k\Omega$, $R_3 = 2k\Omega$, $C_2 = 50.5nF$



Funzione di trasferimento

$$Z_2 = \frac{R_2}{1 + i\omega \cdot R_2 \cdot C_2}$$

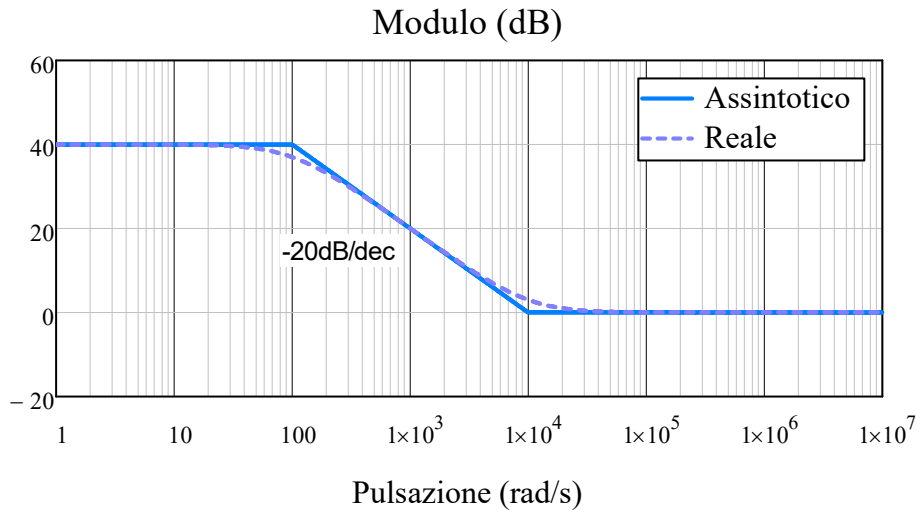
$$v_O = -\frac{R_3 + Z_2}{R_1} = -\left(\frac{R_2 + R_3}{R_1}\right) \cdot \frac{1 + i\omega \cdot \frac{R_3 \cdot R_2}{R_2 + R_3} \cdot C_2}{1 + i\omega \cdot R_2 \cdot C_2}$$

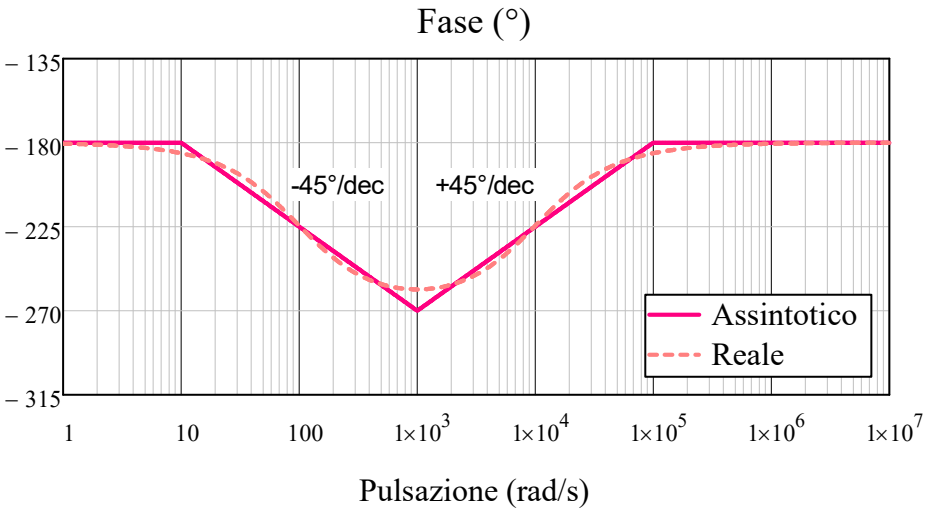
Definiamo:

$$R_P = \frac{R_3 \cdot R_2}{R_2 + R_3} = 1.98 \cdot k\Omega \qquad \boxed{\omega_Z = \frac{1}{C_2 \cdot R_P} = 1 \times 10^4 \cdot s^{-1}} \qquad \boxed{\omega_P = \frac{1}{C_2 \cdot R_2} = 100 \cdot s^{-1}} \qquad \boxed{A = -\left(\frac{R_2 + R_3}{R_1}\right) = -100}$$

$$\boxed{W(s) = A \cdot \frac{1 + \frac{s}{\omega_Z}}{1 + \frac{s}{\omega_P}}}$$

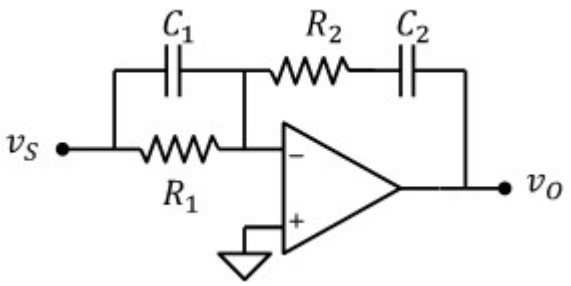
Diagramma di Bode asintotico del modulo e della fase





Esercizio 12

DATI: $R_1 = 10k\Omega$, $R_2 = 5k\Omega$, $C_1 = 1nF$, $C_2 = 20nF$



Funzione di trasferimento

$$Z_1 = \frac{R_1}{1 + i\omega \cdot R_1 \cdot C_1} \qquad Z_2 = \frac{1 + i\omega \cdot R_2 \cdot C_2}{j\omega \cdot C_2}$$

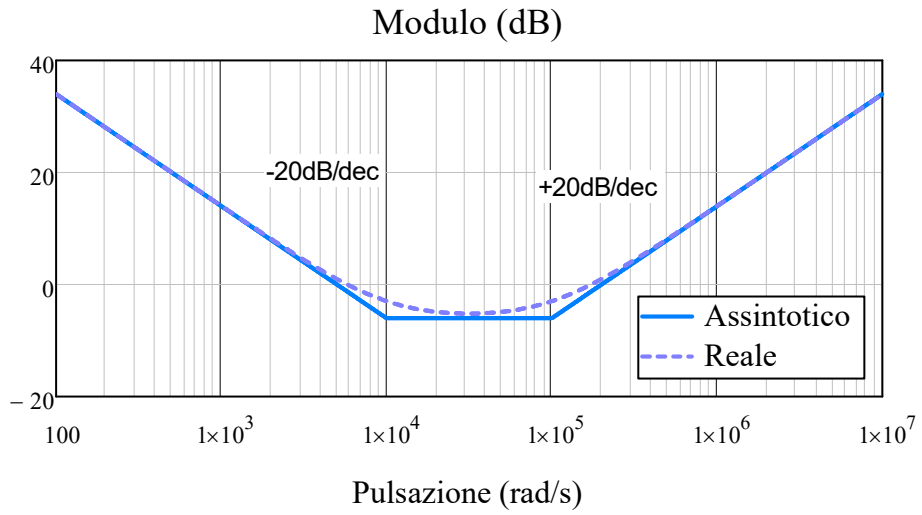
$$v_O = -v_S \cdot \frac{Z_2}{Z_1} = \frac{-R_2}{R_1} \cdot \frac{(1 + i\omega \cdot R_2 \cdot C_2) \cdot (1 + i\omega \cdot R_1 \cdot C_1)}{i\omega \cdot C_2 \cdot R_2} \cdot v_S$$

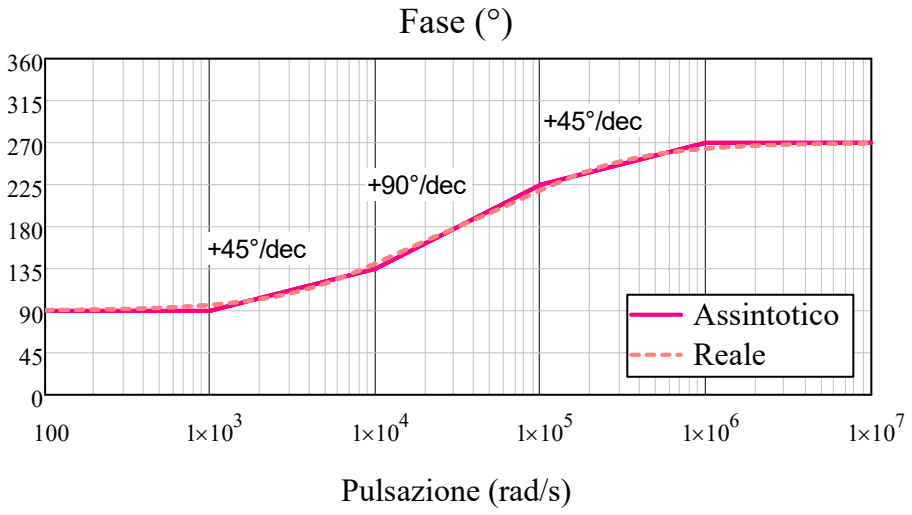
Definiamo:

$$A = \frac{-R_2}{R_1} = -0.5 \qquad \omega_{Z_1} = \frac{1}{C_1 \cdot R_1} = 1 \times 10^5 \cdot s^{-1} \qquad \omega_{Z_2} = \frac{1}{C_2 \cdot R_2} = 1 \times 10^4 \cdot s^{-1}$$

$$W(s) = A \cdot \frac{\left(1 + \frac{s}{\omega_{Z_1}}\right) \cdot \left(1 + \frac{s}{\omega_{Z_2}}\right)}{\frac{s}{\omega_{Z_2}}}$$

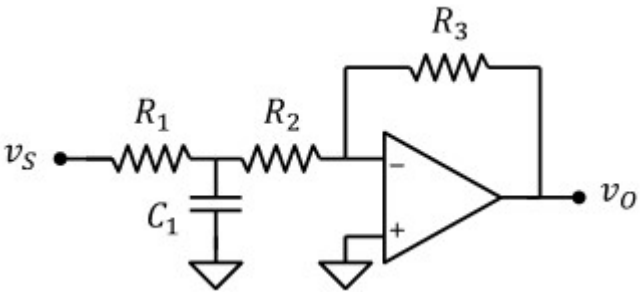
Diagramma di Bode asintotico del modulo e della fase





Esercizio 13

DATI: $R_1 = 1k\Omega$, $R_2 = 1k\Omega$, $R_3 = 200k\Omega$, $C_1 = 20nF$



Funzione di trasferimento

Rete equivalente secondo thevening all'ingresso invertente dell'AO

Tensione a vuoto: $v_{EQ} = v_S \cdot \dfrac{1}{1 + i\omega \cdot R_1 \cdot C_1}$

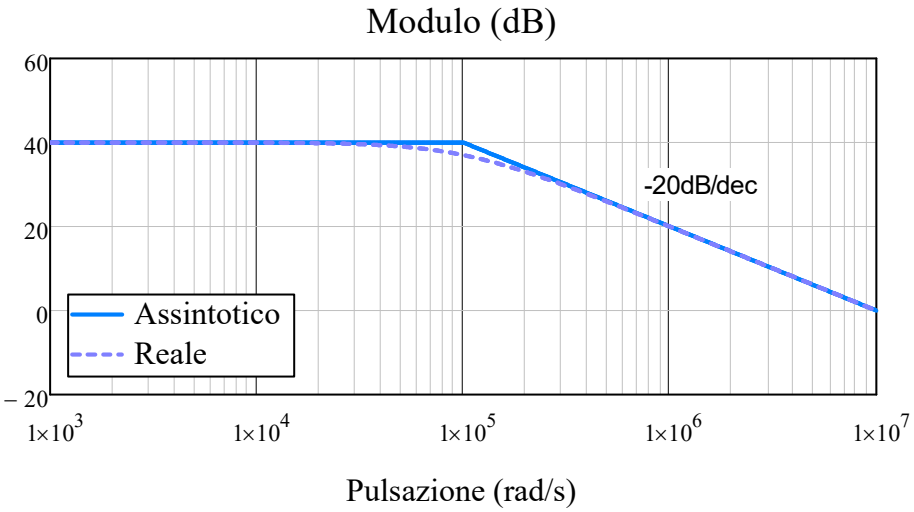
Resistenza equivalente (con v_S=0):

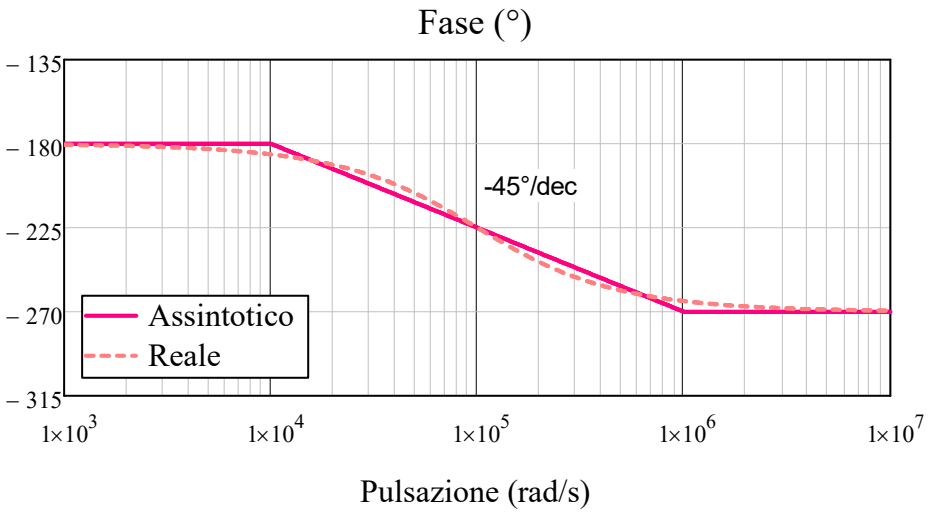
$$Z_{EQ} = R_2 + \frac{R_1}{1 + i\omega \cdot R_1 \cdot C_1} = \frac{R_2 + R_1 + i\omega \cdot R_1 \cdot R_2 \cdot C_1}{(1 + i\omega \cdot R_1 \cdot C_1)} = (R_2 + R_1) \cdot \frac{1 + i\omega \cdot \dfrac{R_1 \cdot R_2}{R_2 + R_1} \cdot C_1}{(1 + i\omega \cdot R_1 \cdot C_1)}$$

$$v_O = \frac{-R_3}{Z_{EQ}} \cdot v_{EQ} = -v_S \cdot \frac{1}{1 + i\omega \cdot R_1 \cdot C_1} \cdot \frac{R_3 \cdot (1 + i\omega \cdot R_1 \cdot C_1)}{(R_2 + R_1) \cdot \left(1 + i\omega \cdot \dfrac{R_1 \cdot R_2}{R_2 + R_1} \cdot C_1\right)} = -v_S \cdot \frac{R_3}{(R_2 + R_1)} \cdot \frac{1}{\left(1 + i\omega \cdot \dfrac{R_1 \cdot R_2}{R_2 + R_1} \cdot C_1\right)}$$

Poniamo: $\boxed{A = \dfrac{-R_3}{R_1 + R_2} = -100}$ $\quad R_P = \dfrac{R_1 \cdot R_2}{R_2 + R_1} = 500\,\Omega$ $\quad \boxed{\omega_P = (R_P \cdot C_1)^{-1} = 1 \times 10^5 \cdot s^{-1}}$ $\quad \boxed{W(s) = \dfrac{A}{1 + \dfrac{s}{\omega_P}}}$

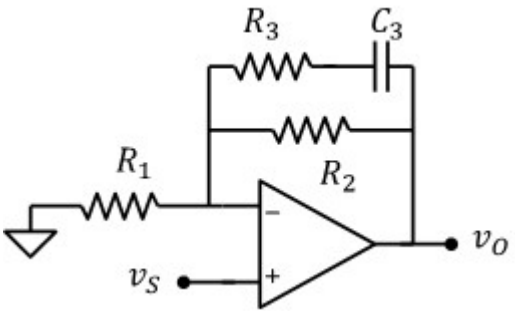
Diagramma di Bode asintotico del modulo e della fase





Esercizio 14

DATI: $R_1 = 1k\Omega$, $R_2 = 99k\Omega$, $R_3 = 1k\Omega$, $C_3 = 100nF$



1) Guadagno in condizioni stazionarie

$$\boxed{A_v = 1 + \frac{R_2}{R_1} = 100}$$

2) Guadagno ad alta frequenza

$$R_P = \frac{R_2 \cdot R_3}{R_2 + R_3} \qquad \boxed{A_v = 1 + \frac{R_P}{R_1} = 1.99}$$

2) Ricavare l'espressione della funzione di trasferimento

$$Z_2 = \frac{R_2 \cdot \left(R_3 + \frac{1}{i\omega \cdot C_3}\right)}{R_2 + \left(R_3 + \frac{1}{i\omega \cdot C_3}\right)} = \frac{R_2 \cdot (i\omega \cdot C_3 \cdot R_3 + 1)}{i\omega \cdot C_3 \cdot (R_2 + R_3) + 1}$$

$$v_O = \left(1 + \frac{Z_2}{R_1}\right) \cdot v_S = 1 + \frac{R_2 \cdot (s \cdot C_3 \cdot R_3 + 1)}{R_1 \cdot \left[s \cdot C_3 \cdot (R_2 + R_3) + 1\right]} = \left(1 + \frac{R_2}{R_1}\right) \cdot \frac{1 + i\omega \cdot C_3 \cdot \left(\frac{R_2 \cdot R_1}{R_1 + R_2} + R_3\right)}{1 + i\omega \cdot C_3 \cdot (R_2 + R_3)} \cdot v_S$$

Poniamo:

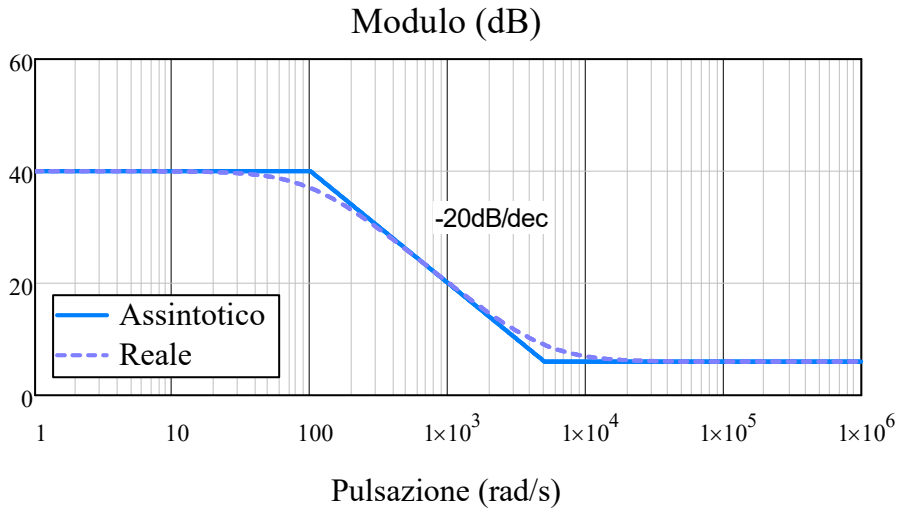
$$\boxed{\omega_{Z_1} = \left[C_3 \cdot \left(\frac{R_2 \cdot R_1}{R_1 + R_2} + R_3\right)\right]^{-1} = 5 \times 10^3 \cdot s^{-1}} \quad \boxed{\omega_{P_1} = \frac{1}{C_3 \cdot (R_2 + R_3)} = 100 \cdot s^{-1}}$$

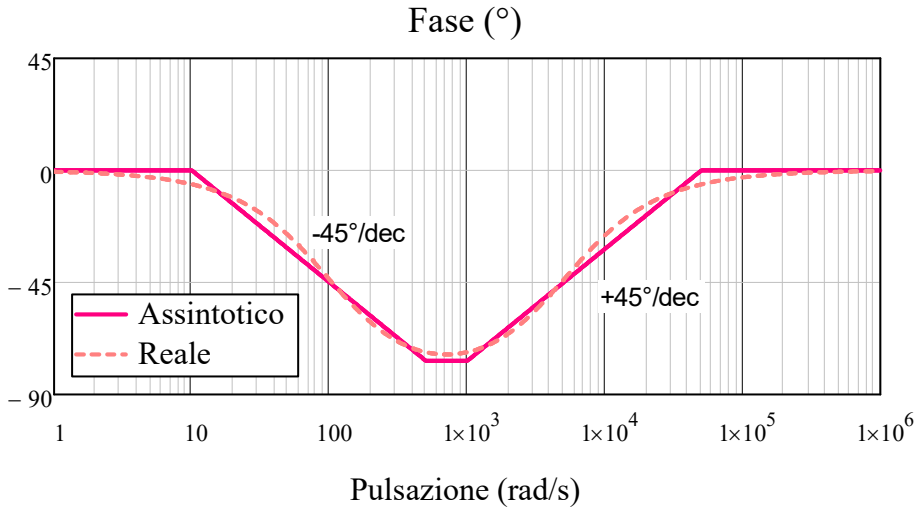
$$\boxed{A = \left(1 + \frac{R_2}{R_1}\right) = 100}$$

Funzione di trasferimento:

$$\boxed{W(s) = A \cdot \frac{1 + \frac{s}{\omega_{Z_1}}}{1 + \frac{s}{\omega_{P_1}}}}$$

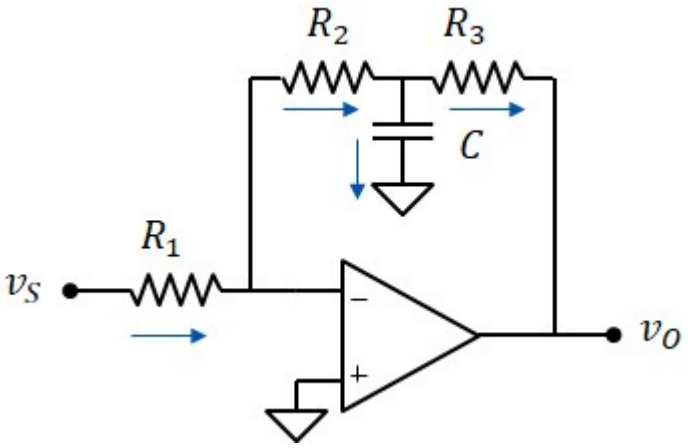
3) Tracciare il diagramma di Bode asintotico del modulo e della fase





Esercizio 15

DATI: $R_1 = 10k\Omega$, $R_2 = 10k\Omega$, $R_3 = 10k\Omega$, $\omega_{Z_1} = 1000s^{-1}$



1) Funzione di trasferimento

Corrente attraverso R_1 e R_2

$$I_{R1} = \frac{v_S}{R_1} \qquad I_{R2} = I_{R1} = \frac{v_S}{R_1}$$

Potenziale ai capi della capacità C:

$$v_C = 0 - V_{R2} = \frac{-R_2}{R_1}\cdot v_S$$

Corrente attraverso C:

$$I_C = \frac{v_C}{Z_C} = i\omega\cdot C\cdot v_C$$

Corrente attraverso R_3

$$I_{R3} = I_{R2} - I_C = \frac{v_S}{R_1} - i\omega\cdot C\cdot v_C = \frac{v_S}{R_1} - i\omega\cdot C\cdot\left(\frac{-R_2}{R_1}\cdot v_S\right) = \left(\frac{1}{R_1} + \frac{i\omega\cdot C\cdot R_2}{R_1}\right)\cdot v_S$$

Tensione di uscita: $$v_O = v_C - V_{R3} = \frac{-R_2}{R_1}\cdot v_S - R_3\cdot\left(\frac{1}{R_1} + \frac{i\omega\cdot C\cdot R_2}{R_1}\right)\cdot v_S = -\left(\frac{R_2}{R_1} + \frac{R_3}{R_1} + \frac{i\omega\cdot C\cdot R_2\cdot R_3}{R_1}\right)\cdot v_S$$

$$W(s) = -\frac{R_2 + R_3}{R_1}\cdot\left(1 + s\cdot C\cdot\frac{R_2\cdot R_3}{R_2 + R_3}\right)$$

Definiamo: $$A = -\frac{R_2 + R_3}{R_1} = -2$$

$$R_P = \frac{R_2\cdot R_3}{R_2 + R_3} = 5\cdot k\Omega$$

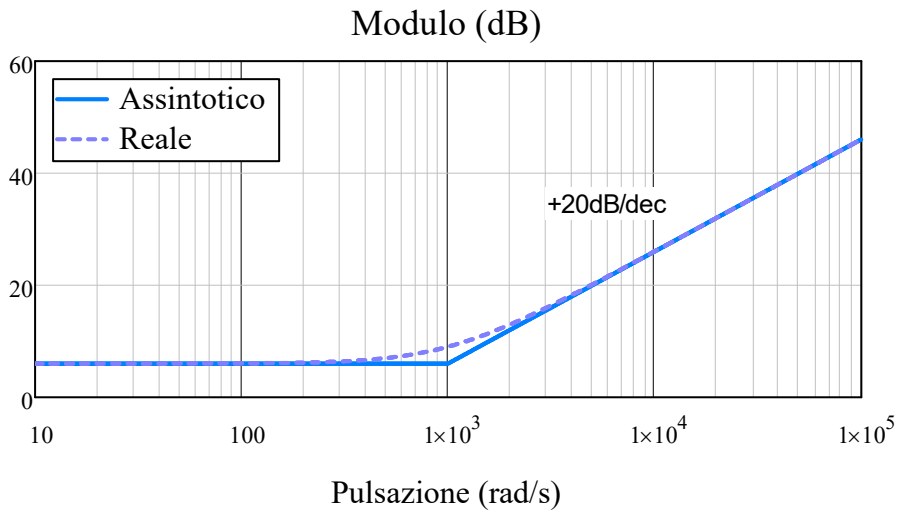
$$\boxed{W(s) = A\cdot\left(1 + \frac{s}{\omega_{Z_1}}\right)}$$

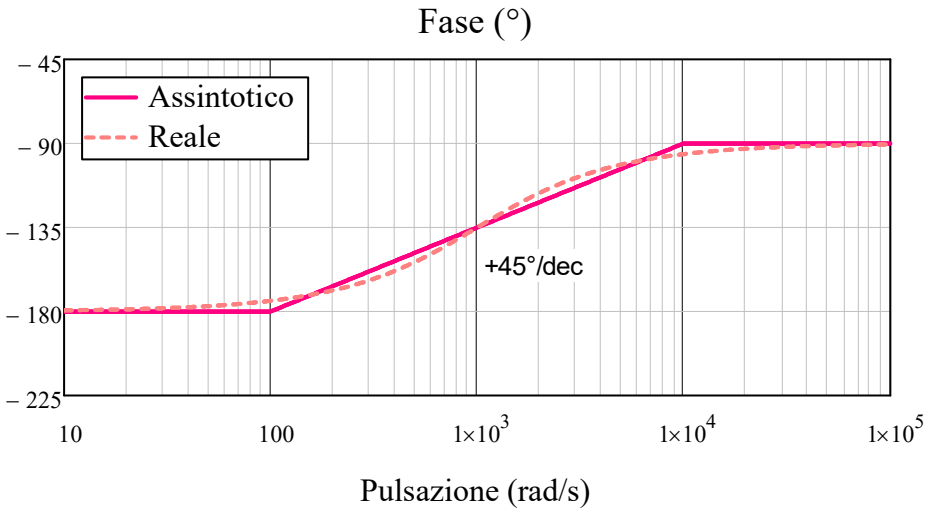
funzione di trasferimento con un solo zero in: $\omega_{Z_1} = \frac{1}{C\cdot R_P}$

2) valore di C per avere $\omega_{Z_1} = 1\times 10^3\cdot s^{-1}$

$$\boxed{C = \frac{1}{\omega_{Z_1}\cdot R_P} = 200\cdot nF}$$

3) Diagramma di Bode asintotico del modulo e della fase

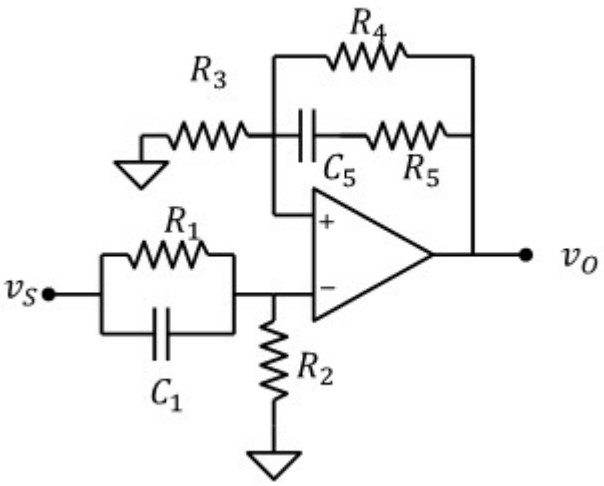




Esercizio 16

DATI:

$R_2 = 1k\Omega, R_3 = 1k\Omega, R_4 = 99k\Omega\ C_1 = 11\,1nF,\ C_5 = 918nF$



1) calcolare R_1 e R_2 affinché $R_{IN} = 10k\Omega$ in ω=0

e |W(0)| = |W(∞)|

$R_{IN} = R_1 + R_2$ $\quad$ $\boxed{R_1 = R_{IN} - R_2 = 9\cdot k\Omega}$

Guadagno in ω=0 $\quad A_0 = \dfrac{R_2}{R_1 + R_2}\cdot\left(1 + \dfrac{R_4}{R_3}\right) = 10$

Guadagno in ω=∞ $\quad A_{HF} = \left(1 + \dfrac{R_P}{R_3}\right)$

con: $\quad R_P = \dfrac{R_4\cdot R_5}{R_4 + R_5}$ $\quad R_P = (A_0 - 1)\cdot R_3 = 9\cdot k\Omega$ $\quad \boxed{R_5 = \left(\dfrac{1}{R_P} - \dfrac{1}{R_4}\right)^{-1} = 9.9\cdot k\Omega}$

2) Funzione di trasferimento

Calcoliamo le impedenze: $\quad Z_1 = \dfrac{R_1}{1 + i\omega\cdot R_1\cdot C_1}$ $\quad Z_5 = \dfrac{1 + i\omega\cdot R_5\cdot C_5}{i\omega\cdot C_5}$ $\quad Z_4 = \dfrac{R_4\cdot Z_5}{R_4 + Z_5} = \dfrac{R_4\cdot(1 + i\omega\cdot R_5\cdot C_5)}{1 + i\omega\cdot(R_5 + R_4)\cdot C_5}$

Potenziale del terminale non invertente: $\quad v_P = v_S\cdot\dfrac{R_2}{R_2 + Z_1} = v_S\cdot\dfrac{R_2}{R_1 + R_2}\cdot\dfrac{(1 + i\omega\cdot R_1\cdot C_1)}{1 + i\omega\cdot\dfrac{R_2\cdot R_1}{R_1 + R_2}\cdot C_1}$

Nodo di uscita:

$$v_O = v_P\cdot\left(1 + \frac{Z_4}{R_3}\right) = v_P\cdot\left[1 + \frac{1}{R_3}\cdot\frac{R_4\cdot(1 + i\omega\cdot R_5\cdot C_5)}{1 + i\omega\cdot(R_5 + R_4)\cdot C_5}\right] = v_P\cdot\left[\frac{1}{R_3}\cdot\frac{R_3 + R_4 + i\omega\cdot(R_5\cdot R_3 + R_4\cdot R_3 + R_5\cdot R_4)\cdot C_5}{1 + i\omega\cdot(R_5 + R_4)\cdot C_5}\right]$$

$$v_O = v_P\cdot\left[\frac{R_3 + R_4}{R_3}\cdot\frac{1 + i\omega\cdot\left(R_5 + \dfrac{R_3\cdot R_4}{R_3 + R_4}\right)\cdot C_5}{1 + i\omega\cdot(R_5 + R_4)\cdot C_5}\right]$$

$$v_O = \frac{R_2}{R_1 + R_2}\cdot\frac{(1 + i\omega\cdot R_1\cdot C_1)}{1 + i\omega\cdot\dfrac{R_2\cdot R_1}{R_1 + R_2}\cdot C_1}\cdot\frac{1 + i\omega\cdot\left(R_5 + \dfrac{R_4\cdot R_3}{R_3 + R_4}\right)\cdot C_5}{1 + i\omega\cdot(R_5 + R_4)\cdot C_5}\cdot v_S$$

$$\frac{R_2\cdot R_1}{R_1 + R_2} = 900\,\Omega \qquad R_4 + R_5 = 108.9\cdot k\Omega \qquad R_5 + \frac{R_4\cdot R_3}{R_3 + R_4} = 10.89\cdot k\Omega$$

Poniamo: $A = \frac{R_2}{R_1 + R_2} \cdot \frac{R_3 + R_4}{R_3} = 10$ che coincide con il guadagno per ω=0

$$\omega_{Z_1} = \left[\left(R_5 + \frac{R_4 \cdot R_3}{R_3 + R_4}\right) \cdot C_5\right]^{-1} = 100 \cdot s^{-1}$$

$$\omega_{Z_2} = \left(R_1 \cdot C_1\right)^{-1} = 1 \times 10^3 \cdot s^{-1}$$

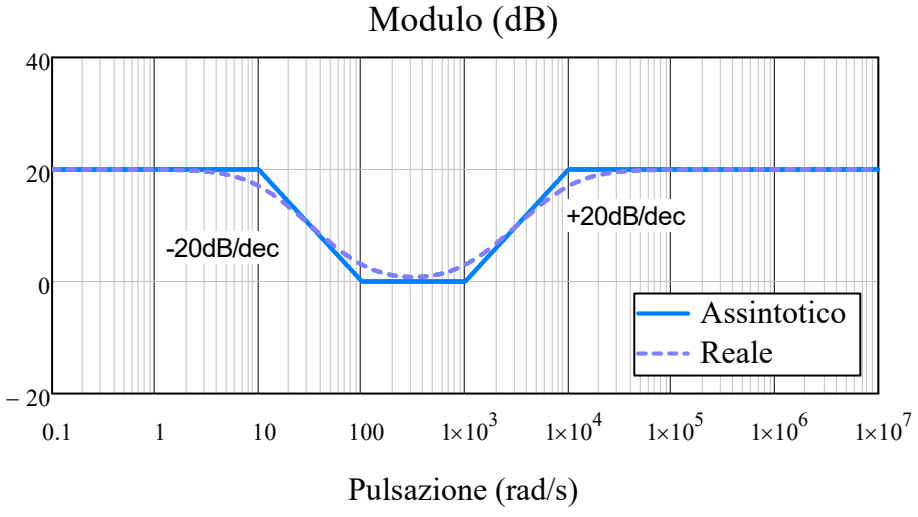
$$\omega_{P_1} = \left[\left(R_5 + R_4\right) \cdot C_5\right]^{-1} = 10 \cdot s^{-1}$$

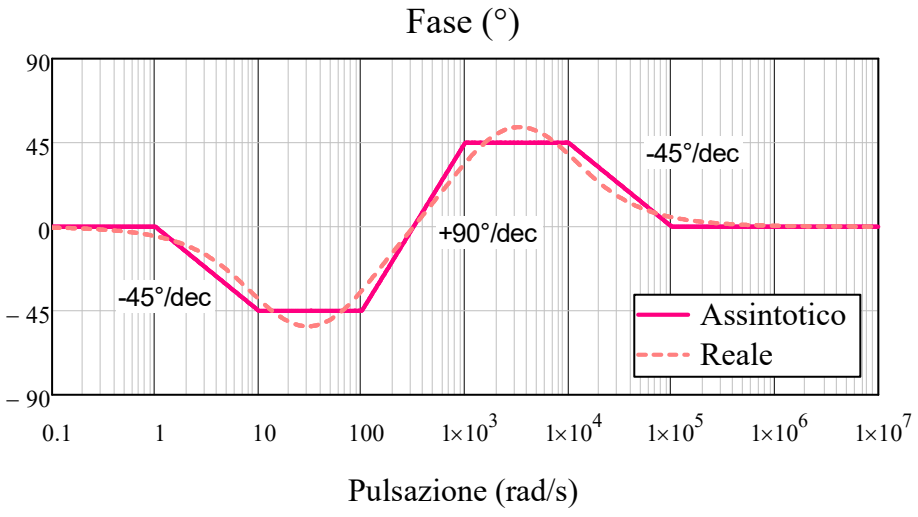
$$\omega_{P_2} = \left(\frac{R_2 \cdot R_1}{R_1 + R_2} \cdot C_1\right)^{-1} = 1 \times 10^4 \cdot s^{-1}$$

Otteniamo la seguente funzione di trasferimento:

$$W(s) = A \cdot \frac{\left(1 + \frac{s}{\omega_{Z_1}}\right) \cdot \left(1 + \frac{s}{\omega_{Z_2}}\right)}{\left(1 + \frac{s}{\omega_{P_1}}\right) \cdot \left(1 + \frac{s}{\omega_{P_2}}\right)}$$

3) Diagramma di Bode asintotico del modulo e della fase





5) Ampiezza e fase del segnale di uscita, con segnale di ingresso:

$V_{S1} = 10mV, \quad \omega_1 = 10^6 s^{-1}, \quad \varphi_1 = 180°$

$V_{S2} = 1mV, \quad \omega_2 = 690 s^{-1}, \quad \varphi_2 = 90°$

$v_S(t) = V_{S1} \cdot \sin(\omega_1 \cdot t + \varphi_1) + V_{S2} \cdot \sin(\omega_2 \cdot t + \varphi_2)$

Guadagno in ω_1

$W_1 = 20dB \quad A_1 = 10^{\frac{W_1}{20}} = 10$

$\Delta\varphi_1 = 0°$

$V_{O1} = A_1 \cdot V_{S1} = 0.1\,V$

$\varphi_{O1} = \varphi_1 + \Delta\varphi_1 = 180 \cdot °$

Guadagno in ω_2

$W_2 = 0dB \quad A_2 = 10^{\frac{W_2}{20}} = 1$

$\Delta\varphi_2 = -45° + 90° \cdot \log\left(\frac{\omega_2}{\omega_{Z_1}}\right) = 30 \cdot °$

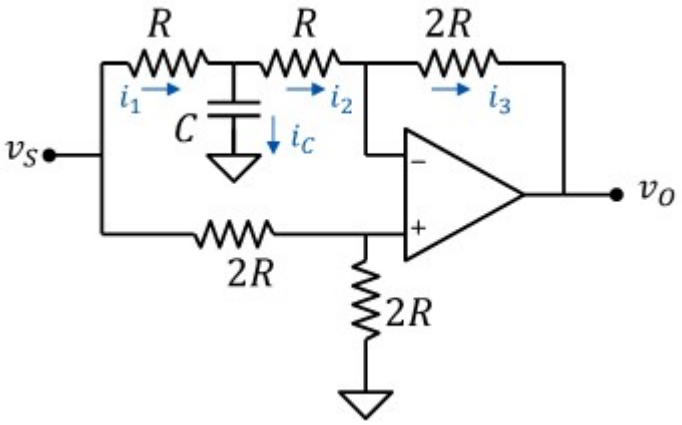
$V_{O2} = A_2 \cdot V_{S2} = 1 \cdot mV$

$\varphi_{O2} = \varphi_2 + \Delta\varphi_2 = 120 \cdot °$

$v_O(t) = V_{O1} \cdot \sin(\omega_1 \cdot t + \varphi_{O1}) + V_{O2} \cdot \sin(\omega_2 \cdot t + \varphi_{O2})$

Esercizio 17

DATI: $R = 2k\Omega$, $C = 100nF$
Segnale di ingresso: $V_S = 5V$, $\varphi_S = 45°$



1) Funzione di trasferimento

$$v_P = \frac{v_S}{2} = v_N$$

Dalle leggi di kirchhoff ricaviamo:

$$i_1 = i_C + i_2 \qquad i_2 = i_3$$

Sia v_C la tensione ai capi del condensatore:

$$i_1 = \frac{v_S - v_C}{R} \qquad i_2 = \frac{v_C - \frac{v_S}{2}}{R} \qquad i_3 = \frac{v_O - \frac{v_S}{2}}{2R} \qquad i_C = v_C \cdot i\omega \cdot C$$

$$\frac{v_S - v_C}{R} = \frac{v_C - \frac{v_S}{2}}{R} + v_C \cdot i\omega \cdot C$$

Ricaviamo v_C:

$$v_C = \frac{3v_S}{2\cdot(2 + i\omega\cdot R\cdot C)}$$

Calcoliamo $i_3 = i_2$

$$i_3 = i_2 = \frac{v_C - \frac{v_S}{2}}{R} = \frac{1 - i\omega\cdot R\cdot C}{2 + i\omega\cdot R\cdot C}\cdot\frac{v_S}{2R}$$

e infine v_O:

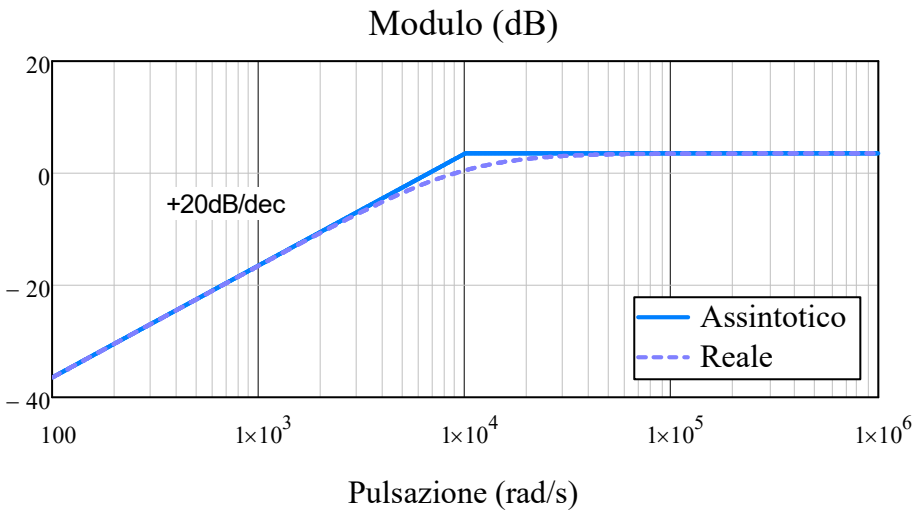
$$v_O = \frac{v_S}{2} - 2R\cdot i_3 = \frac{v_S}{2} - \left(\frac{1 - i\omega\cdot R\cdot C}{2 + i\omega\cdot R\cdot C}\cdot v_S\right) = v_S\cdot\left[\frac{i\omega\cdot 3R\cdot C}{4\cdot\left(1 + i\omega\cdot\frac{R}{2}\cdot C\right)}\right]$$

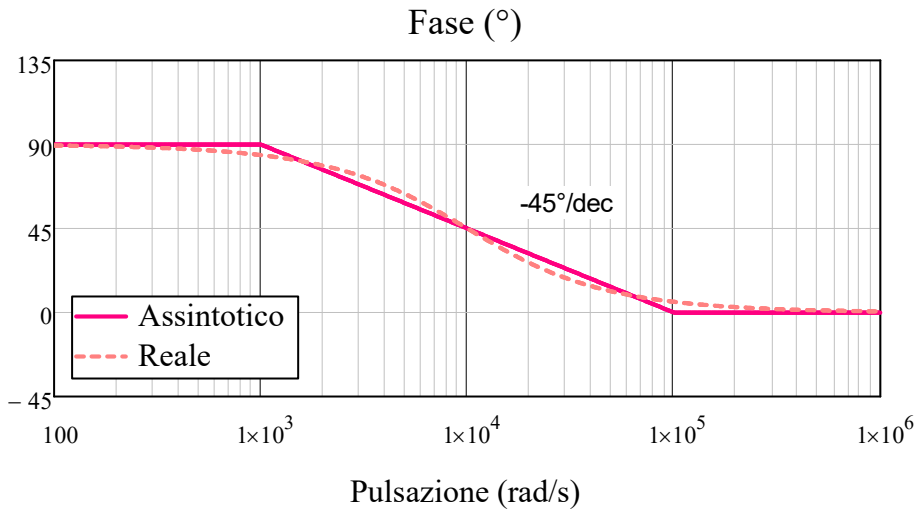
Poniamo: $A = \frac{3}{2} \qquad \omega_{P_1} = \frac{2}{R\cdot C} = 1\times 10^4\cdot s^{-1}$

Otteniamo la seguente funzione di trasferimento:

$$\boxed{W(s) = A\cdot\frac{\frac{s}{\omega_{P_1}}}{1 + \frac{s}{\omega_{P_1}}}}$$

2) diagramma di Bode asintotico del modulo e della fase





3a) Ampiezza e fase del segnale di uscita, con segnale di ingresso a pulsazione: $\omega_S = 400s^{-1}$

$$v_S(t) = V_S \cdot \sin(\omega_S \cdot t + \varphi_S)$$

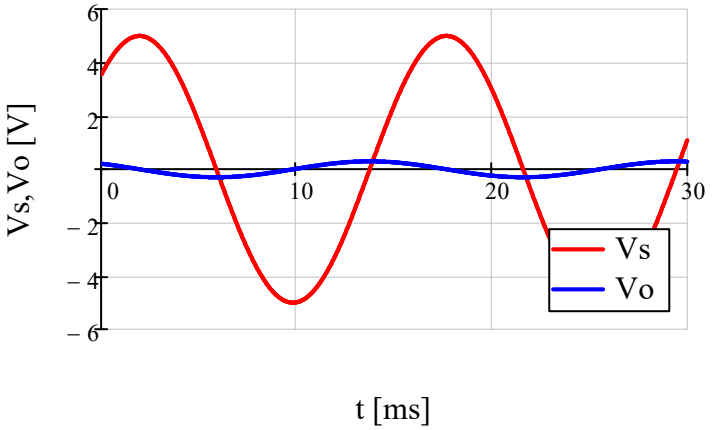
Modulo del guadagno asintotico per $\omega=\omega_S$:

$$A_{S_{dB}} = \left(\left|W(i\omega_S)\right|\right)_{dB} = \left(\left|W(i\omega_{P_1})\right|\right)_{dB} + 20\cdot\log\left(\frac{\omega_S}{\omega_{P_1}}\right) \qquad A_{S_{dB}} = -24.5dB \qquad A_S = 10^{\frac{A_{S_{dB}}}{20}} = 0.06$$

Fase del guadagno asintotico per $\omega=\omega_S$: $\Delta\varphi_S = 90°$

$$\boxed{V_O = A_S \cdot V_S = 0.3\,V} \qquad \boxed{\varphi_O = \varphi_S + \Delta\varphi_S = 135\cdot°}$$

$$\boxed{v_O(t) = V_O \cdot \sin(\omega_S \cdot t + \varphi_O)}$$



3b) Ampiezza e fase del segnale di uscita, con segnale di ingresso a pulsazione: $\omega_S = 5000 s^{-1}$

Modulo del guadagno asintotico per $\omega=\omega_S$:

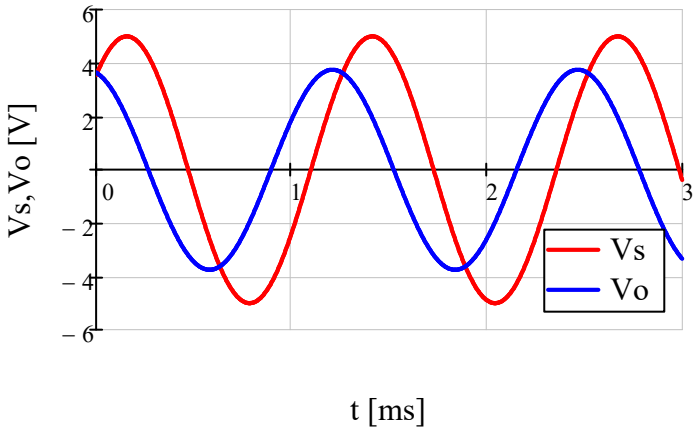
$$A_{S_{dB}} = \left(\left|W(i\omega_S)\right|\right)_{dB} = \left(\left|W(i\omega_{P_1})\right|\right)_{dB} + 20\cdot\log\left(\frac{\omega_S}{\omega_{P_1}}\right) \qquad A_{S_{dB}} = -2.5dB \qquad A_S = 10^{\frac{A_{S_{dB}}}{20}} = 0.75$$

Fase del guadagno asintotico per $\omega=\omega_S$:

$$\Delta\varphi_S = 90° - 45°\cdot\log\left(\frac{\omega_S}{0.1\,\omega_{P_1}}\right) = 58.5\cdot°$$

$$V_O = A_S\cdot V_S = 3.75\,V \qquad \varphi_O = \varphi_S + \Delta\varphi_S = 103.5\cdot°$$

$$v_O(t) = V_O\cdot\sin(\omega_S\cdot t + \varphi_O)$$



3c) Ampiezza e fase del segnale di uscita, con segnale di ingresso a pulsazione: $\omega_S = 20000 s^{-1}$

Modulo del guadagno asintotico per $\omega=\omega_S$:

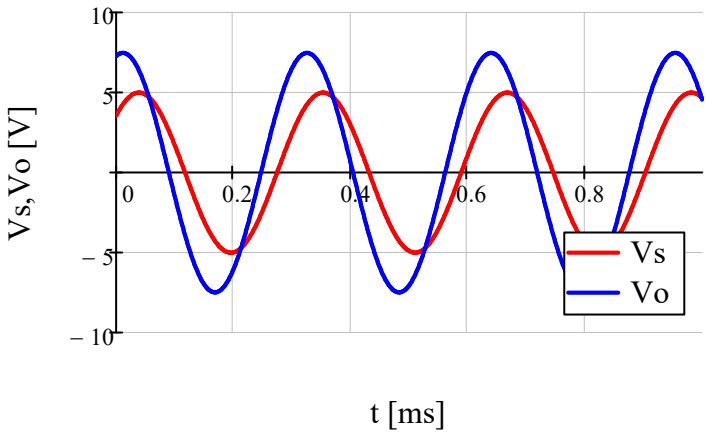
$$A_{S_{dB}} = 3.5dB \qquad A_S = 10^{\frac{A_{S_{dB}}}{20}} = 1.5$$

Fase del guadagno asintotico per $\omega=\omega_S$:

$$\Delta\varphi_S = 90° - 45°\cdot\log\left(\frac{\omega_S}{0.1\,\omega_{P_1}}\right) = 31.5\cdot°$$

$$V_O = A_S\cdot V_S = 7.5\,V \qquad \varphi_O = \varphi_S + \Delta\varphi_S = 76.5\cdot°$$

$$v_O(t) = V_O\cdot\sin(\omega_S\cdot t + \varphi_O)$$



Esercizio 18

DATI:

$R_1 = 10k\Omega$,

$R_2 = 200k\Omega$,

$R_3 = 1k\Omega$,

$R_4 = 5k\Omega$

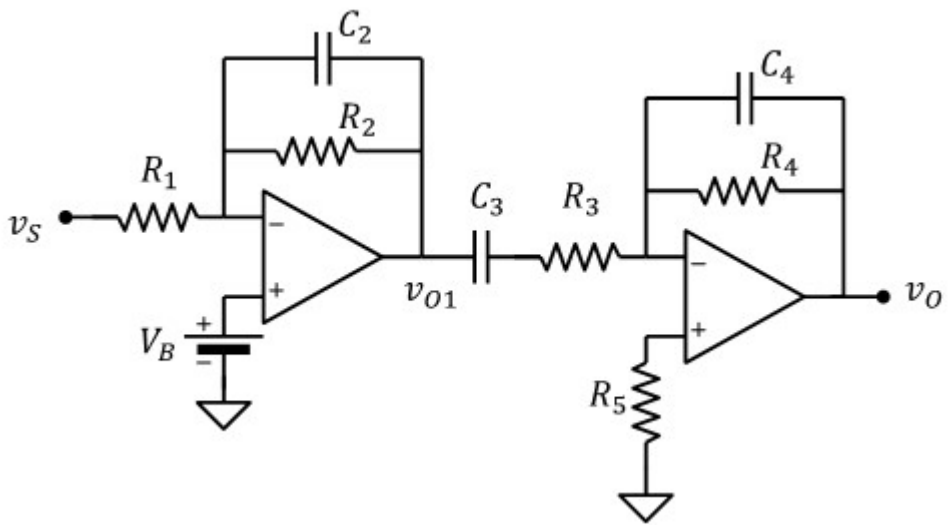
$R_5 = 3.3k\Omega$

$C_2 = 50pF$,

$C_3 = 10\mu F$,

$C_4 = 20nF$,

$V_B = 1V$



1) resistenza di ingresso

$R_{IN} = R_1 = 10 \cdot k\Omega$

2) La tensione di uscita in condizioni stazionarie con $v_S = 200mV$

Tutte le capacità sono circuiti aperti.

$v_O = 0$

3) Funzione di trasferimento

Applichiamo la sovrapposizione degli effetti:
con $v_S = 0V$ e V_B ON

$$v_{O1} = V_B \cdot \left(1 + \frac{Z_2}{R_2}\right)$$

poichè VB è costante, l'uscita è costante pari a:

$$v_{O1} = V_B \cdot \left(1 + \frac{R_2}{R_1}\right) = 21\,V$$

il secondo stadio in presenza di un ingresso costante ha $v_O = 0$ poichè all suo ingresso la capacità C_3 è in aperto.
Di conseguenza $v_N = v_P = v_O = 0$

con $V_B = 0$ e v_S ON:

Primo stadio

$$Z_2 = \frac{R_2}{1 + i\omega \cdot R_2 \cdot C_2}$$

$$v_{O1} = -v_S \cdot \frac{Z_2}{R_1} = \frac{-R_2}{R_1} \cdot \frac{1}{1 + i\omega \cdot R_2 \cdot C_2} \cdot v_S$$

Secondo stadio

$$v_O = -\frac{Z_4}{Z_3} \cdot v_{O1} \qquad Z_3 = \frac{1 + i\omega \cdot R_3 C_3}{i\omega \cdot C_3} \qquad Z_4 = \frac{R_4}{1 + i\omega \cdot R_4 \cdot C_4}$$

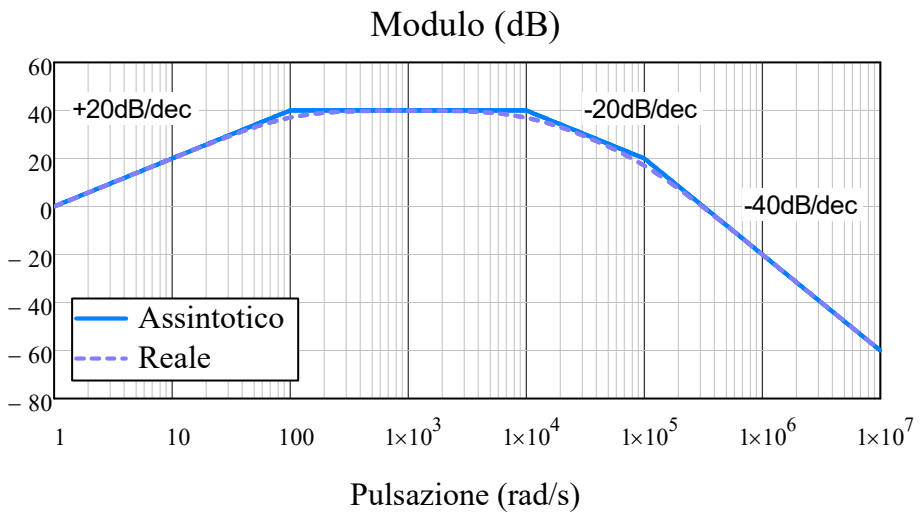
$$v_O = -\frac{i\omega \cdot C_3 \cdot R_4}{\left(1 + i\omega \cdot R_3 C_3\right)\cdot\left(1 + i\omega \cdot R_4 \cdot C_4\right)} \cdot v_{O1} = \frac{i\omega \cdot C_3 \cdot R_4}{\left(1 + i\omega \cdot R_3 C_3\right)\cdot\left(1 + i\omega \cdot R_4 \cdot C_4\right)} \cdot \left(\frac{R_2}{R_1} \cdot \frac{1}{1 + i\omega \cdot R_2 \cdot C_2} \cdot v_S\right)$$

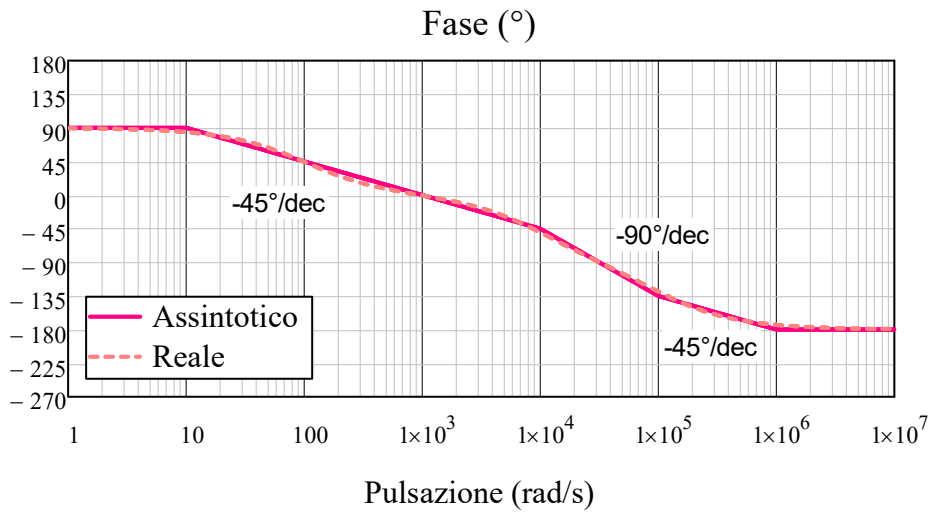
$$v_O = \frac{R_2 \cdot R_4}{R_1 \cdot R_3} \cdot \frac{i\omega \cdot R_3 \cdot C_3}{\left(1 + i\omega \cdot R_2 \cdot C_2\right)\cdot\left(1 + i\omega \cdot R_3 C_3\right)\cdot\left(1 + i\omega \cdot R_4 \cdot C_4\right)} \cdot v_S$$

Poniamo: $A = \frac{R_2 \cdot R_4}{R_1 \cdot R_3} = 100$ $\quad \omega_{P_1} = \frac{1}{R_3 \cdot C_3} = 100 \cdot s^{-1}$ $\quad \omega_{P_2} = \frac{1}{R_4 \cdot C_4} = 1 \times 10^4 \cdot s^{-1}$ $\quad \omega_{P_3} = \frac{1}{R_2 \cdot C_2} = 1 \times 10^5 \cdot s^{-1}$

$$W(s) = A \cdot \frac{\frac{s}{\omega_{P_1}}}{\left(1 + \frac{s}{\omega_{P_1}}\right)\cdot\left(1 + \frac{s}{\omega_{P_2}}\right)\cdot\left(1 + \frac{s}{\omega_{P_3}}\right)}$$

4) Diagramma di Bode asintotico del modulo e della fase





5) Ampiezza e fase del segnale di uscita, con segnale di ingresso:

$V_{S0} = 1V$

$V_{S1} = 10mV, \quad \omega_1 = 10^3 s^{-1}, \quad \varphi_1 = 180°$

$V_{S2} = 300mV, \quad \omega_2 = 10^6 s^{-1}, \quad \varphi_2 = 90°$

$v_S(t) = V_{S0} + V_{S1} \cdot \sin(\omega_1 \cdot t + \varphi_1) + V_{S2} \cdot \sin(\omega_2 \cdot t + \varphi_2)$

Guadagno in DC (ω=0): $W_0 = 0$ $\quad V_{O0} = W_0 \cdot V_{S0} = 0$

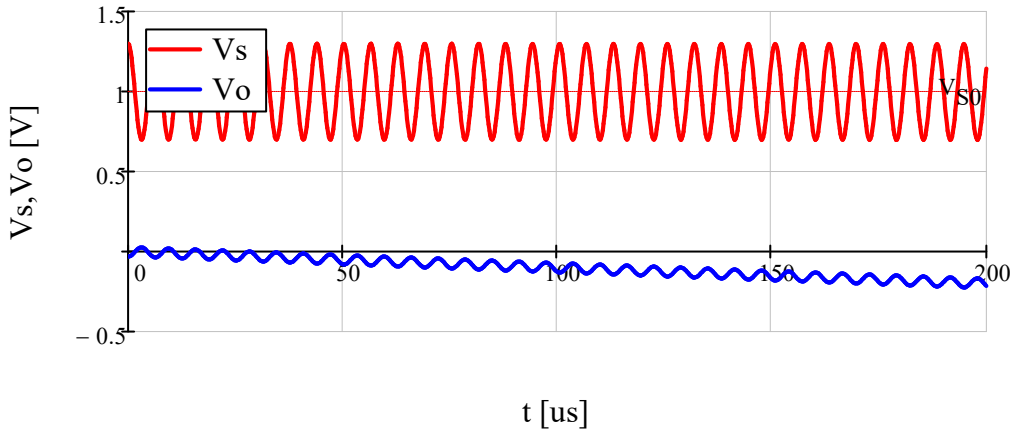
Guadagno in ω_1: $W_1 = 40dB \quad A_1 = 10^{\frac{W_1}{20}} = 100$ $\quad V_{O1} = A_1 \cdot V_{S1} = 1\,V$

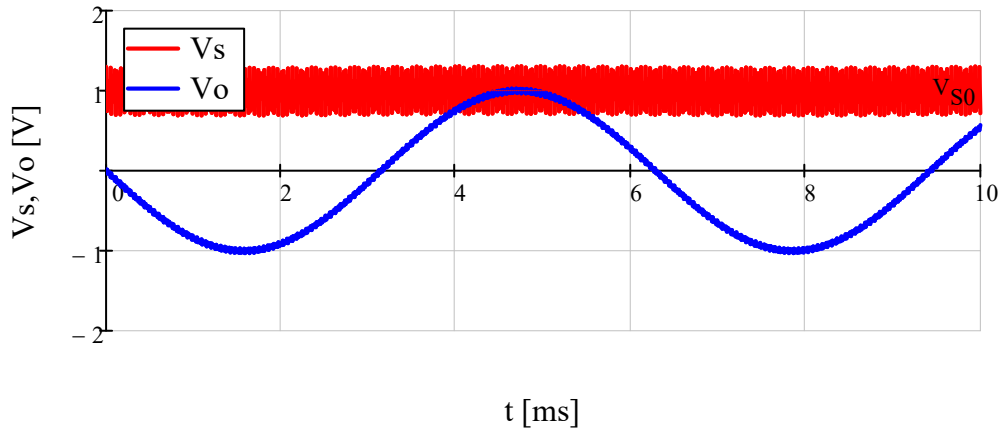
$\Delta\varphi_1 = 0°$ $\quad \varphi_{O1} = \varphi_1 + \Delta\varphi_1 = 180 \cdot °$

Guadagno in ω_2: $W_2 = -20dB \quad A_2 = 10^{\frac{W_2}{20}} = 0.1$ $\quad V_{O2} = A_2 \cdot V_{S2} = 30 \cdot mV$

$\Delta\varphi_2 = -180°$ $\quad \varphi_{O2} = \varphi_2 + \Delta\varphi_2 = -90 \cdot °$

$v_O(t) = V_{O0} + V_{O1} \cdot \sin(\omega_1 \cdot t + \varphi_{O1}) + V_{O2} \cdot \sin(\omega_2 \cdot t + \varphi_{O2})$





Esercizio 19

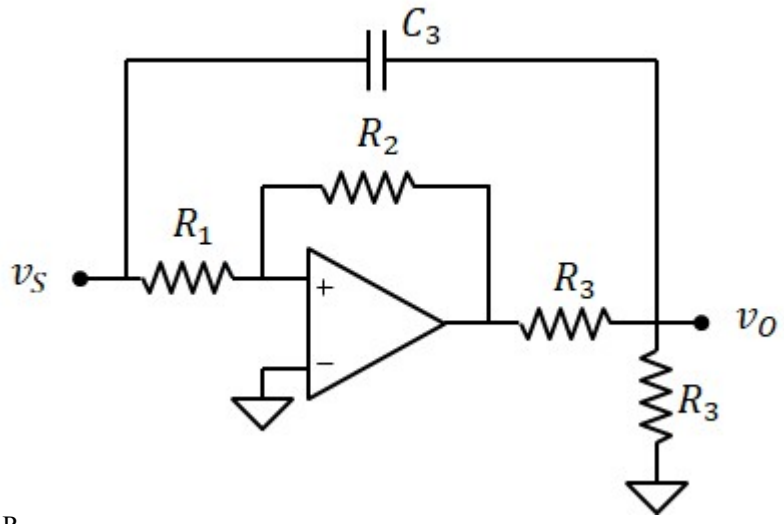
DATI:

$R_1 = 1k\Omega$

$R_2 = 20k\Omega$

$R_3 = 5k\Omega$

$C_3 = 400nF$



1) Funzione di trasferimento

Tensione di uscita dell'operazionale: $v_A = \frac{-R_2}{R_1} \cdot v_S$

Tensione di uscita (legge di kirchhoff):

$$\frac{v_O}{R_3} + (v_O - v_S) \cdot i\omega \cdot C_3 + \frac{v_O - v_A}{R_3} = 0$$

$$\frac{v_O}{R_3} + i\omega \cdot C_3 \cdot v_O + \frac{v_O}{R_3} = \frac{v_A}{R_3} + i\omega \cdot C_3 \cdot v_S$$

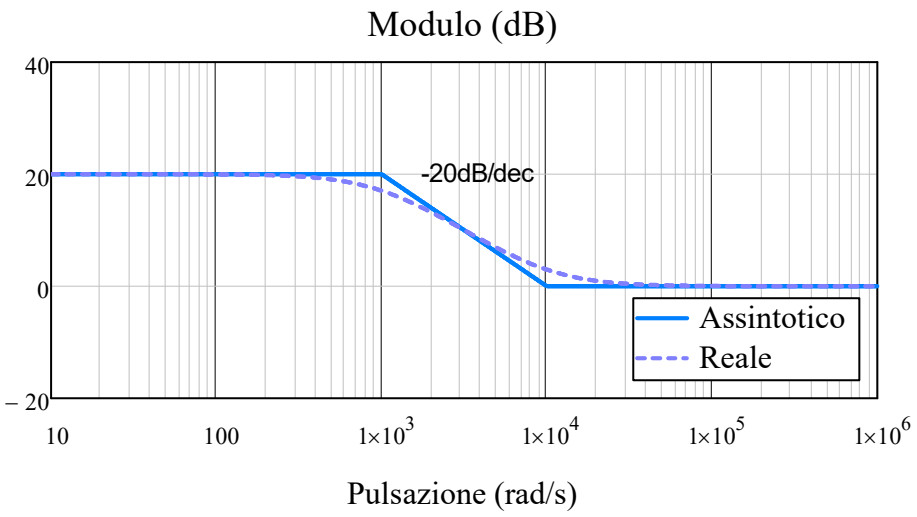
$$v_O = \frac{R_3}{2 + i\omega \cdot C_3 \cdot R_3} \cdot \left(\frac{v_A}{R_3} + i\omega \cdot C_3 \cdot v_S \right) = \frac{R_3}{2 + i\omega \cdot C_3 \cdot R_3} \cdot \left(\frac{-R_2}{R_1 \cdot R_3} + i\omega \cdot C_3 \right) \cdot v_S = \frac{R_3}{2 + i\omega \cdot C_3 \cdot R_3} \cdot \left(\frac{-R_2 + i\omega \cdot C_3 \cdot R_1 \cdot R_3}{R_1 \cdot R_3} \right) \cdot v_S$$

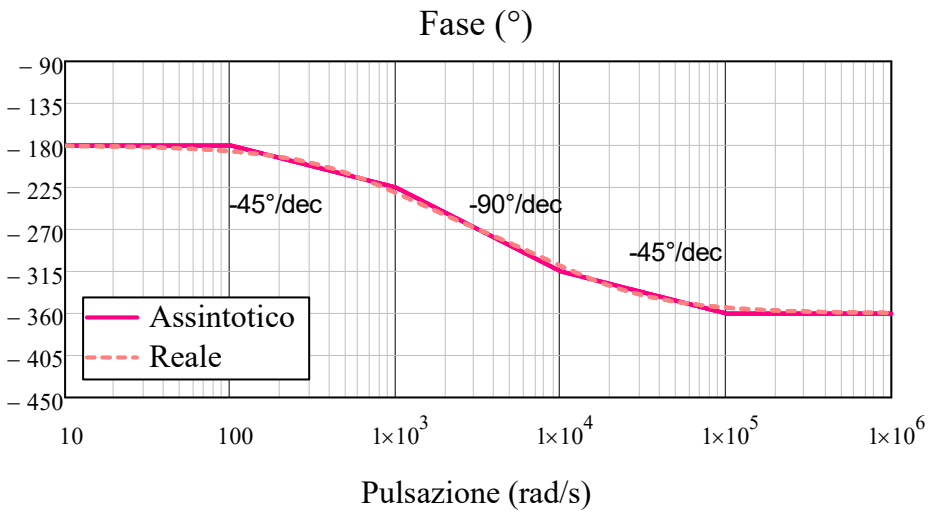
$$\frac{v_O}{v_S} = \frac{-R_2}{2 \cdot R_1} \cdot \frac{\left(1 - i\omega \cdot C_3 \cdot R_1 \cdot \frac{R_3}{R_2} \right)}{1 + i\omega \cdot \frac{C_3 \cdot R_3}{2}} \cdot v_S$$

$\omega_{P_1} = \frac{2}{R_3 \cdot C_3} = 1 \times 10^3 \cdot s^{-1}$ $\quad \omega_{Z_1} = \frac{R_2}{C_3 \cdot R_1 \cdot R_3} = 1 \times 10^4 \cdot s^{-1}$ (zero a parte reale positiva) $\quad A = \frac{-R_2}{2 \cdot R_1} = -10$

$$\boxed{W(s) = A \cdot \frac{\left(1 - \frac{s}{\omega_{Z_1}} \right)}{\left(1 + \frac{s}{\omega_{P_1}} \right)}}$$

2) diagramma di Bode asintotico del modulo e della fase





3) Ampiezza e fase del segnale di uscita, con segnale di ingresso:

$V_{S1} = 1V, \qquad \omega_1 = 2\cdot 10^3 s^{-1}, \quad \varphi_1 = 45°$

$v_S(t) = V_{S1}\cdot \sin(\omega_1 \cdot t + \varphi_1)$

Guadagno in ω_1 $\qquad W_1 = 20dB - 20dB\cdot \log\left(\frac{\omega_1}{1000s^{-1}}\right) = 13.979 \qquad A_1 = 10^{\frac{W_1}{20}} = 5$

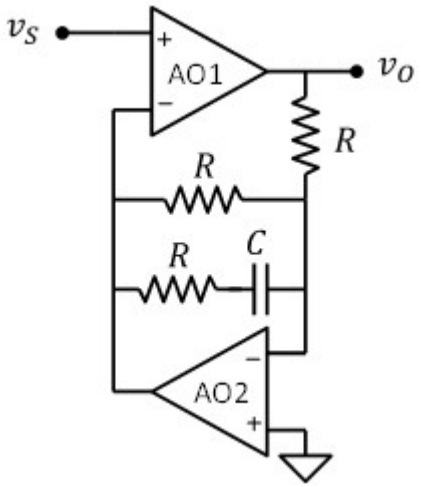
Fase in ω_1: $\qquad \Delta\varphi_1 = -180° - 45° - 90°\cdot \log\left(\frac{\omega_1}{1000s^{-1}}\right) = -252\cdot °$

$V_{O1} = A_1 \cdot V_{S1} = 5\,V \qquad \varphi_{O1} = \varphi_1 + \Delta\varphi_1 = -207\cdot °$

$v_O(t) = V_{O1}\cdot \sin(\omega_1 \cdot t + \varphi_{O1})$

Esercizio 20

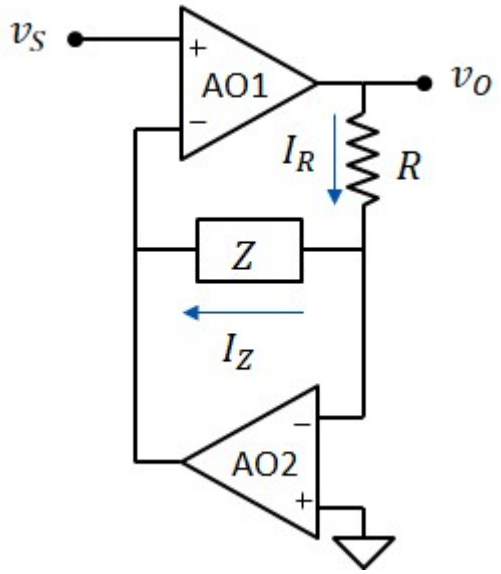
DATI: $R = 10k\Omega$, $C = 100nF$



1) Funzione di trasferimento

Impedenza della rete di retroazione comune ai due operazionali

$$Z = \frac{R\left(R + \frac{1}{i\omega \cdot C}\right)}{R + R + \frac{1}{i\omega \cdot C}} = R \cdot \frac{1 + i\omega \cdot R \cdot C}{1 + i\omega \cdot 2R \cdot C}$$



Legge di kirchhoff tra R e Z:

$$I_R = I_Z$$

$$\frac{v_{O1} - v_{N2}}{R} = \frac{v_{N2} - v_{N1}}{Z}$$

Per il principio del cortocircuito virtuale:

$$v_{N1} = v_S \qquad v_{N2} = 0$$

$$\frac{v_{O1}}{R} = \frac{-v_S}{Z}$$

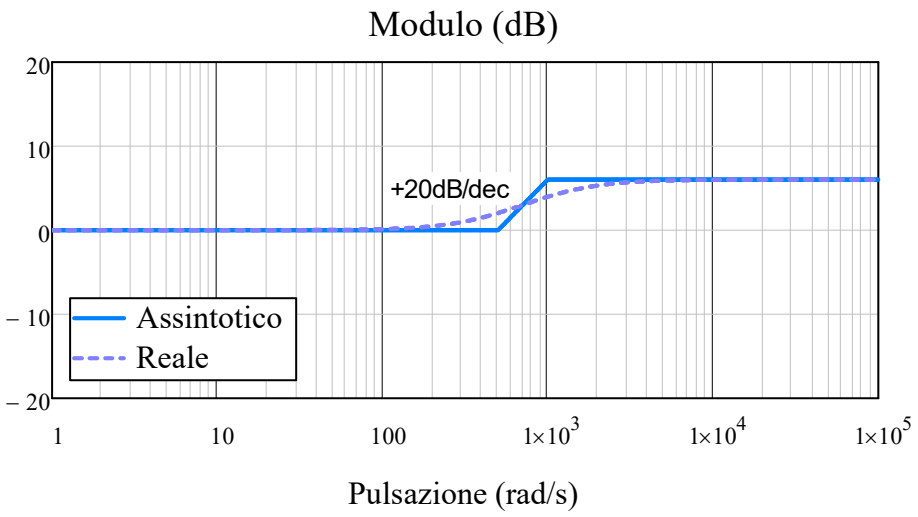
$$v_O = v_{O1} = \frac{-R}{Z} \cdot v_S$$

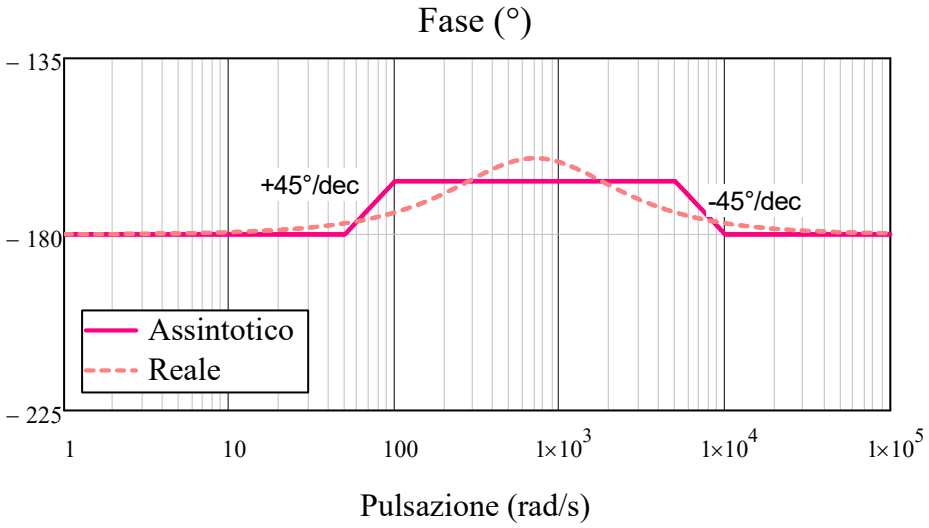
$$W(\omega) = \frac{-R}{Z} = \frac{1 + i\omega \cdot 2R \cdot C}{1 + i\omega \cdot R \cdot C}$$

$$\boxed{\omega_{Z_1} = \frac{1}{2 \cdot R \cdot C} = 500 \cdot s^{-1}} \qquad \boxed{\omega_{P_1} = \frac{1}{R \cdot C} = 1 \times 10^3 \cdot s^{-1}}$$

$$\boxed{W(s) = -\frac{1 + \frac{s}{\omega_{Z_1}}}{1 + \frac{s}{\omega_{P_1}}}}$$

2) Tracciare il diagramma di Bode asintotico del modulo e della fase





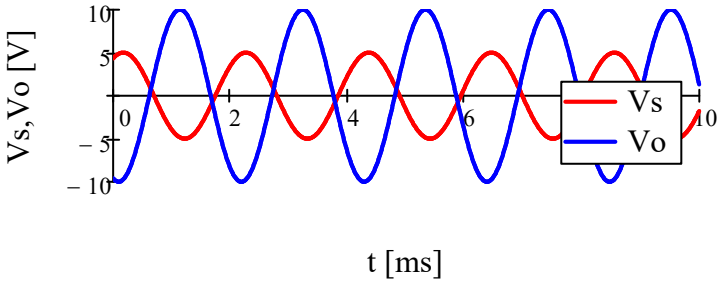
3) Ampiezza e fase del segnale di uscita, quando all'ingresso è presente il segnale: $v_S = V_S \cdot \sin(\omega_1 \cdot t + \varphi_1)$

con $V_S = 5V$, $\omega_1 = 3 \cdot 10^3 s^{-1}$, $\varphi_1 = 60°$

Modulo del guadagno asintotico per $\omega=\omega_1$: $W_1 = 0 + 20 \cdot \log\left(\frac{\omega_{P_1}}{\omega_{Z_1}}\right) = 6 \cdot dB$ $\quad A_1 = 10^{\frac{W_1}{20}} = 2$ $\quad \boxed{V_O = A_1 \cdot V_S = 10\,V}$

Fase del guadagno asintotico per $\omega=\omega_1$: $\Delta\varphi_1 = -180° + 45° \cdot \left(\log\left(\frac{\omega_{P_1} \cdot 0.1}{\omega_{Z_1} \cdot 0.1}\right)\right) = -166 \cdot °$ $\quad \boxed{\varphi_O = \varphi_1 + \Delta\varphi_1 = -106 \cdot °}$

$v_S(t) = V_S \cdot \sin(\omega_1 \cdot t + \varphi_1) \qquad v_O(t) = V_O \cdot \sin(\omega_1 \cdot t + \varphi_O)$



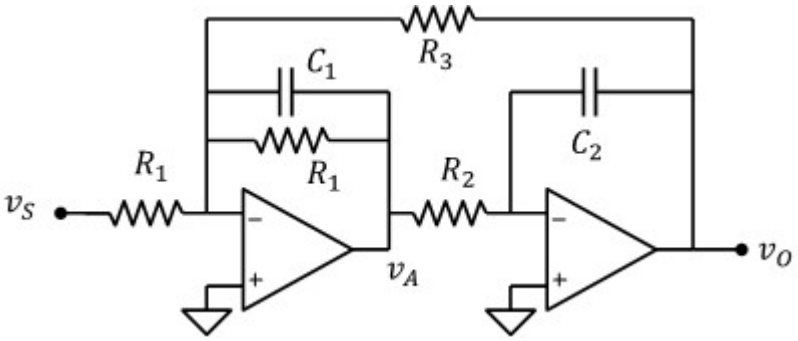
Esercizio 21

DATI:

$R_1 = 1k\Omega$, $R_2 = 1k\Omega$

$C_1 = 250nF$, $C_2 = 80nF$

$R_3 = 10k\Omega$



Funzione di trasferimento

$$Z_1 = \frac{R_1}{1 + i\omega \cdot R_1 \cdot C_1}$$

$$v_O = v_A \cdot \left(-\frac{1}{i\omega \cdot R_2 \cdot C_2}\right) \qquad v_A = -v_O \cdot \left(i\omega \cdot R_2 \cdot C_2\right)$$

Legge di kirchhoff al terminale invertente del primo stadio: $\frac{v_S}{R_1} + \frac{v_A}{Z_1} + \frac{v_O}{R_3} = \frac{v_S}{R_1} - \frac{v_O \cdot \left(i\omega \cdot R_2 \cdot C_2\right)}{Z_1} + \frac{v_O}{R_3} = 0$

$$v_O = -\frac{R_3}{R_1} \cdot \frac{v_S}{\left[1 - i\omega \cdot \frac{R_3 \cdot R_2}{R_1} \cdot C_2 - (i\omega)^2 \cdot R_3 \cdot R_2 \cdot C_1 \cdot C_2\right]}$$

Ponendo s = iω, il denominatore della frazione è: $1 + b \cdot s + a \cdot s^2$

$$b = -\left(\frac{R_3 \cdot R_2}{R_1} \cdot C_2\right) = -8 \times 10^{-4}\, s$$

$$a = -R_3 \cdot R_2 \cdot C_1 \cdot C_2 = -2 \times 10^{-7}\, s^2$$

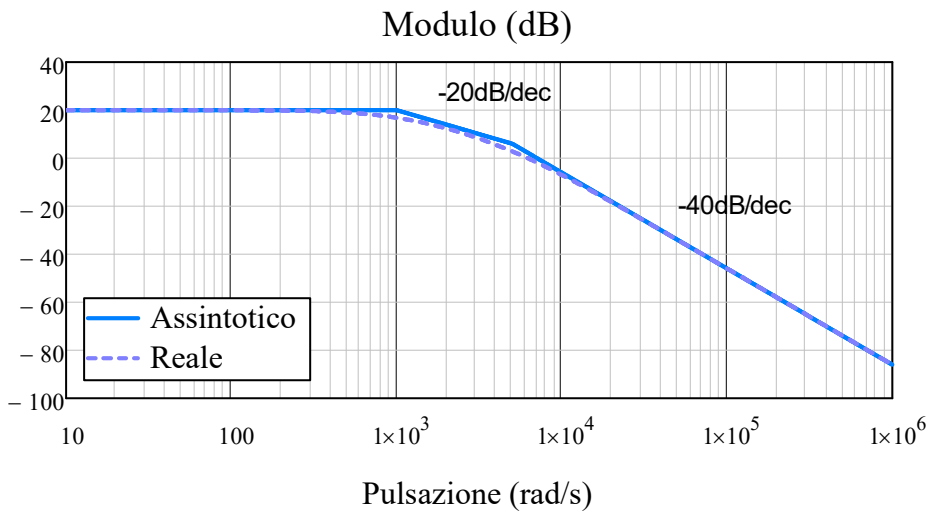
e ha due radici reali una positiva e una negativa:

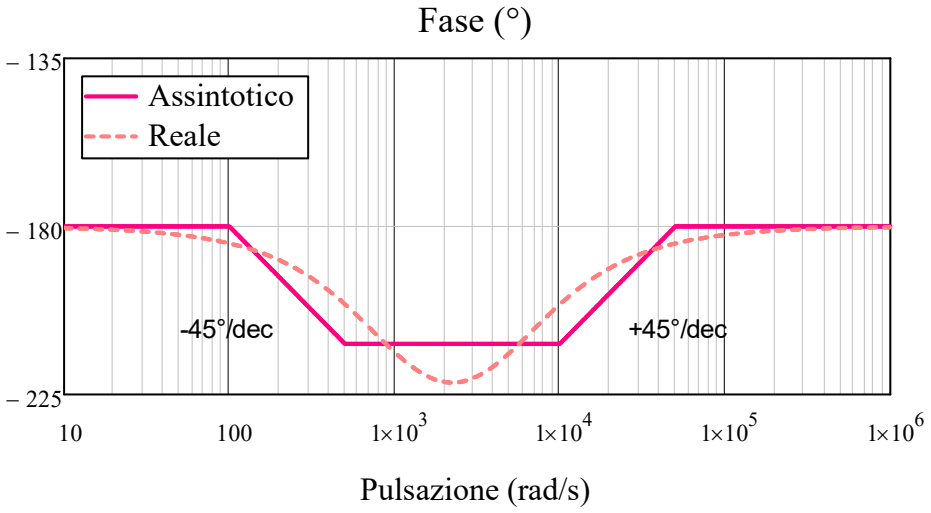
$$\boxed{\omega_{P_1} = \frac{-b - \sqrt{b^2 - 4a}}{2a} = 1 \times 10^3 \cdot s^{-1}} \qquad \boxed{\omega_{P_2} = \frac{-b + \sqrt{b^2 - 4a}}{2a} = -5 \times 10^3 \cdot s^{-1}}$$

Ponendo $A = \frac{-R_3}{R_2} = -10$, la funzione di trasferimento è:

$$\boxed{W(s) = \frac{A}{\left(1 + \frac{s}{\omega_{P_1}}\right) \cdot \left(1 + \frac{s}{\omega_{P_2}}\right)}}$$

Diagramma di Bode asintotico del modulo e della fase





Esercizio 22A

DATI: $B = 40,\ a = 10^{-6}s^2,\ b = 10^{-3}\cdot s,\ c = 4$

$$W(s) = \frac{B}{a\cdot s^2 + b\cdot s + c}$$

Avendo un polinomio di secondo grado al denominatore, è necessario procedere come segue:

1) trovare le radici ω_1 e ω_2.

Se sono reali (positive o negative) si fattorizza il polinomio scrivendolo nella forma:

$$\left(1 + \frac{s}{\omega_1}\right)\cdot\left(1 + \frac{s}{\omega_2}\right)\cdot C$$

e si ottengono due poli reali. Si può procedere con il diagramma di Bode come negli esercizi precedenti.
Se le radici sono complesse coniugate bisogna riscrivere il polinomi in questo modo:

$$\left[1 + 2\delta\cdot\frac{s}{\omega_P} + \left(\frac{s}{\omega_P}\right)^2\right]\cdot C$$

in questo caso le radici sono:

$$s_1 = -b - \sqrt{b^2 - 4\cdot a\cdot c} = \left(-1\times 10^{-3} - 3.873i\times 10^{-3}\right)s$$

$$s_2 = -b + \sqrt{b^2 - 4\cdot a\cdot c} = \left(-1\times 10^{-3} + 3.873i\times 10^{-3}\right)s$$

Essendo complesse coniugate riscriviamo il polinomio nel modo seguente:

$$a\cdot s^2 + b\cdot s + c = c\cdot\left(1 + \frac{b}{c}\cdot s + \frac{a}{c}\cdot s^2\right) = c\cdot\left[1 + \frac{b}{c}\cdot s + \left(\frac{s}{\omega_P}\right)^2\right] \qquad \text{con: } \omega_P = \sqrt{\frac{c}{a}} = 2\times 10^3\cdot s^{-1}$$

Poniamo infine: $\frac{2\delta}{\omega_P} = \frac{b}{c}$ ovvero: $\delta = \frac{1}{2}\cdot\frac{b}{c}\cdot\omega_P = 0.25$

In alternativa, dato il polinomio $as^2 + bs + c$, lo scriviamo nella forma:

$$\left[1 + 2\delta\cdot\frac{s}{\omega_P} + \left(\frac{s}{\omega_P}\right)^2\right]\cdot c \qquad \text{con: } \omega_P = \sqrt{\frac{c}{a}},\ \delta = \frac{1}{2}\cdot\frac{b}{c}\cdot\omega_P$$

Il discriminante del polinomio è:

$$\Delta = \frac{\delta^2}{\omega_P{}^2} - \frac{1}{\omega_P{}^2} = \frac{\delta^2 - 1}{\omega_P\cdot 2}$$

le radici sono reali se $\Delta \geq 0$ ovvero $|\delta| \geq 1$
le radici sono complesse coniugate se $|\delta| < 1$ (come in questo caso)

Incorporiamo la costante c insieme a B in un'unica costante $A = \frac{B}{c} = 10$ e otteniamo:

$$W(s) = \frac{A}{\left[1 + 2\delta\cdot\frac{s}{\omega_P} + \left(\frac{s}{\omega_P}\right)^2\right]}$$

ω_P rappresenta la posizione del doppio zero o del doppio polo (in questo caso, essendo il denominatore, si tratta di un polo).
Il valore di δ (in modulo minore di 1) è legato alla presenza di picchi di risonanza nel modulo intorno al polo (o zero) e alla pendenza dello sfasamento in un intorno del polo (o zero).

Ponendo $x = \omega/\omega_P$, il modulo del polinomio è: $\left|1 + 2\delta\cdot ix + (ix)^2\right| = \sqrt{\left(1 - x^2\right)^2 + 4\delta^2 x^2} = \sqrt{1 + \left(4\delta^2 - 2\right)x^2 + x^4}$

Derivando l'argomernto della radice otteniamo la posizione del minimo $\left(4\delta^2 - 2\right)2x + 4\cdot x^3 = 0$ $\quad x^2 = 1 - 2\delta^2$

Se $1 - 2\delta^2 < 0$ --> $|\delta| > \frac{\sqrt{2}}{2}$ la derivata non si annulla mai e la funzione non ha un minimo

Se $1 - 2\delta^2 > 0$ --> $|\delta| \geq \frac{\sqrt{2}}{2}$ la derivata si annulla e il valore del minimo è:

$$\sqrt{1 + \left(4\delta^2 - 2\right)x^2 + x^4} = \sqrt{1 - \left(1 - 2\delta^2\right)^2} = 2\cdot\delta\cdot\sqrt{1 - 2\delta^2}$$

in questo caso, il valore minimo in decibel vale:

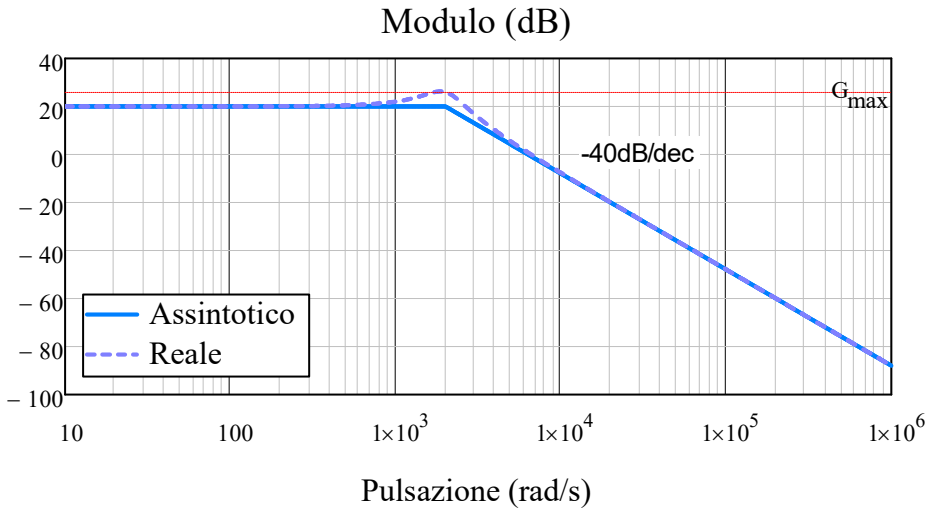
$$\Delta_{dB} = 20\cdot\log\left(2\cdot\delta\cdot\sqrt{1 - 2\delta^2}\right) = -6.6\cdot dB$$

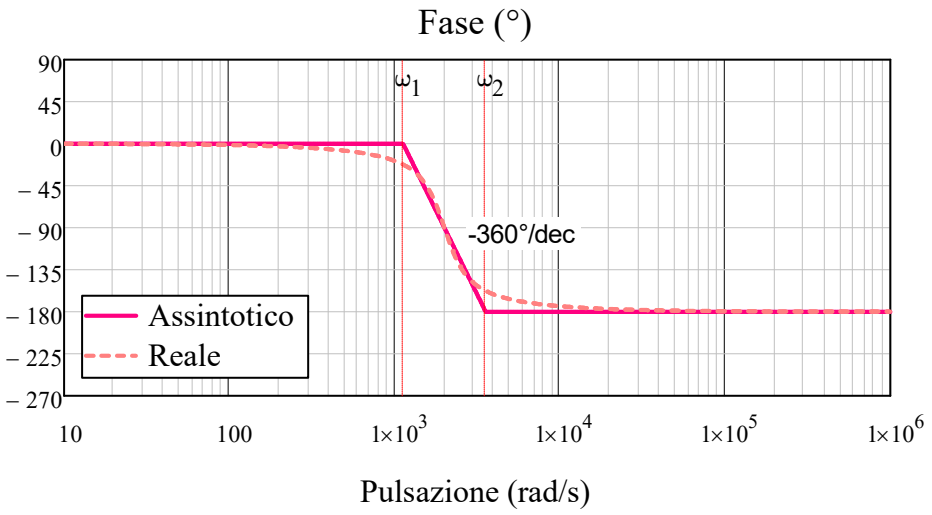
Trattandosi di un polo il minimo del polinomio si traduce in un massimo nel guadagno pari a: $G_{max} = 20\log(A) - \Delta_{dB}$

Per quanto riguarda la fase, il polo produce uno sfasamento di -180° (polo doppio) approssimativamente tra le pulsazioni:

$$\omega_1 = \frac{\omega_P}{10^\delta} = 1.125\times 10^3\cdot s^{-1} \qquad \omega_2 = \omega_P\cdot 10^\delta = 3.557\times 10^3\cdot s^{-1}$$

La pendenza del diagramma asintotico della fase è: $\frac{-180°}{\log(\omega_2) - \log(\omega_2)} = \frac{-180°}{2\delta} = \frac{-90°}{\delta}$





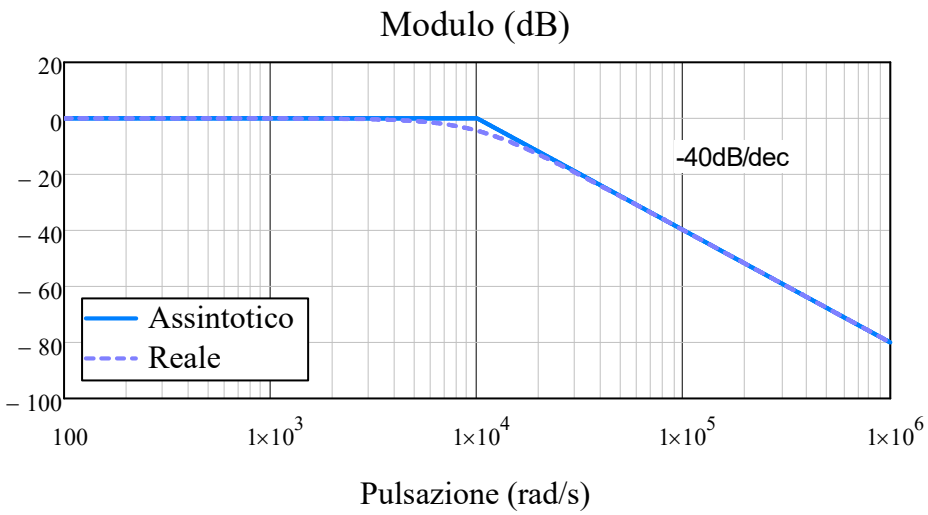
Esercizio 22B

DATI: $A = 1,\ \omega_A = 6\cdot 10^3 s^{-1},\ \omega_B = 10^4 s^{-1}$

$$W(s) = \frac{A}{1 - \frac{s}{\omega_A} + \left(\frac{s}{\omega_B}\right)^2}$$

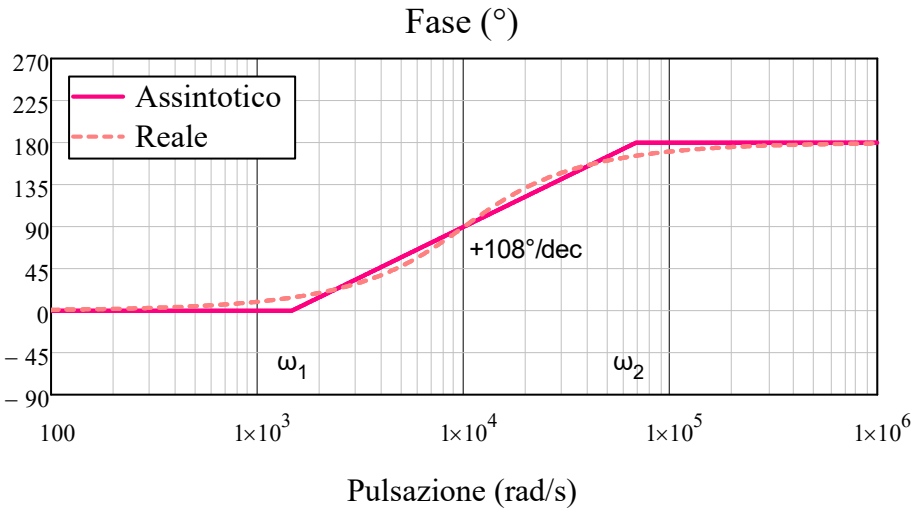
Poniamo: $\omega_P = \omega_B$ $\quad \delta = \frac{1}{2}\cdot\left(\frac{-1}{\omega_A}\right)\cdot\omega_B = -0.833$

Riscriviamo la funzione di trasferimento: $W(s) = \frac{A}{1 + 2\delta\cdot\frac{s}{\omega_P} + \left(\frac{s}{\omega_P}\right)^2}$



$\delta > \frac{\sqrt{2}}{2}$

nessun picco



Lo sfasamento avviene tra:

$$\omega_P\cdot 10^{-\delta} = 6.8\times 10^4\cdot s^{-1}$$

$$\omega_P\cdot 10^{\delta} = 1.5\times 10^3\cdot s^{-1}$$

e ha pendenza:

$$\frac{-90°}{\delta} = 108\cdot\frac{°}{dec}$$

Esercizio 22C

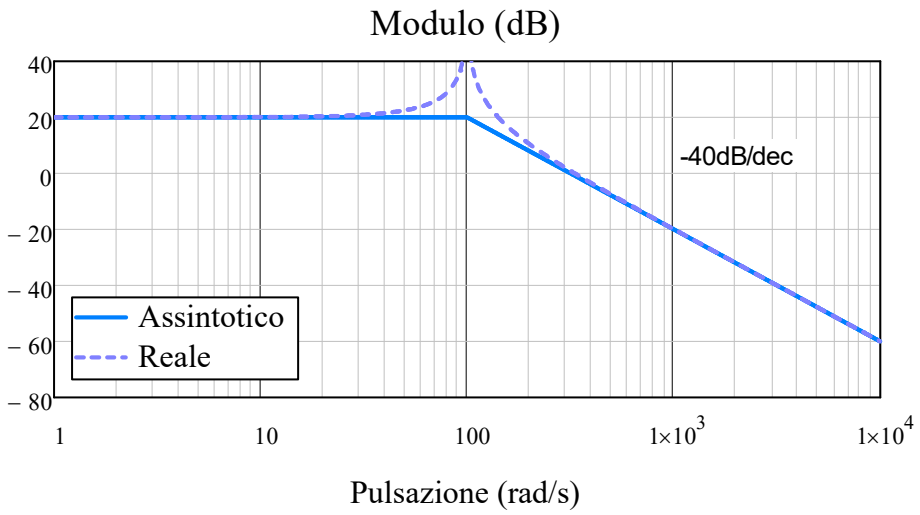
DATI: $A = -10,\ \omega_P = 100 s^{-1}$

$$W(s) = \frac{A}{1 + \left(\frac{s}{\omega_P}\right)^2}$$

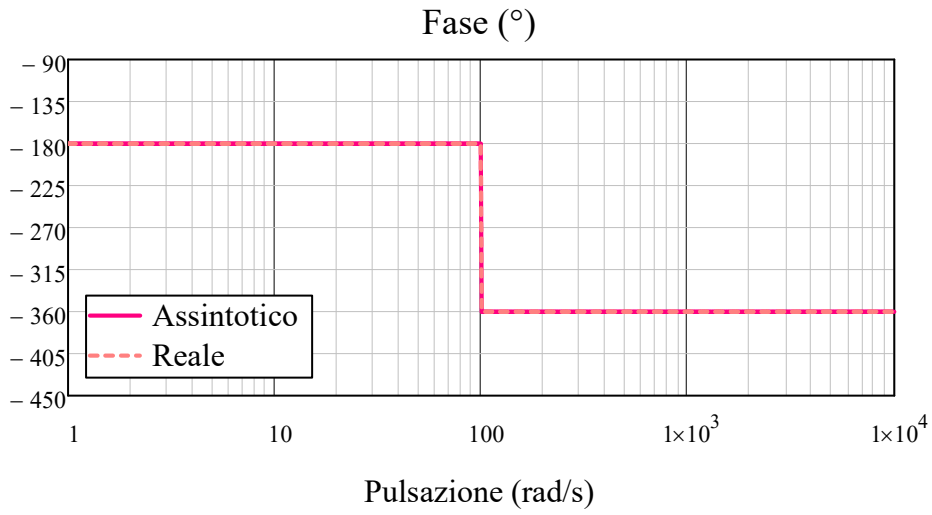
Riscriviamo la funzione di trasferimento:

$$W(s) = \frac{A}{1 + 2\delta \cdot \frac{s}{\omega_P} + \left(\frac{s}{\omega_P}\right)^2}$$

con δ=0



con δ=0, la funzione di trasferimento ha un guadagno infinito in prossimità del polo



Lo sfasamento avviene istantaneamente alla pulsazione ω_P con pendenza infinita

Esercizio 23A

DATI: $K = 10,\ \omega_0 = 500 rad \cdot s^{-1},\ \omega_1 = 2000 rad \cdot s^{-1},\ \omega_2 = 1000 rad \cdot s^{-1}$

$$W(s) = \frac{K \cdot \left(\frac{s}{\omega_0}\right)^2}{1 + \left(\frac{s}{\omega_1}\right) + \left(\frac{s}{\omega_2}\right)^2}$$

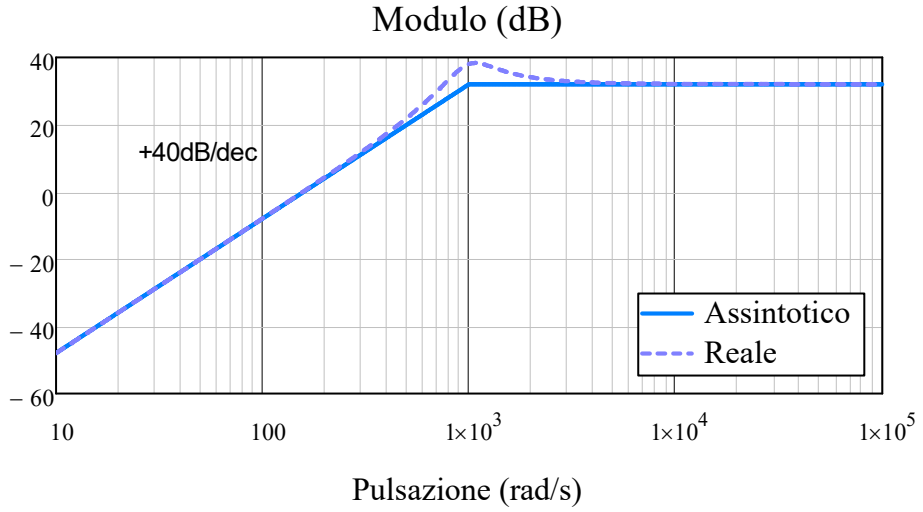
Poniamo, $\omega_P = \omega_2$,

riscriviamo la funzione di trasferimento:

$$W(s) = \frac{K \cdot \frac{\omega_2^2}{\omega_0^2} \cdot \left(\frac{s}{\omega_P}\right)^2}{1 + \frac{\omega_P}{\omega_1} \cdot \left(\frac{s}{\omega_P}\right) + \left(\frac{s}{\omega_P}\right)^2}$$

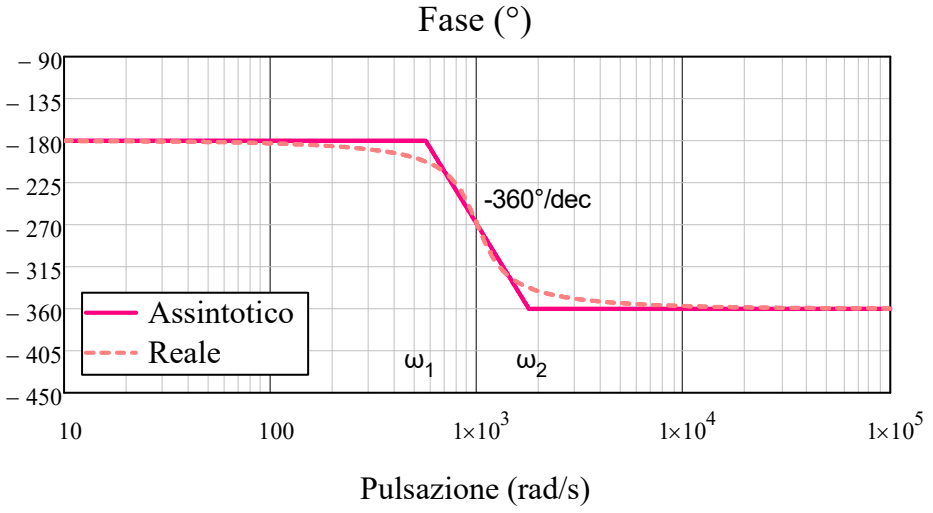
Poniamo: $\delta = \frac{1}{2} \cdot \frac{\omega_P}{\omega_1} = 0.25$ $\quad A = K \cdot \frac{\omega_2^2}{\omega_0^2} = 40$

$$W(s) = \frac{A \cdot \left(\frac{s}{\omega_P}\right)^2}{1 + 2 \cdot \delta \cdot \left(\frac{s}{\omega_P}\right) + \left(\frac{s}{\omega_P}\right)^2}$$



picco positivo in prossimità del polo

$$20 \cdot \log\left(\frac{1}{2 \cdot \delta \cdot \sqrt{1 - 2\delta^2}}\right) = 6.6 \cdot dB$$



$$\omega_1 = \omega_P \cdot 10^{-\delta} = 562.3 \cdot s^{-1}$$

$$\omega_2 = \omega_P \cdot 10^{\delta} = 1.8 \times 10^3 \cdot s^{-1}$$

Esercizio 23B

DATI: $\omega_1 = 1.6\cdot 10^5 \mathrm{rad}\cdot s^{-1}$, $\omega_2 = 8\cdot 10^4 \mathrm{rad}\cdot s^{-1}$, $\omega_3 = 125 \mathrm{rad}\cdot s^{-1}$, $\omega_4 = 200 \mathrm{rad}\cdot s^{-1}$

$$W(s) = \frac{1 + \frac{s}{\omega_1} + \left(\frac{s}{\omega_2}\right)^2}{1 + \frac{s}{\omega_3} + \left(\frac{s}{\omega_4}\right)^2}$$

Poniamo, $\omega_P = \omega_4$ e $\omega_Z = \omega_2$

riscriviamo la funzione di trasferimento:

$$W(s) = \frac{1 + \frac{s\cdot\omega_Z}{\omega_Z\cdot\omega_1} + \left(\frac{s}{\omega_Z}\right)^2}{1 + \frac{s\cdot\omega_P}{\omega_P\cdot\omega_3} + \left(\frac{s}{\omega_P}\right)^2}$$

Poniamo: $\delta_Z = \frac{1}{2}\cdot\frac{\omega_Z}{\omega_1} = 0.25$ $\quad \delta_P = \frac{1}{2}\cdot\frac{\omega_P}{\omega_3} = 0.8$

$$W(s) = \frac{1 + 2\delta_Z\cdot\frac{s}{\omega_Z} + \left(\frac{s}{\omega_Z}\right)^2}{1 + 2\delta_P\cdot\frac{s}{\omega_P} + \left(\frac{s}{\omega_P}\right)^2}$$

W ha due poli e due zeri entrambi complessi coniugati:

$\left|\delta_Z\right| < \frac{\sqrt{2}}{2}$ il numeratore ha un minimo in prossimità dello zero. il picco è: $20\cdot\log\left(2\cdot\delta_Z\cdot\sqrt{1 - 2\delta_Z^2}\right) = -6.6\cdot dB$

$\left|\delta_P\right| < \frac{\sqrt{2}}{2}$ il denominatore non ha il minimo in prossimità del polo. Nessun picco

Lo sfasamento introdotto dallo zero è complessivamente +180° e avviene tra le pulsazioni:

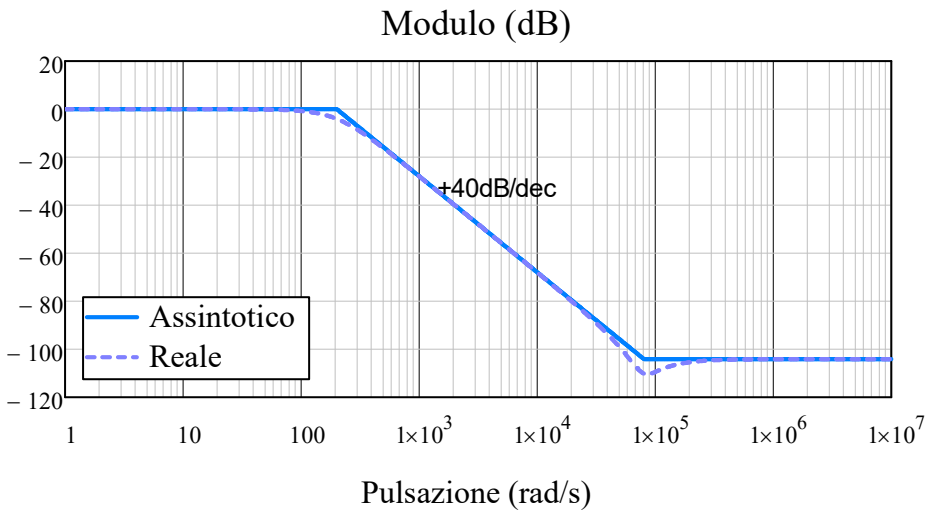
$$\omega_{Z1} = \omega_Z\cdot 10^{-\delta_Z} = 4.5\times 10^4\cdot s^{-1} \qquad \omega_{Z2} = \omega_Z\cdot 10^{\delta_Z} = 1.4\times 10^5\cdot s^{-1}$$

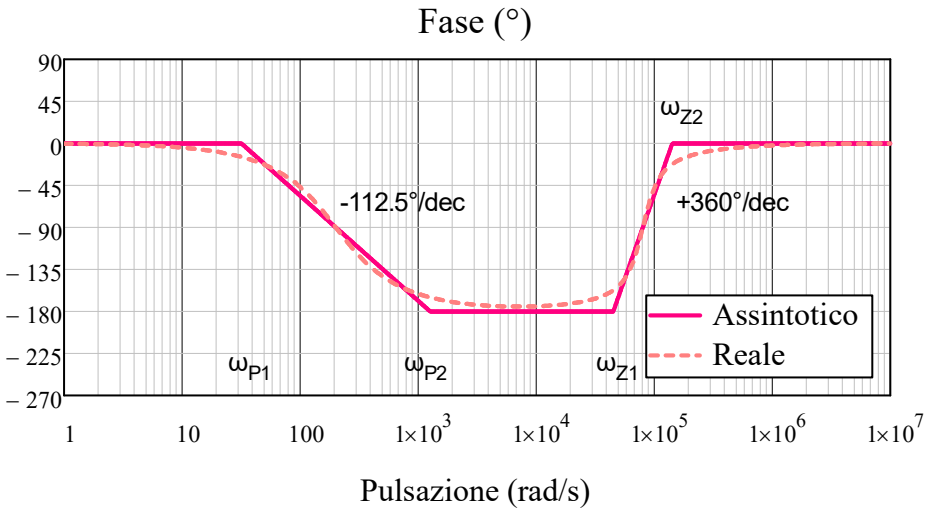
con una pendenza pari a: $\frac{90°}{\delta_Z} = 360\cdot\frac{°}{dec}$

Lo sfasamento introdotto dal polo è complessivamente -180° e avviene tra le pulsazioni:

$$\omega_{P1} = \omega_P\cdot 10^{-\delta_P} = 31.7\cdot s^{-1} \qquad \omega_{P2} = \omega_P\cdot 10^{\delta_P} = 1.3\times 10^3\cdot s^{-1}$$

con una pendenza pari a: $\frac{90°}{\delta_P} = 112.5\cdot\frac{°}{dec}$


Modulo (dB)
20
0
– 20
– 40
– 60
– 80
– 100
– 120
+40dB/dec
Assintotico
Reale
1
10
100
1×10^3
1×10^4
1×10^5
1×10^6
1×10^7
Pulsazione (rad/s)


Fase (°)
90
45
0
– 45
– 90
– 135
– 180
– 225
– 270
-112.5°/dec
+360°/dec
ωZ2
ωP1
ωP2
ωZ1
Assintotico
Reale
1
10
100
1×10^3
1×10^4
1×10^5
1×10^6
1×10^7
Pulsazione (rad/s)

Esercizio 23C

DATI: $A = -20$, $\omega_1 = 100 rad \cdot s^{-1}$, $\omega_2 = 5000 rad \cdot s^{-1}$, $\omega_3 = 1000 rad \cdot s^{-1}$, $\omega_4 = 1000 rad \cdot s^{-1}$

$$W(s) = A \cdot \frac{1 + \frac{s}{\omega_1} + \left(\frac{s}{\omega_2}\right)^2}{1 + \frac{s}{\omega_3} + \left(\frac{s}{\omega_4}\right)^2}$$

Poniamo, $\omega_P = \omega_4$ e $\omega_Z = \omega_2$

riscriviamo la funzione di trasferimento:

$$W(s) = A \cdot \frac{1 + \frac{s \cdot \omega_Z}{\omega_Z \cdot \omega_1} + \left(\frac{s}{\omega_Z}\right)^2}{1 + \frac{s \cdot \omega_P}{\omega_P \cdot \omega_3} + \left(\frac{s}{\omega_P}\right)^2}$$

Poniamo: $\delta_Z = \frac{1}{2} \cdot \frac{\omega_Z}{\omega_1} = 25$ $\quad \delta_P = \frac{1}{2} \cdot \frac{\omega_P}{\omega_3} = 0.5$

$$W(s) = \frac{1 + 2\delta_Z \cdot \frac{s}{\omega_Z} + \left(\frac{s}{\omega_Z}\right)^2}{1 + 2\delta_P \cdot \frac{s}{\omega_P} + \left(\frac{s}{\omega_P}\right)^2}$$

W ha due poli complessi coniugati:

$\left|\delta_P\right| < \frac{\sqrt{2}}{2}$ il denominatore ha il minimo in prossimità del polo. il picco ha vale: $20 \cdot \log\left(\frac{1}{2 \cdot \delta_P \cdot \sqrt{1 - 2\delta_P^2}}\right) = 3.01 \cdot dB$

W ha due zeri reali poichè $\delta_Z > 1$ il valore degli zeri si ricavano risolvendo l'equazine di secondo grado:

$$1 + 2\delta_Z \cdot \frac{s}{\omega_Z} + \left(\frac{s}{\omega_Z}\right)^2 = 0$$

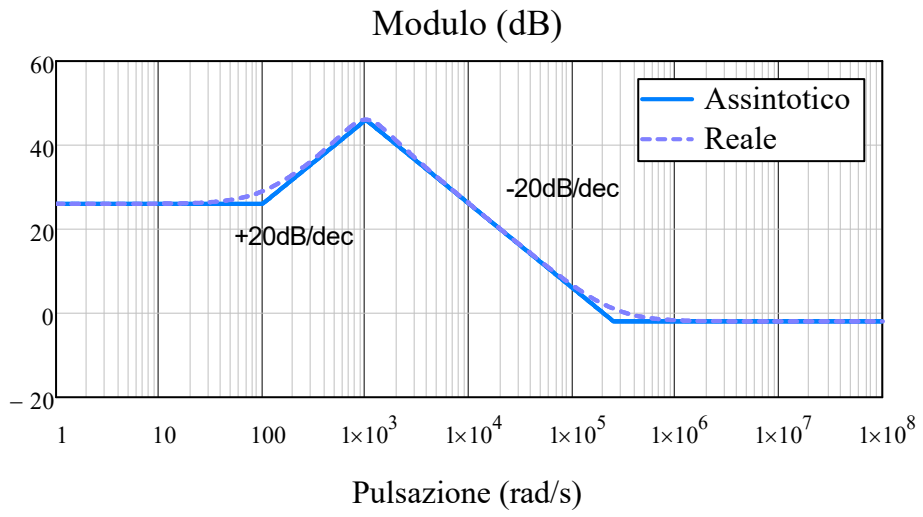
$$\omega_{Z1} = -\omega_Z \cdot \left(-\delta_Z + \sqrt{\delta_Z^2 - 1}\right) = 100 \cdot s^{-1}$$

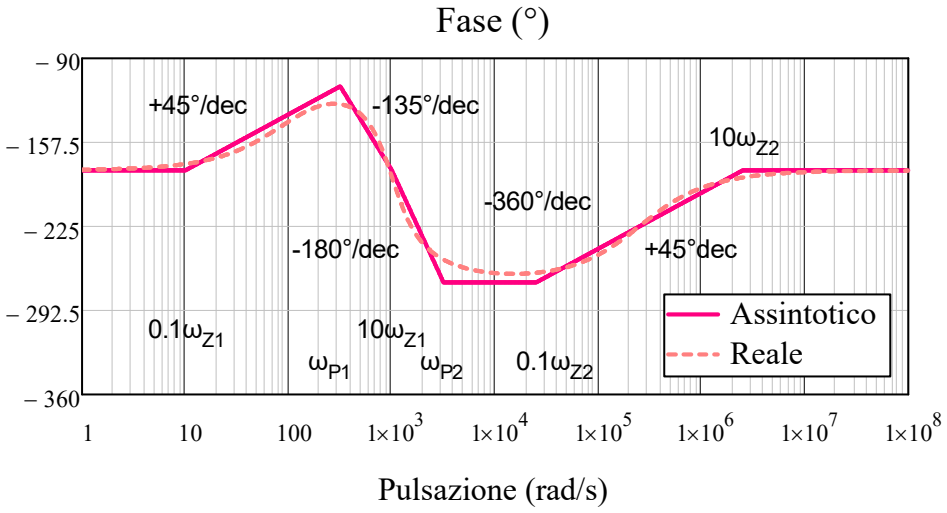
$$\omega_{Z2} = -\omega_Z \cdot \left(-\delta_Z - \sqrt{\delta_Z^2 - 1}\right) = 2.5 \times 10^5 \cdot s^{-1}$$

Oppure in modo equivalente, risolvendo l'equazione di partenza:

$$1 + \frac{s}{\omega_1} + \left(\frac{s}{\omega_2}\right)^2 = 0$$

$$-\frac{-\frac{1}{\omega_1} + \sqrt{\left(\frac{1}{\omega_1}\right)^2 - 4 \cdot \left(\frac{1}{\omega_2^2}\right)}}{\frac{2}{\omega_2^2}} = 100 \cdot s^{-1} \qquad -\frac{-\frac{1}{\omega_1} - \sqrt{\left(\frac{1}{\omega_1}\right)^2 - 4 \cdot \left(\frac{1}{\omega_2^2}\right)}}{\frac{2}{\omega_2^2}} = 2.5 \times 10^5 \cdot s^{-1}$$





$\omega_{Z1} \cdot 0.1 = 10 \cdot s^{-1}$

$\omega_{P1} = \omega_P \cdot 10^{-\delta_P} = 316.2 \cdot s^{-1}$

$\omega_{Z1} \cdot 10 = 1 \times 10^3 \cdot s^{-1}$

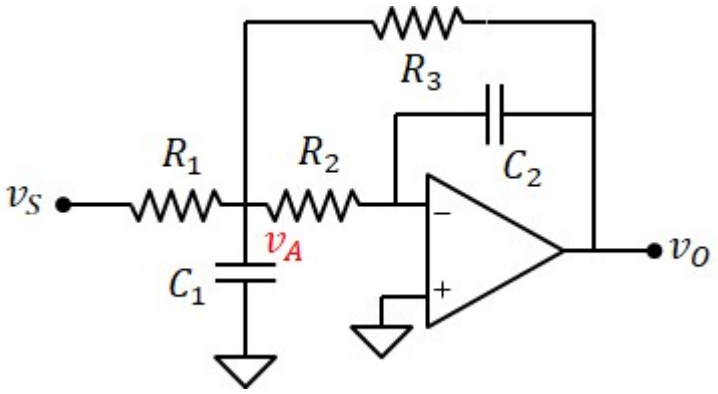
$\omega_{P2} = \omega_P \cdot 10^{\delta_P} = 3.2 \times 10^3 \cdot s^{-1}$

$\omega_{Z2} \cdot 0.1 = 2.5 \times 10^4 \cdot s^{-1}$

$\omega_{Z2} \cdot 10 = 2.5 \times 10^6 \cdot s^{-1}$

Esercizio 24

DATI: $R_1 = 4k\Omega$, $C_1 = 500nF$, $R_2 = 2k\Omega$, $C_2 = 1nF$, $R_3 = 40k\Omega$,



1) trovare la funzione di trasferimento

$$v_O = -v_A \cdot \frac{1}{i\omega \cdot R_2 \cdot C_2}$$

$$v_A = \frac{\frac{v_S}{R_1} + \frac{v_O}{R_3}}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + i\omega \cdot C_1} = \frac{\frac{v_S}{R_1} + \frac{v_O}{R_3}}{\frac{1}{R_P} + i\omega \cdot C_1}$$

$$-v_O \cdot i\omega \cdot R_2 \cdot C_2 = \frac{\frac{v_S}{R_1} + \frac{v_O}{R_3}}{\frac{1}{R_P} + i\omega \cdot C_1}$$

Definiamo: $R_P = \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right)^{-1} = 1.29 \cdot k\Omega$

$$v_S = \frac{-R_1}{R_3} \cdot \left[1 + \frac{i\omega \cdot R_2 \cdot R_3 \cdot C_2}{R_P} + (i\omega)^2 \cdot C_1 \cdot R_2 \cdot R_3 \cdot C_2\right] \cdot v_O$$

Poniamo: $A = \frac{-R_3}{R_1} = -10$ $\quad \omega_P = \frac{1}{\sqrt{C_1 \cdot R_2 \cdot R_3 \cdot C_2}} = 5 \times 10^3 \cdot s^{-1}$ $\quad \delta = \frac{1}{2} \cdot \frac{R_2 \cdot R_3 \cdot C_2}{R_P} \cdot \omega_P = 0.155$

$$\boxed{W(s) = \frac{A}{1 + 2\delta \cdot \frac{s}{\omega_P} + \left(\frac{s}{\omega_P}\right)^2}}$$

2) calcolare il valore massimo del modulo del guadagno vicino al polo

il punto di massimo si ha in corrispondenza del punto di minimo del denominatore:

$$1 + 2\delta \cdot \frac{s}{\omega_P} + \left(\frac{s}{\omega_P}\right)^2$$

Per il calcolo del punto di minimo poniamo $x = \omega/\omega_P$ e $s = ix$. $\quad 1 + 2\delta \cdot ix - x^2$

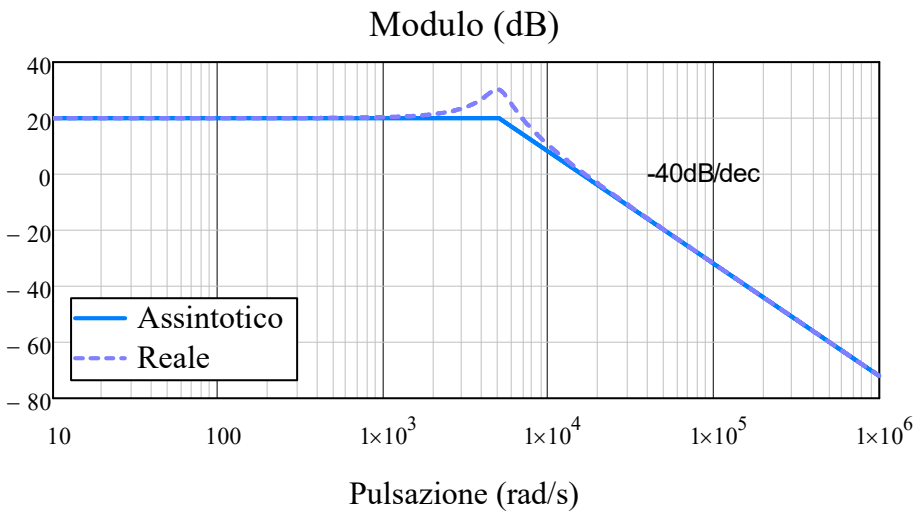
Calcoliamo il modulo $\quad f(x)^2 = \left(1 - x^2\right)^2 + (2 \cdot \delta \cdot x)^2$

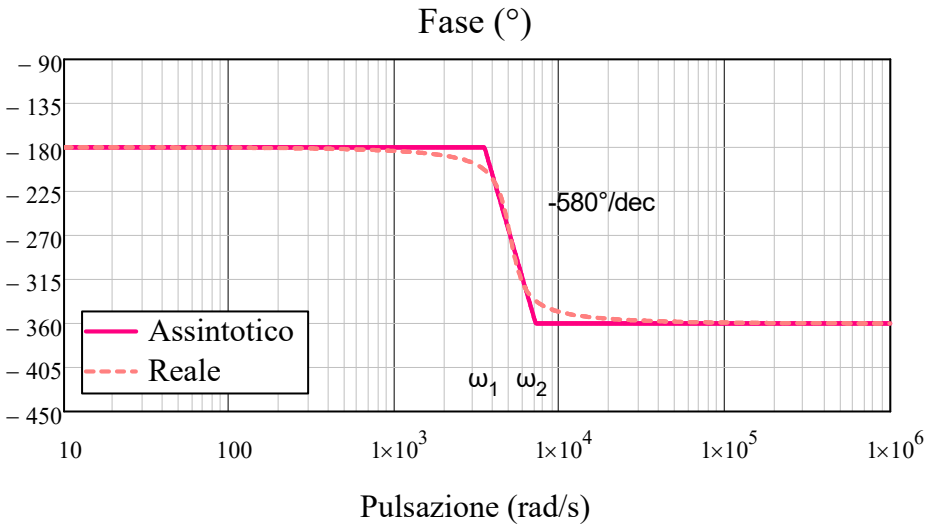
Posizione del minimo: $\quad \frac{d\left(f(x)^2\right)}{dx} = -4 \cdot \left(1 - x^2\right) \cdot x + 8 \cdot \delta^2 \cdot x = 0$ $\quad x = \sqrt{1 - 2 \cdot \delta^2} = 0.976$

Valore del minimo (del denominatore): $\quad f_{min} = 2 \cdot \delta \cdot \sqrt{1 - \delta^2} = 0.306$

Massimo del modulo: $\quad \boxed{W_{max} = \frac{|A|}{\left(2 \cdot \delta \cdot \sqrt{1 - \delta^2}\right)} = 32.7}$ $\quad \boxed{W_{max.dB} = 20 \cdot \log\left[\frac{|A|}{\left(2 \cdot \delta \cdot \sqrt{1 - \delta^2}\right)}\right] = 30.3}$

3) Tracciare il diagramma di Bode asintotico del modulo e della fase





$$\omega_1 = \omega_P \cdot 10^{-\delta} = 3.5 \times 10^3 \cdot s^{-1}$$

$$\omega_2 = \omega_P \cdot 10^{\delta} = 7.14 \times 10^3 \cdot s^{-1}$$

4) Sapendo che all'ingresso è applicato il segnale $v_S = V_{S1}\sin(\omega_S t)$ con $V_{S1} = 0.2V$ e $\omega_S = 4000s^{-1}$, calcolare ampiezza e fase del segnale di uscita <u>dal diagramma di bode asintotico</u>

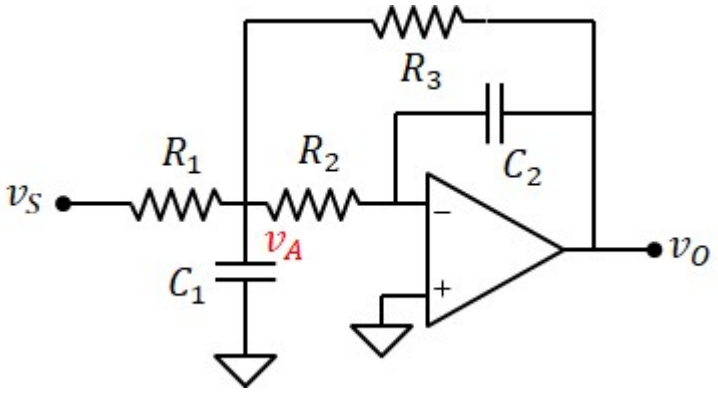
$$W_{dB} = 20dB \qquad \Phi = -180° - \frac{180°}{|2\delta|} \cdot \log\left(\frac{\omega_S}{\omega_1}\right) = -213.73 \cdot °$$

$$\boxed{V_O = V_{S1} \cdot 10^{\frac{W_{dB}}{20}} = 2\,V}$$

$$\boxed{\Phi_O = \Phi = -214 \cdot °}$$

Esercizio 25

DATI: $R_1 = 10k\Omega$, $R_2 = 10k\Omega$,



1) trovare la funzione di trasferimento

$$v_O = -v_A \cdot \frac{1}{i\omega \cdot R_2 \cdot C_2}$$

$$v_A = \frac{\frac{v_S}{R_1} + \frac{v_O}{R_3}}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + i\omega \cdot C_1} = \frac{\frac{v_S}{R_1} + \frac{v_O}{R_3}}{\frac{1}{R_P} + i\omega \cdot C_1}$$

$$-v_O \cdot i\omega \cdot R_2 \cdot C_2 = \frac{\frac{v_S}{R_1} + \frac{v_O}{R_3}}{\frac{1}{R_P} + i\omega \cdot C_1}$$

Definiamo: $R_P = \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right)^{-1}$

$$v_S = \frac{-R_1}{R_3} \cdot \left[1 + \frac{i\omega \cdot R_2 \cdot R_3 \cdot C_2}{R_P} + (i\omega)^2 \cdot C_1 \cdot R_2 \cdot R_3 \cdot C_2\right] \cdot v_O$$

Poniamo: $A = \frac{-R_3}{R_1}$ $\quad \omega_P = \frac{1}{\sqrt{C_1 \cdot R_2 \cdot R_3 \cdot C_2}}$ $\quad \delta = \frac{1}{2} \cdot \frac{R_2 \cdot R_3 \cdot C_2}{R_P} \cdot \omega_P = \frac{1}{2} \cdot \sqrt{\frac{C_2}{C_1}} \cdot \frac{\sqrt{R_2 \cdot R_3}}{R_P}$

$$\boxed{W(s) = \frac{A}{1 + 2\delta \cdot \frac{s}{\omega_P} + \left(\frac{s}{\omega_P}\right)^2}}$$

2) trovare il valore di C_1, C_2 e R_3 in modo tale che il guadagno sia $A = -4$, il polo sia $\omega_P = 2 \cdot 10^4 s^{-1}$, $Q = \frac{1}{\sqrt{2}}$

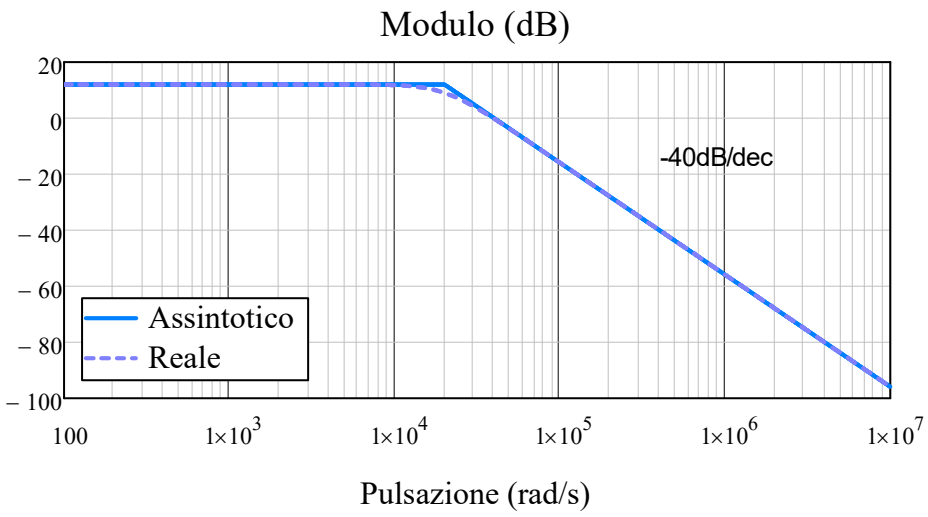
Noto il guadagno e R_1, ricaviamo: $\boxed{R_3 = -R_1 \cdot A = 40 \cdot k\Omega}$ da cui risulta: $R_P = \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right)^{-1} = 4.44 \cdot k\Omega$

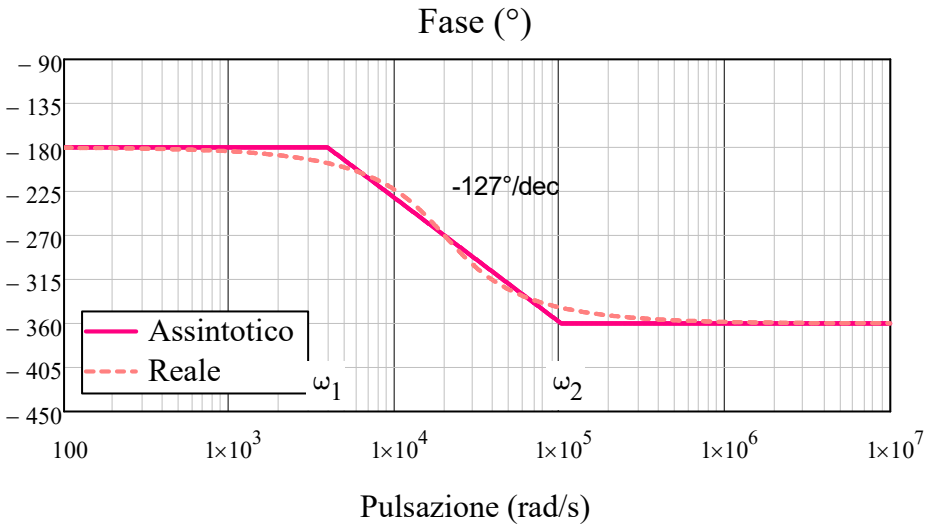
Da Q ricaviamo: $\delta = \frac{1}{2 \cdot Q} = 0.707$

Nota R_2, R_3 e δ, ricaviamo: $\boxed{C_2 = \frac{2\delta \cdot R_P}{\omega_P \cdot R_2 \cdot R_3} = 0.786 \cdot nF}$

Note R_2, R_3, C_2 e ω_P, ricaviamo: $\boxed{C_1 = \frac{1}{\omega_P^2 \cdot R_2 \cdot R_3 \cdot C_2} = 7.95 \cdot nF}$

3) Tracciare il diagramma di Bode asintotico del modulo e della fase



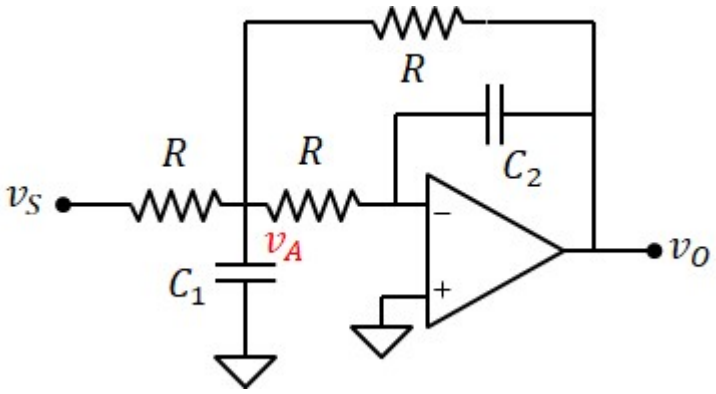


$$\omega_1 = \omega_P \cdot 10^{-\delta} = 3.93 \times 10^3 \cdot s^{-1}$$

$$\omega_2 = \omega_P \cdot 10^{\delta} = 1.02 \times 10^5 \cdot s^{-1}$$

Esercizio 26

DATI: $R = 10k\Omega$,



1) trovare la funzione di trasferimento

$$v_O = -v_A \cdot \frac{1}{i\omega \cdot R \cdot C_2}$$

$$v_A = \frac{\frac{v_S}{R} + \frac{v_O}{R}}{\frac{1}{R} + \frac{1}{R} + \frac{1}{R} + i\omega \cdot C_1} = \frac{v_S + v_O}{3 + i\omega \cdot R \cdot C_1}$$

$$-v_O \cdot i\omega \cdot R \cdot C_2 = \frac{v_S + v_O}{3 + i\omega \cdot R \cdot C_1}$$

$$v_O = \frac{-v_S}{\left(1 + i\omega \cdot 3 \cdot R \cdot C_2 + i\omega^2 \cdot R^2 \cdot C_1 \cdot C_2\right)}$$

Poniamo: $A = -1$ $\quad \omega_P = \frac{1}{R \cdot \sqrt{C_1 \cdot C_2}}$ $\quad \delta = \frac{3}{2} \cdot R \cdot C_2 \cdot \omega_P = \frac{3}{2} \cdot \sqrt{\frac{C_2}{C_1}}$

$$\boxed{W(s) = \frac{A}{1 + 2\delta \cdot \frac{s}{\omega_P} + \left(\frac{s}{\omega_P}\right)^2}}$$

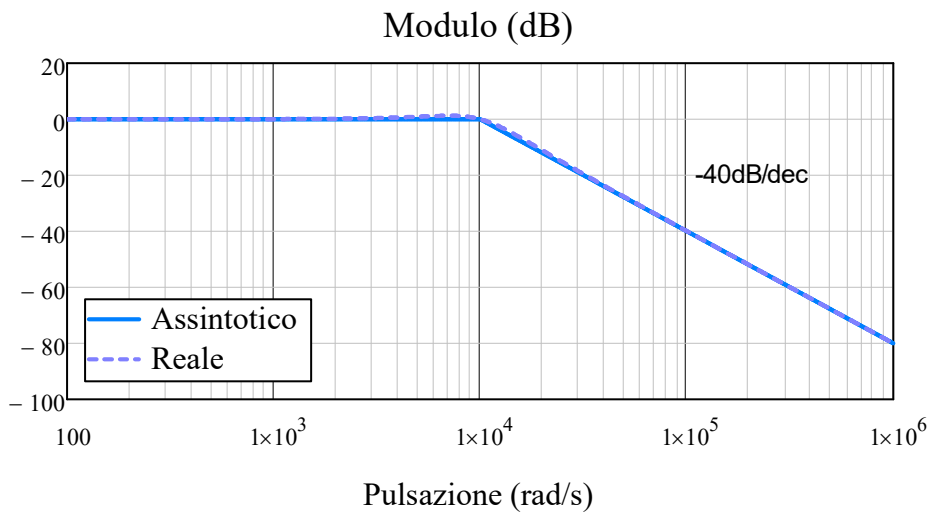
2) trovare il valore di C_1 e C_2 in modo tale che il polo sia $\omega_P = 10^4 s^{-1}$ e che il diagramma asintotico della fase in prossimità del polo abbia pendenza -180°/decade

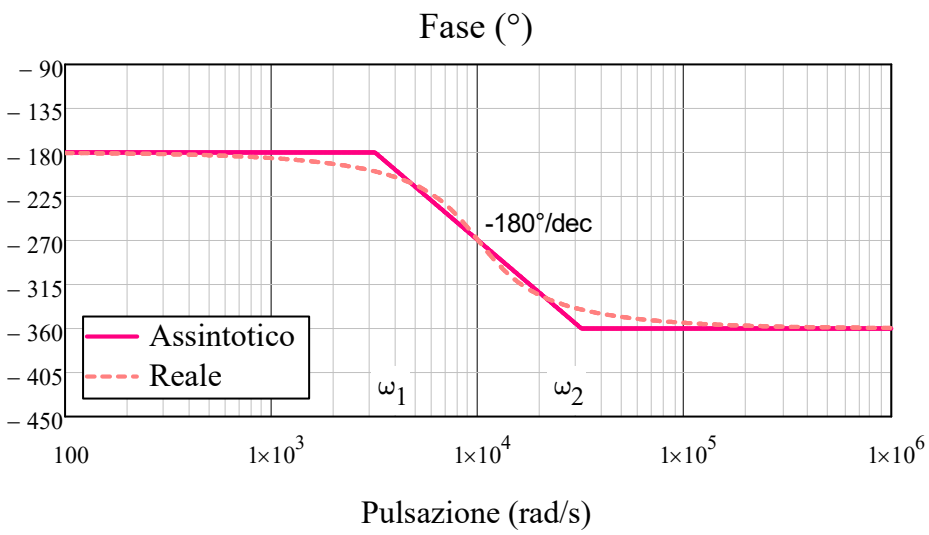
Pendenza del diagramma della fase (asintotico) nell'intorno del polo: $\frac{180°}{2 \cdot \delta} = \frac{180°}{dec}$ $\quad \delta = 0.5$

$$\boxed{C_2 = \frac{2}{3} \cdot \frac{\delta}{R \cdot \omega_P} = 3.33 \cdot nF} \qquad \boxed{C_1 = C_2 \cdot \left(\frac{3}{2\delta}\right)^2 = 30 \cdot nF}$$

3) Tracciare il diagramma di Bode asintotico del modulo e della fase

$$W(s) = \frac{A}{1 + 2\delta \cdot \frac{s}{\omega_P} + \left(\frac{s}{\omega_P}\right)^2}$$



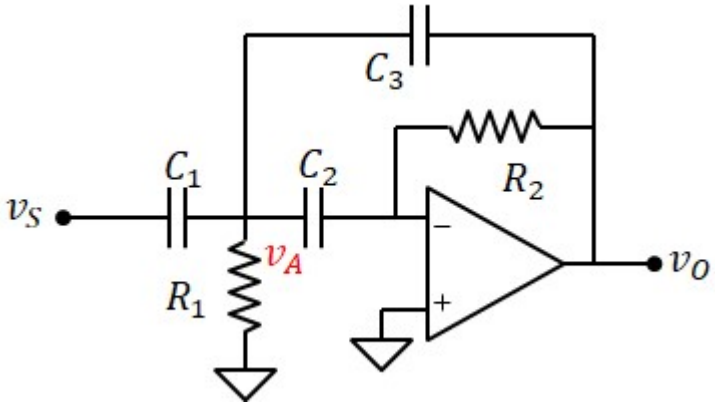


$$\omega_1 = \omega_P \cdot 10^{-\delta} = 3.16 \times 10^3 \cdot s^{-1}$$

$$\omega_2 = \omega_P \cdot 10^{\delta} = 3.16 \times 10^4 \cdot s^{-1}$$

Esercizio 27

DATI: $R_1 = 2k\Omega$, $C_1 = 400nF$, $R_2 = 200k\Omega$, $C_2 = 1\mu F$, $C_3 = 40nF$,



1) trovare la funzione di trasferimento

$$v_O = -v_A \cdot (i\omega \cdot R_2 \cdot C_2)$$

$$v_A = \frac{i\omega \cdot C_1 \cdot v_S + i\omega \cdot C_3 \cdot v_O}{\frac{1}{R_1} + i\omega \cdot C_1 + i\omega \cdot C_2 + i\omega \cdot C_3}$$

$$v_O = -\frac{i\omega \cdot R_1 \cdot C_1 \cdot v_S + i\omega \cdot R_1 \cdot C_3 \cdot v_O}{1 + i\omega \cdot R_1 \cdot C_P} \cdot (i\omega \cdot R_2 \cdot C_2)$$

Definiamo: $C_P = C_1 + C_2 + C_3 = 1.44 \times 10^3 \cdot nF$

$$v_O = \frac{-(i\omega)^2 \cdot R_1 \cdot C_1 \cdot R_2 \cdot C_2 \cdot v_S}{1 + i\omega \cdot R_1 \cdot C_P + (i\omega)^2 \cdot R_1 \cdot C_3 \cdot R_2 \cdot C_2}$$

Poniamo: $\omega_P = \frac{1}{\sqrt{R_1 \cdot R_2 \cdot C_2 \cdot C_3}} = 250 \cdot s^{-1}$ $\quad \delta = \frac{1}{2} \cdot R_1 \cdot C_P \cdot \omega_P = 0.36$ $\quad A = -\frac{C_1}{C_3} = -10$

$$\boxed{W(s) = \frac{A \cdot \left(\frac{s}{\omega_P}\right)^2}{1 + 2\delta \cdot \frac{s}{\omega_P} + \left(\frac{s}{\omega_P}\right)^2}}$$

2) calcolare il valore massimo del modulo del guadagno vicino al polo

il punto di massimo si ha in corrispondenza del punto di minimo del denominatore:

$$1 + 2\delta \cdot \frac{s}{\omega_P} + \left(\frac{s}{\omega_P}\right)^2$$

Per il calcolo del punto di minimo poniamo $x = \omega/\omega_P$ e $s = ix$. $\quad 1 + 2\delta \cdot ix - x^2$

Calcoliamo il modulo $\quad f(x)^2 = \left(1 - x^2\right)^2 + (2 \cdot \delta \cdot x)^2$

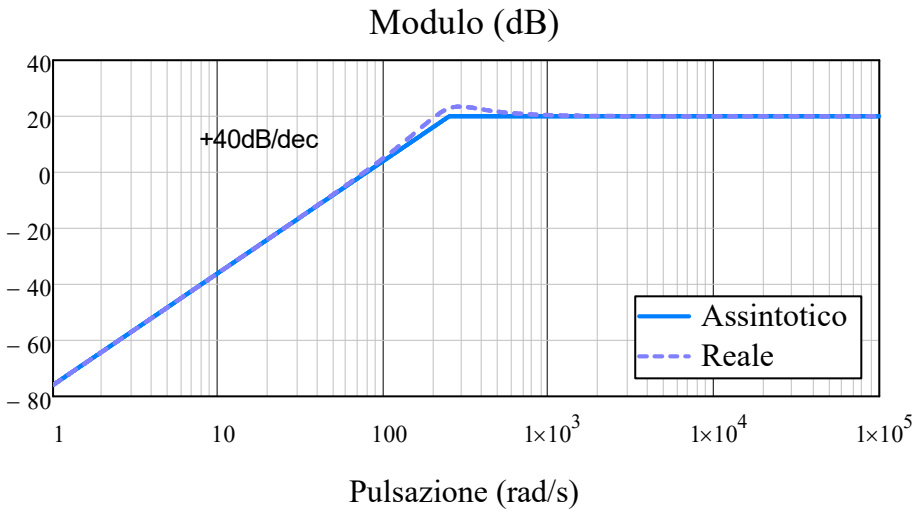
Posizione del minimo: $\quad \frac{d\left(f(x)^2\right)}{dx} = -4 \cdot \left(1 - x^2\right) \cdot x + 8 \cdot \delta^2 \cdot x = 0 \quad x = \sqrt{1 - 2 \cdot \delta^2} = 0.861$

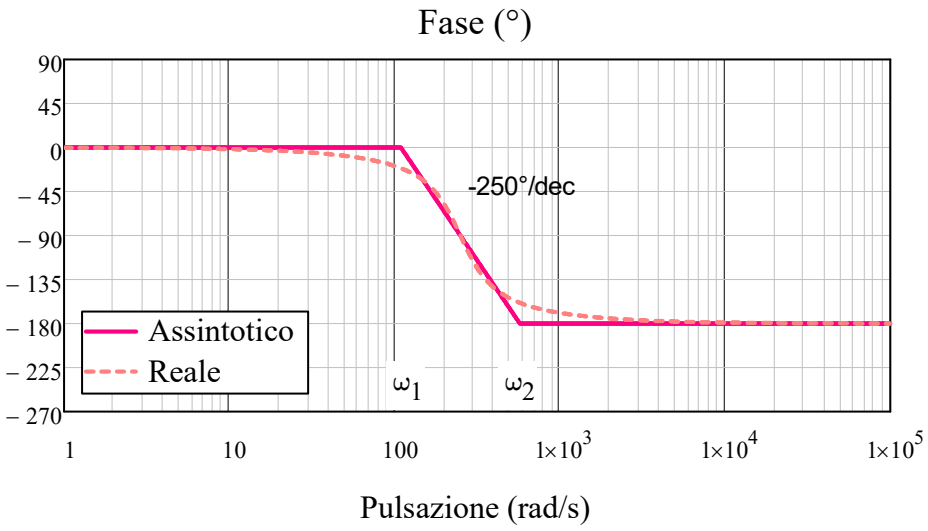
Valore del minimo (del denominatore): $\quad f_{min} = 2 \cdot \delta \cdot \sqrt{1 - \delta^2} = 0.672$

Massimo del modulo: $\quad W_{max} = \frac{|A|}{\left(2 \cdot \delta \cdot \sqrt{1 - \delta^2}\right)} = 14.887 \quad W_{max.dB} = 20 \cdot \log\left[\frac{|A|}{\left(2 \cdot \delta \cdot \sqrt{1 - \delta^2}\right)}\right] = 23.456$

3) Tracciare il diagramma di Bode asintotico del modulo e della fase

$$W(s) = \frac{A\cdot\left(\frac{s}{\omega_P}\right)^2}{1 + 2\delta\cdot\frac{s}{\omega_P} + \left(\frac{s}{\omega_P}\right)^2}$$





$$\omega_1 = \omega_P\cdot 10^{-\delta} = 109.13\cdot s^{-1}$$

$$\omega_2 = \omega_P\cdot 10^{\delta} = 572.72\cdot s^{-1}$$

4) Sapendo che all'ingresso è applicato il segnale $v_S = V_{S1}\sin(\omega_S t)$ con $V_{S1} = 1V$ e $\omega_S = 100s^{-1}$, calcolare ampiezza e fase del segnale di uscita <u>dal diagramma di bode asintotico</u>

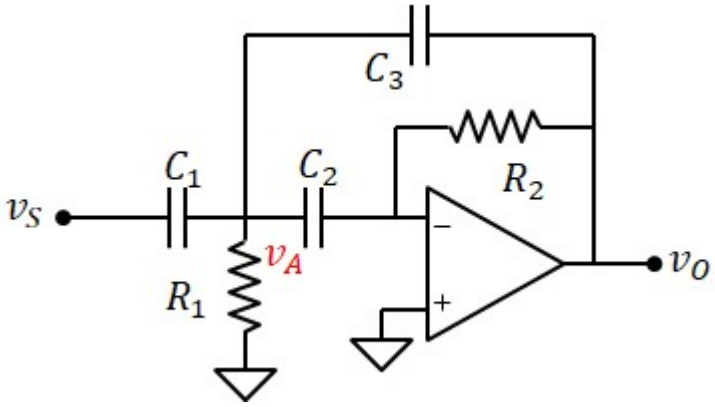
$$W_{dB} = 20dB + 40\cdot\log\left(\frac{\omega_S}{\omega_P}\right) = 4.082 \qquad \Phi = 0°$$

$$\boxed{V_O = V_{S1}\cdot 10^{\frac{W_{dB}}{20}} = 1.6\,V}$$

$$\boxed{\Phi_O = \Phi = 0\cdot°}$$

Esercizio 28

DATI: $R_1 = 5k\Omega$, $C_1 = C_2 = C$



1) trovare la funzione di trasferimento

$$v_O = -v_A\cdot(i\omega\cdot R_2\cdot C)$$

$$v_A = \frac{i\omega\cdot C\cdot v_S + i\omega\cdot C\cdot v_O}{\frac{1}{R_1} + i\omega\cdot 2C + i\omega\cdot C_3}$$

$$v_O = -\frac{i\omega\cdot R_1\cdot C\cdot v_S + i\omega\cdot R_1\cdot C_3\cdot v_O}{1 + i\omega\cdot R_1\cdot C_P}\cdot(i\omega\cdot R_2\cdot C)$$

Definiamo: $C_P = 2C + C_3$

$$v_O = \frac{-(i\omega)^2\cdot R_1\cdot R_2\cdot C^2\cdot v_S}{1 + i\omega\cdot R_1\cdot C_P + (i\omega)^2\cdot R_1\cdot C_3\cdot R_2\cdot C}$$

Poniamo: $\omega_P = \frac{1}{\sqrt{R_1\cdot R_2\cdot C\cdot C_3}}$ $\quad \delta = \frac{1}{2}\cdot R_1\cdot C_P\cdot\omega_P$ $\quad A = -\frac{C}{C_3}$

$$\boxed{W(s) = \frac{A\cdot\left(\frac{s}{\omega_P}\right)^2}{1 + 2\delta\cdot\frac{s}{\omega_P} + \left(\frac{s}{\omega_P}\right)^2}}$$

2) trovare il valore di C_1, C_2, C_3 e R_2 in modo tale che il polo sia $\omega_P = 20s^{-1}$, che il diagramma asintotico della fase in prossimità del polo abbia pendenza -360°/decade e che il guadagno sia $A = -2$

Per avere una pendenza di 360°/dec dobbiamo fissare $\delta = \frac{90°}{360°} = 0.25$

Da A, ricaviamo: $C = -A\cdot C_3$

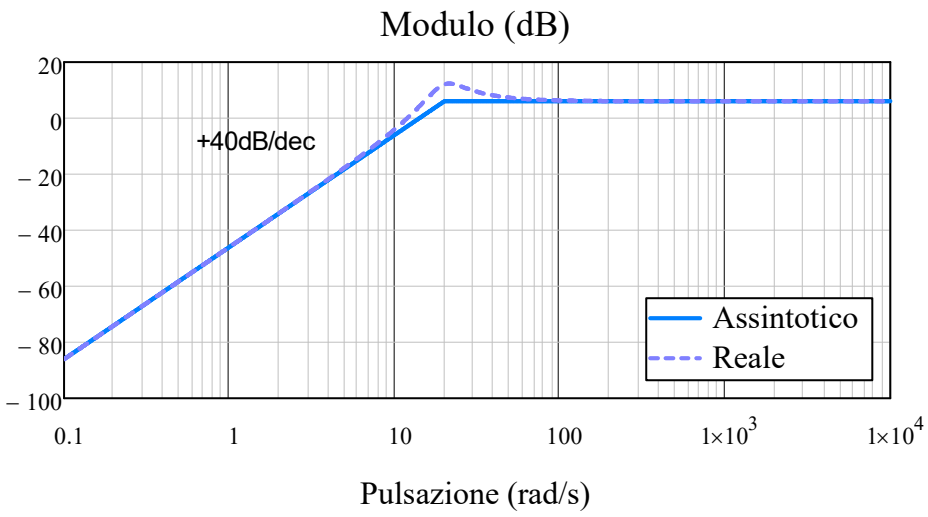
Da δ, ω_P e R_1, calcoliamo: $C_P = 2C + C_3 = C_3\cdot(1 - 2A) = \frac{2\cdot\delta}{R_1\cdot\omega_P}$

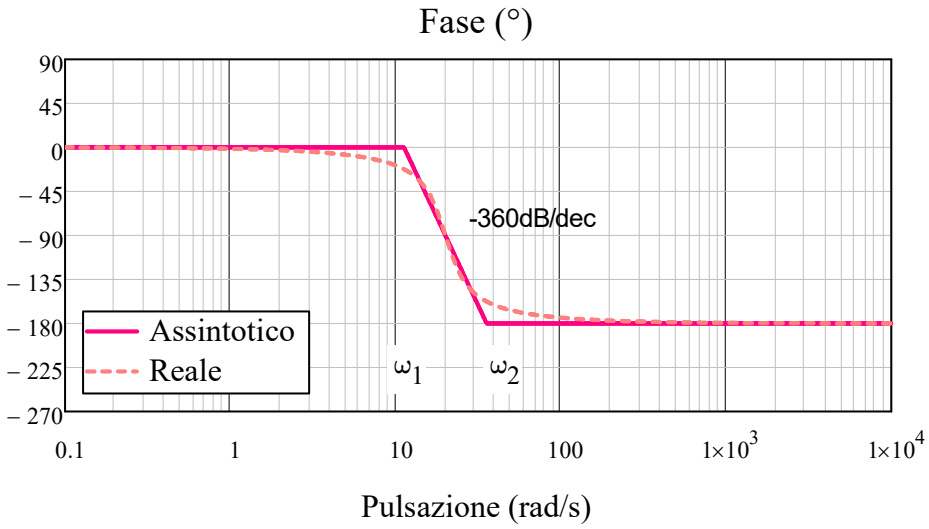
$$\boxed{C_3 = \frac{2\cdot\delta}{(1 - 2A)\cdot R_1\cdot\omega_P} = 1\cdot\mu F}$$

$$\boxed{C = -A\cdot C_3 = 2\cdot\mu F}$$

Da C, C3, ω_P e R_1, calcoliamo: $R_2 = \frac{1}{\omega_P^2\cdot R_1\cdot C\cdot C_3} = 250\cdot k\Omega$

3) Tracciare il diagramma di Bode asintotico del modulo e della fase



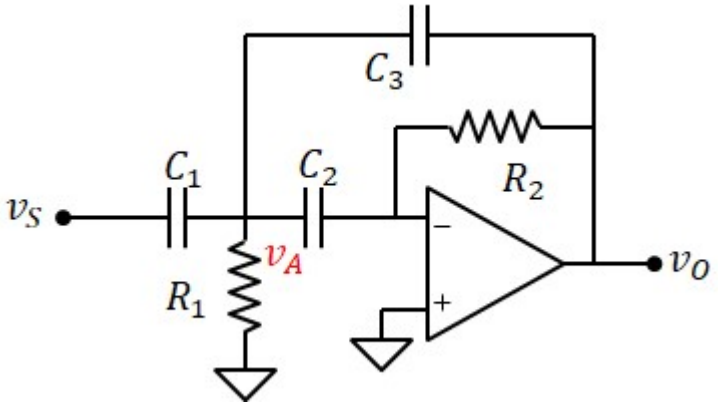


$$\omega_1 = \omega_P \cdot 10^{-\delta} = 11.25 \cdot s^{-1}$$

$$\omega_2 = \omega_P \cdot 10^{\delta} = 35.57 \cdot s^{-1}$$

Esercizio 29

DATI: $R_1 = 25k\Omega$, $C_1 = 2\mu F$, $R_2 = 50k\Omega$, $C_2 = 2\mu F$, $C_3 = 1\mu F$,



1) trovare la funzione di trasferimento

$$v_O = -v_A \cdot (i\omega \cdot R_2 \cdot C_2)$$

$$v_A = \frac{i\omega \cdot C_1 \cdot v_S + i\omega \cdot C_3 \cdot v_O}{\frac{1}{R_1} + i\omega \cdot C_1 + i\omega \cdot C_2 + i\omega \cdot C_3}$$

$$v_O = -\frac{i\omega \cdot R_1 \cdot C_1 \cdot v_S + i\omega \cdot R_1 \cdot C_3 \cdot v_O}{1 + i\omega \cdot R_1 \cdot C_P} \cdot (i\omega \cdot R_2 \cdot C_2)$$

Definiamo: $C_P = C_1 + C_2 + C_3 = 5 \cdot \mu F$

$$v_O = \frac{-(i\omega)^2 \cdot R_1 \cdot C_1 \cdot R_2 \cdot C_2 \cdot v_S}{1 + i\omega \cdot R_1 \cdot C_P + (i\omega)^2 \cdot R_1 \cdot C_3 \cdot R_2 \cdot C_2}$$

Poniamo: $\omega_P = \frac{1}{\sqrt{R_1 \cdot R_2 \cdot C_2 \cdot C_3}} = 20 \cdot s^{-1}$ $\quad \delta = \frac{1}{2} \cdot R_1 \cdot C_P \cdot \omega_P = 1.25$ $\quad A = -\frac{C_1}{C_3} = -2$

il denominatore, riscritto come $1 + 2\delta(s/\omega_P) + (s/\omega_P)^2$, ammette come soluzioni:

$$S_1 = \omega_P \cdot \left(-\delta + \sqrt{\delta^2 - 1}\right) = -10 \cdot s^{-1}$$

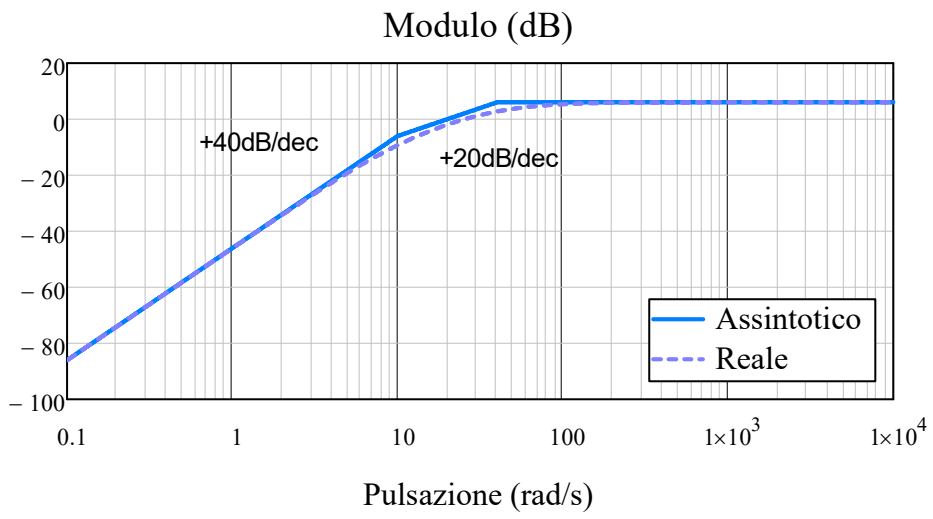
$$S_2 = \omega_P \cdot \left(-\delta - \sqrt{\delta^2 - 1}\right) = -40 \cdot s^{-1}$$

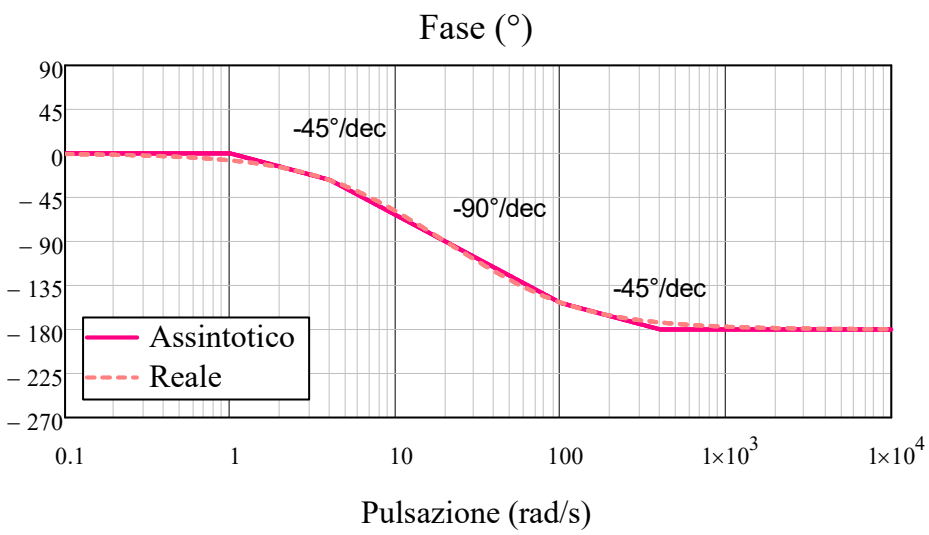
Quindi abbiamo due poli reali negativi e non una coppia di poli complessi coniugati.

$$\omega_{P_1} = -S_1 = 10 \cdot s^{-1} \quad \omega_{P_2} = -S_2 = 40 \cdot s^{-1} \quad \omega_o = \frac{1}{\sqrt{R_1 \cdot C_3 \cdot R_2 \cdot C_2}} = 20 \cdot s^{-1}$$

$$\boxed{W(s) = \frac{A \cdot \left(\frac{s}{\omega_o}\right)^2}{\left(1 + \frac{s}{\omega_{P_1}}\right) \cdot \left(1 + \frac{s}{\omega_{P_2}}\right)}}$$

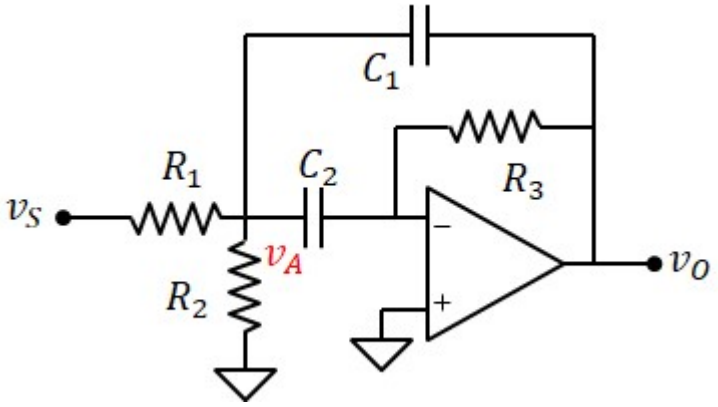
2) Tracciare il diagramma di Bode asintotico del modulo e della fase





Esercizio 30

DATI: $R_1 = 20k\Omega$, $C_1 = 10nF$, $R_2 = 20k\Omega$, $C_2 = 10nF$, $R_3 = 40k\Omega$,



1) trovare la funzione di trasferimento

$$v_O = -v_A \cdot i\omega \cdot R_3 \cdot C_2$$

$$v_A = \frac{\dfrac{v_S}{R_1} + i\omega \cdot C_1 \cdot v_O}{\dfrac{1}{R_1} + \dfrac{1}{R_2} + i\omega \cdot C_2 + i\omega \cdot C_1}$$

$$v_O = -\left(i\omega \cdot R_3 \cdot C_2\right) \cdot \frac{\dfrac{v_S}{R_1} + i\omega \cdot C_1 \cdot v_O}{\dfrac{1}{R_P} + i\omega \cdot C_P}$$

Definiamo:

$$R_P = \left(\frac{1}{R_1} + \frac{1}{R_2}\right)^{-1} = 10 \cdot k\Omega$$

$$C_P = C_1 + C_2 = 0.02 \cdot \mu F$$

$$v_O = -\left[\frac{\dfrac{i\omega \cdot R_3 \cdot R_P \cdot C_2}{R_1}}{1 + i\omega \cdot C_P \cdot R_P + (i\omega)^2 \cdot R_3 \cdot R_P \cdot C_2 \cdot C_1}\right] \cdot v_S$$

Poniamo:

$$\omega_P = \frac{1}{\sqrt{R_3 \cdot R_P \cdot C_2 \cdot C_1}} = 5 \times 10^3 \cdot s^{-1} \qquad \delta = \frac{1}{2} \cdot R_P \cdot C_P \cdot \omega_P = 0.5$$

e riscriviamo l'espressione in questo modo:

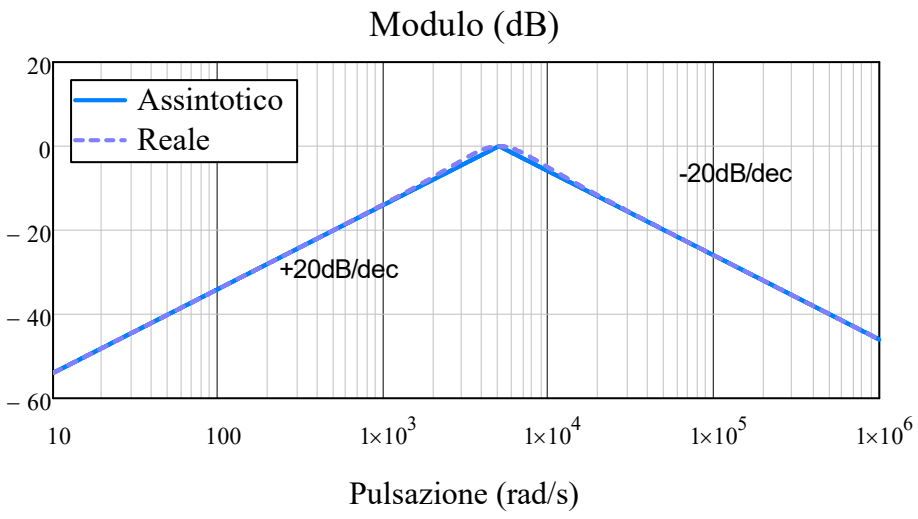
$$v_O = \frac{-\left(\dfrac{i\omega}{\omega_P}\right) \cdot \dfrac{R_3 \cdot R_P \cdot C_2}{R_1} \cdot \omega_P}{1 + 2\delta \cdot \dfrac{i\omega}{\omega_P} + \left(\dfrac{i\omega}{\omega_P}\right)^2} \cdot v_S$$

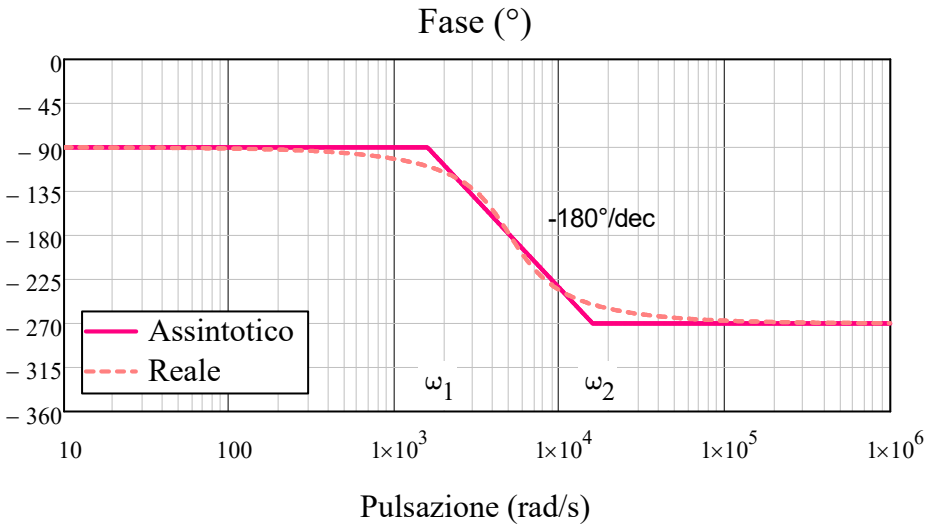
Poniamo infine:

$$A = -\left(\frac{R_3 \cdot R_P \cdot C_2}{R_1} \cdot \omega_P\right) = -1$$

$$\boxed{W(s) = \frac{A \cdot \dfrac{s}{\omega_P}}{1 + 2\delta \cdot \dfrac{s}{\omega_P} + \left(\dfrac{s}{\omega_P}\right)^2}}$$

2) Tracciare il diagramma di Bode asintotico del modulo e della fase



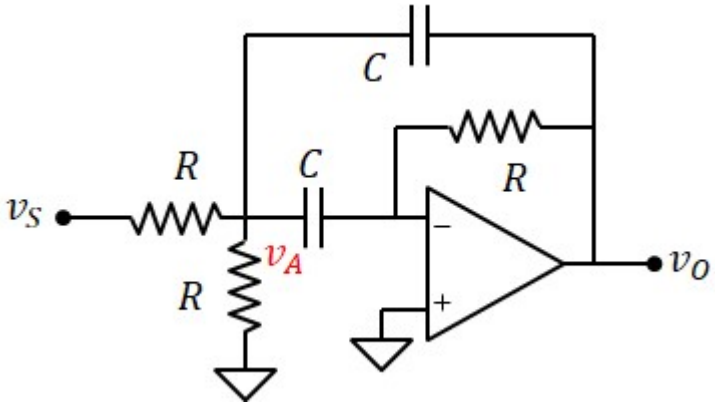


$$\omega_1 = \omega_P \cdot 10^{-\delta} = 1.58 \times 10^3 \cdot s^{-1}$$

$$\omega_2 = \omega_P \cdot 10^{\delta} = 1.58 \times 10^4 \cdot s^{-1}$$

Esercizio 31

DATI: $C = 10nF$



1) trovare la funzione di trasferimento

$$v_O = -v_A \cdot i\omega \cdot R \cdot C$$

$$v_A = \frac{\frac{v_S}{R} + i\omega \cdot C \cdot v_O}{\frac{1}{R} + \frac{1}{R} + i\omega \cdot C + i\omega \cdot C}$$

$$v_O = -(i\omega \cdot R \cdot C) \cdot \frac{\frac{v_S}{R} + i\omega \cdot C \cdot v_O}{2 \cdot \left(\frac{1}{R} + i\omega \cdot C\right)}$$

$$v_O = \frac{-\left(i\omega \cdot \frac{R \cdot C}{2}\right)}{\left[1 + i\omega \cdot R \cdot C + (i\omega)^2 \cdot \frac{R^2 \cdot C^2}{2}\right]} \cdot v_S$$

Poniamo: $\omega_P = \frac{\sqrt{2}}{R \cdot C}$ e $2\delta = \omega_P \cdot R \cdot C = \sqrt{2}$

$$v_O = \frac{-\left(\frac{i\omega}{\omega_P}\right) \cdot \omega_P \cdot \frac{R \cdot C}{2}}{\left[1 + 2\delta \cdot \frac{i\omega}{\omega_P} + \left(\frac{i\omega}{\omega_P}\right)^2\right]} \cdot v_S$$

e poi: $A = -\frac{(\omega_P \cdot R \cdot C)}{2} = \frac{\sqrt{2}}{2}$

Ottenaimo:

$$\boxed{W(s) = \frac{A \cdot \frac{s}{\omega_P}}{1 + 2\delta \cdot \frac{s}{\omega_P} + \left(\frac{s}{\omega_P}\right)^2}}$$

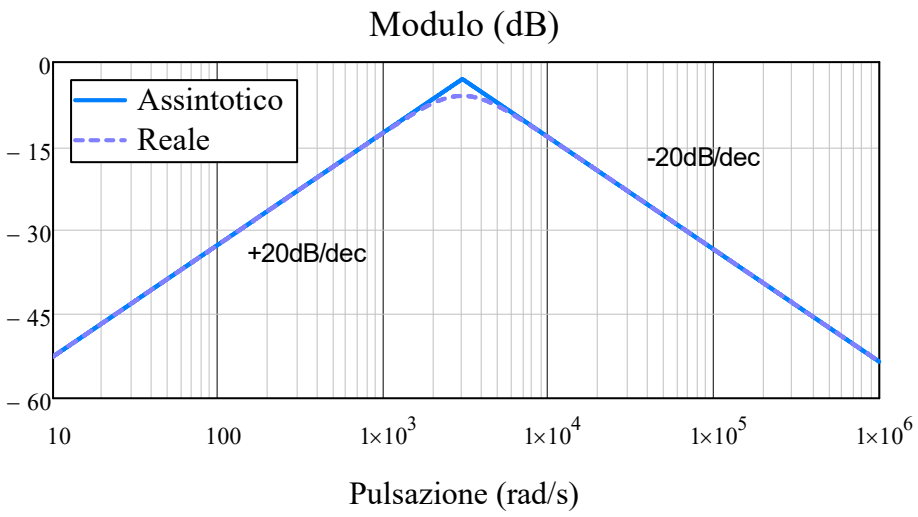
con: $A = -\frac{\sqrt{2}}{2} = -0.707$ e $\delta = \frac{\sqrt{2}}{2} = 0.707$

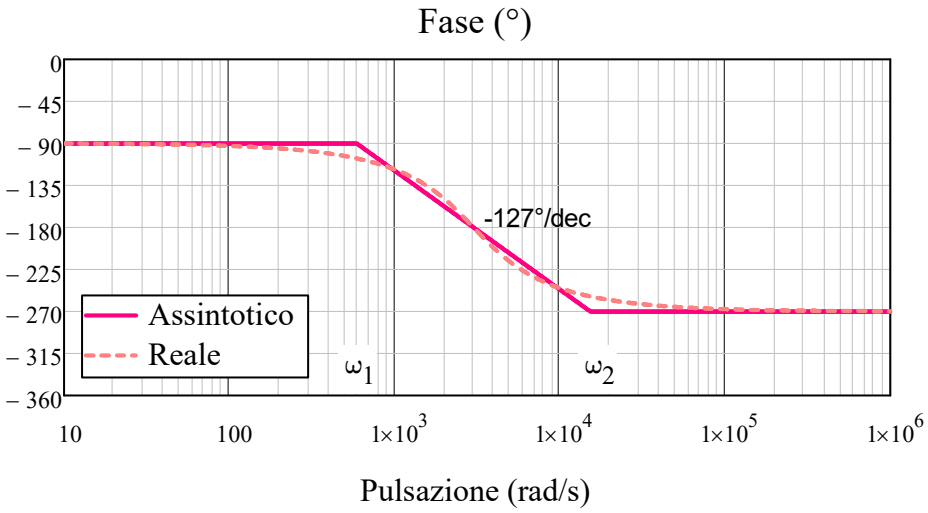
2) trovare il valore di R affinchè $\omega_P = 3000 s^{-1}$

$$\boxed{R = \frac{\sqrt{2}}{\omega_P \cdot C} = 47.1 \cdot k\Omega}$$

$$\delta = \frac{1}{2} \cdot R \cdot C \cdot \omega_P = 0.707$$

3) Tracciare il diagramma di Bode asintotico del modulo e della fase





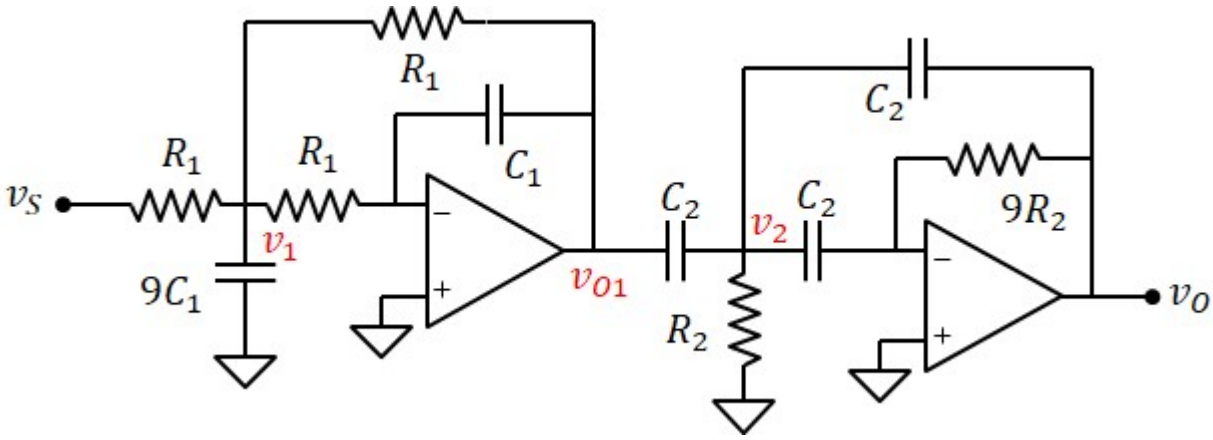
$$\omega_1 = \omega_P \cdot 10^{-\delta} = 588.86 \cdot s^{-1}$$

$$\omega_2 = \omega_P \cdot 10^{\delta} = 1.53 \times 10^4 \cdot s^{-1}$$

Esercizio 32

DATI: $C_1 = 3.33\text{nF}$, $R_1 = 10\text{k}\Omega$, $C_2 = 33.3\mu\text{F}$
, $R_2 = 100\Omega$

1) trovare la funzione di trasferimento



Primo stadio:

$$v_{O1} = \frac{-v_1}{(i\omega \cdot R_1 \cdot C_1)}$$

$$v_1 = \frac{\frac{v_S}{R_1} + \frac{v_{O1}}{R_1}}{\frac{1}{R_1} + \frac{1}{R_1} + \frac{1}{R_1} + i\omega \cdot 9C_1} = \frac{v_S + v_{O1}}{3 + i\omega \cdot 9R_1 \cdot C_1}$$

$$v_{O1} = \frac{-1}{(i\omega \cdot R_1 \cdot C_1)} \cdot \frac{v_S + v_{O1}}{3 + i\omega \cdot 9R_1 \cdot C_1}$$

$$v_{O1} = \frac{-v_S}{1 + i\omega \cdot 3R_1 \cdot C_1 + (i\omega)^2 \cdot 9R_1^2 \cdot C_1^2}$$

Poniamo: $p_1 = \frac{1}{3R_1 \cdot C_1} = 1 \times 10^4 \cdot s^{-1}$

$$\delta_1 = \frac{1}{2} \cdot 3R_1 \cdot C_1 \cdot p_1 = 0.5$$

$$v_{O1} = \frac{-v_S}{1 + 2\delta \cdot \frac{i\omega}{p_1} + \left(\frac{i\omega}{p_1}\right)^2}$$

Secondo stadio:

$$v_O = -v_2 \cdot (i\omega \cdot 9R_2 \cdot C_2)$$

$$v_2 = \frac{v_{O1} \cdot i\omega \cdot C_2 + v_O \cdot i\omega \cdot C_2}{i\omega \cdot C_2 + i\omega \cdot C_2 + i\omega \cdot C_2 + \frac{1}{R_2}} = \frac{(v_{O1} + v_O) \cdot i\omega \cdot R_2 \cdot C_2}{1 + i\omega \cdot 3R_2 \cdot C_2}$$

$$v_O = -i\omega \cdot 9R_2 \cdot C_2 \cdot \frac{(v_{O1} + v_O) \cdot i\omega \cdot R_2 \cdot C_2}{1 + i\omega \cdot 3R_2 \cdot C_2}$$

$$v_O = -v_{O1} \cdot \frac{(i\omega)^2 \cdot 9R_2^2 \cdot C_2^2}{1 + i\omega \cdot 3R_2 \cdot C_2 + (i\omega)^2 \cdot 9R_2^2 \cdot C_2^2}$$

Poniamo: $p_2 = \frac{1}{3R_2 \cdot C_2} = 100 \cdot s^{-1}$

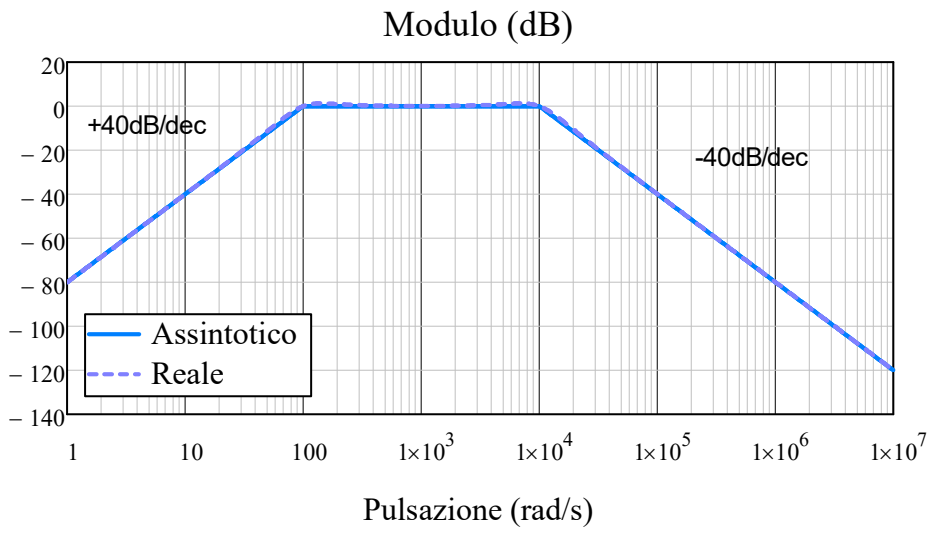
$$\delta_2 = \frac{1}{2} \cdot 3R_2 \cdot C_2 \cdot p_2 = 0.5$$

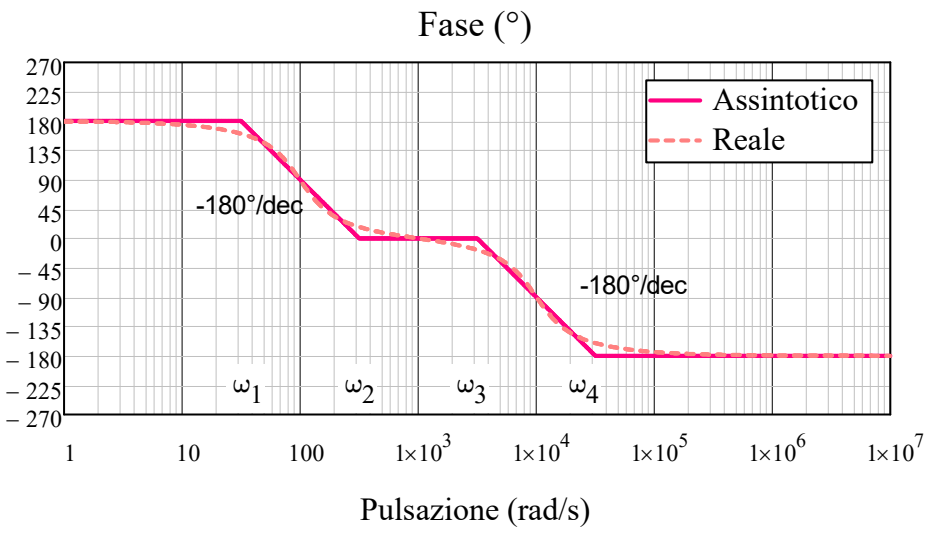
$$v_O = -v_{O1} \cdot \frac{\left(\frac{i\omega}{p_2}\right)^2}{1 + 2\delta \cdot \frac{i\omega}{p_2} + \left(\frac{i\omega}{p_2}\right)^2}$$

Unendo i due stadi otteniamo la funzione di trasferimento:

$$\boxed{W(s) = \frac{\left(\frac{s}{p_2}\right)^2}{\left[1 + 2\delta_1 \cdot \frac{s}{p_1} + \left(\frac{s}{p_1}\right)^2\right] \cdot \left[1 + 2\delta_2 \cdot \frac{s}{p_2} + \left(\frac{s}{p_2}\right)^2\right]}}$$

2) Tracciare il diagramma di Bode asintotico del modulo e della fase





$$\omega_1 = p_2 \cdot 10^{-\delta_2} = 31.65 \cdot s^{-1}$$

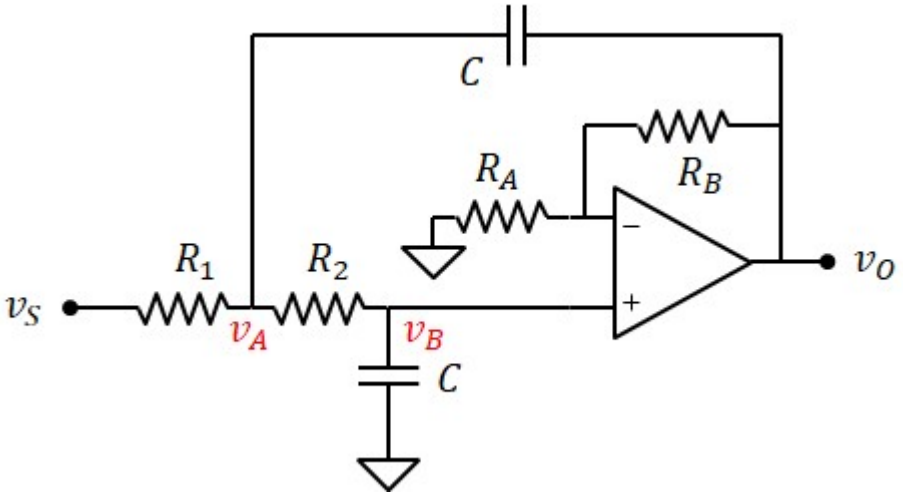
$$\omega_2 = p_2 \cdot 10^{\delta_2} = 316.54 \cdot s^{-1}$$

$$\omega_3 = p_1 \cdot 10^{-\delta_1} = 3.17 \times 10^3 \cdot s^{-1}$$

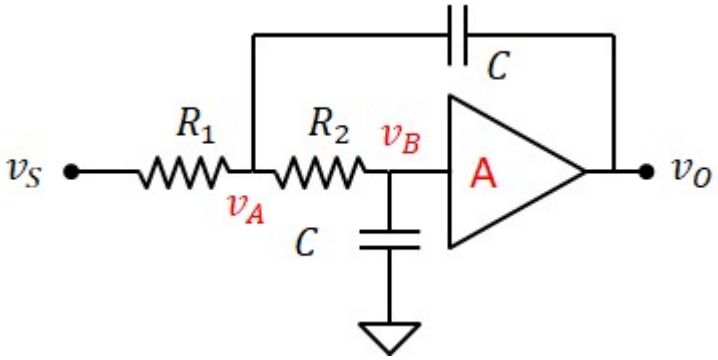
$$\omega_4 = p_1 \cdot 10^{\delta_1} = 3.17 \times 10^4 \cdot s^{-1}$$

Esercizio 33

DATI:
$R_A = 1k\Omega$, $R_B = 2k\Omega$
$C = 10nF$,
$R_1 = 20k\Omega$, $R_2 = 80k\Omega$



1) Funzione di trasferimento



L'AO è in configurazione non invertente e ha un guadagno:

$$A = 1 + \frac{R_B}{R_A} = 3$$

Noto il guadagno, esprimiamo v_B in funzione di v_O

$$v_B = \frac{v_O}{A}$$

e poi v_A in funzione di v_O, mediante la legge del partitore di tensione

$$v_B = v_A \cdot \frac{\frac{1}{i\omega \cdot C}}{R + \frac{1}{i\omega \cdot C}} = v_A \cdot \frac{1}{1 + i\omega \cdot R_2 \cdot C} \qquad v_A = \left(1 + i\omega \cdot R_2 \cdot C\right) \cdot v_B = \left(1 + i\omega \cdot R_2 \cdot C\right) \cdot \frac{v_O}{A}$$

Legge di kirchhoff al nodo v_A

$$v_A = \frac{\frac{v_S}{R_1} + i\omega \cdot C \cdot v_O}{\frac{1}{R_1} + \frac{1}{Z_2} + i\omega \cdot C} = \frac{v_S + i\omega \cdot R_1 \cdot C \cdot v_O}{1 + \frac{R_1}{Z_2} + i\omega \cdot R_1 \cdot C} = \frac{v_S + i\omega \cdot R_1 \cdot C \cdot v_O}{1 + \frac{i\omega \cdot C \cdot R_1}{1 + i\omega \cdot R_2 \cdot C} + i\omega \cdot R_1 \cdot C}$$

con: $Z_2 = \frac{1 + i\omega \cdot R_2 \cdot C}{i\omega \cdot C}$

$$\left(1 + i\omega \cdot R_2 \cdot C\right) \cdot \frac{v_O}{A} = \frac{v_S + i\omega \cdot R_1 \cdot C \cdot v_O}{1 + \frac{i\omega \cdot C \cdot R_1}{1 + i\omega \cdot R_2 \cdot C} + i\omega \cdot R_1 \cdot C}$$

$$\left[1 + i\omega \cdot C \cdot \left(R_2 + 2R_1\right) + R_2 \cdot R_1 \cdot C^2 \cdot (i\omega)^2\right] \cdot \frac{v_O}{A} = v_S + i\omega \cdot R_1 \cdot C \cdot v_O$$

$$\left[1 + i\omega \cdot \left[R_2 + R_1 \cdot (2 - A)\right] \cdot C + (i\omega)^2 \cdot R_2 \cdot R_1 \cdot C^2\right] \cdot \frac{v_O}{A} = v_S$$

Poniamo: $$\omega_P = \frac{1}{C \cdot \sqrt{R_1 \cdot R_2}} = 2.5 \times 10^3 \cdot s^{-1}$$

$$\left[1 + \frac{i\omega}{\omega_P}\cdot\omega_P\cdot\left[R_2 + 2R_1\cdot(1 - A)\right]\cdot C + \left(\frac{i\omega}{\omega_P}\right)^2\right]\cdot\frac{v_O}{A} = v_S$$

Poniamo: $2\delta = \omega_P\cdot\left[R_2 + 2R_1\cdot(1 - A)\right]\cdot C$ $\qquad \delta = \frac{1}{2}\cdot\omega_P\cdot\left[R_2 + R_1\cdot(2 - A)\right]\cdot C = 0.75$

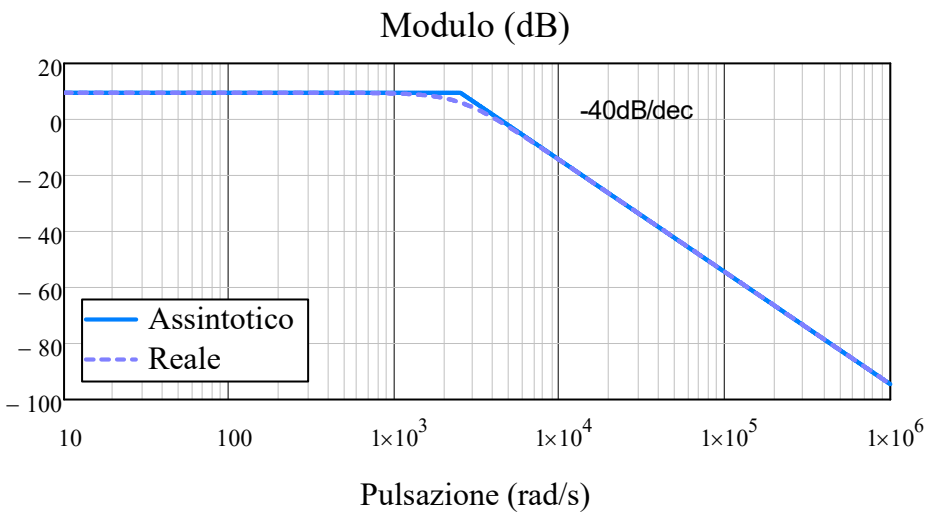
$$v_O = \frac{A}{1 + 2\delta\cdot\frac{i\omega}{\omega_P} + \left(\frac{i\omega}{\omega_P}\right)^2}\cdot v_S$$

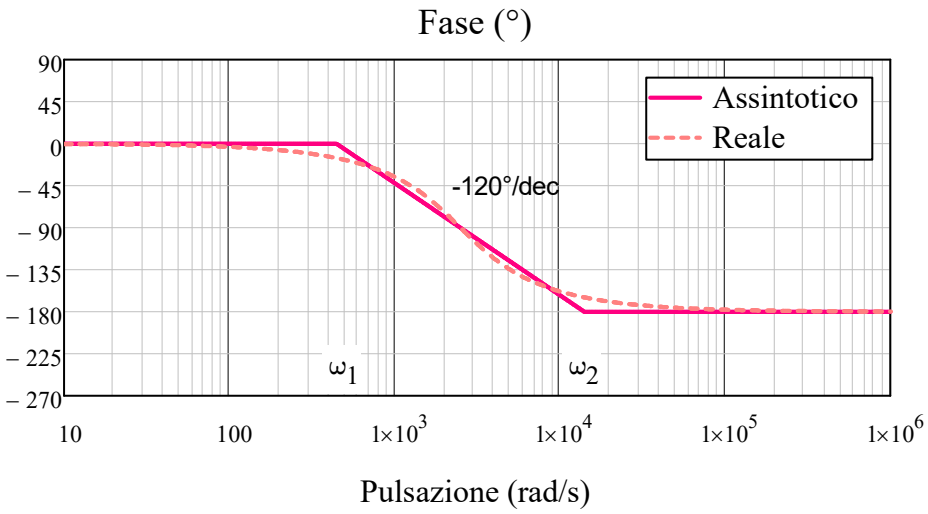
Funzione di trasferimento:

$$\boxed{W(s) = \frac{A}{1 + 2\delta\cdot\frac{s}{\omega_P} + \left(\frac{s}{\omega_P}\right)^2}}$$

$\boxed{\omega_P = 2.5\times 10^3\cdot s^{-1}}$ $\boxed{\delta = 0.75}$ $\boxed{A = 3}$

2) Diagramma di bode



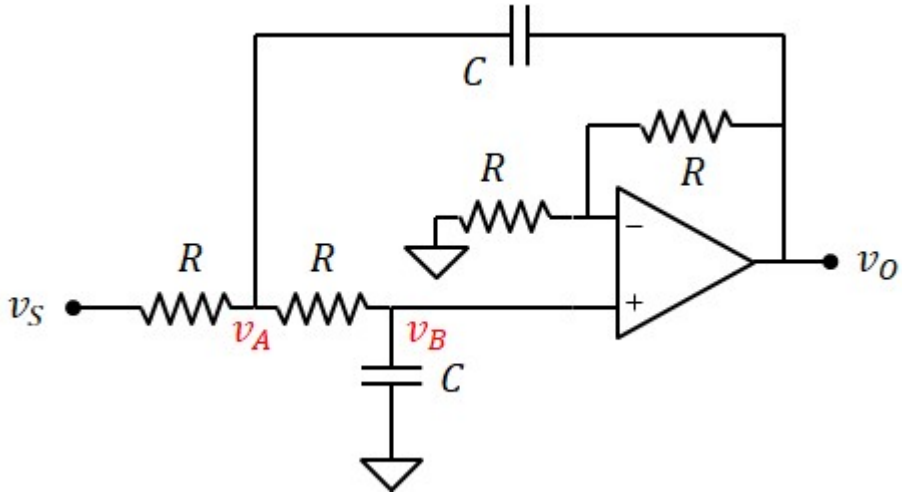


$$\omega_1 = \omega_P\cdot 10^{-\delta} = 444.57\cdot s^{-1}$$

$$\omega_2 = \omega_P\cdot 10^{\delta} = 1.41\times 10^4\cdot s^{-1}$$

Esercizio 34

DATI:
$R = 10k\Omega$



1) Funzione di trasferimento

Tensione di uscita: $v_O = A \cdot v_B \qquad A = \left(1 + \frac{R}{R}\right) = 2$

Tensione v_B (partitore di tensione): $v_B = v_A \cdot \dfrac{\frac{1}{i\omega \cdot C}}{R + \frac{1}{i\omega \cdot C}} = v_A \cdot \dfrac{1}{1 + i\omega \cdot R \cdot C}$

tensione v_A $\qquad v_A = (1 + i\omega \cdot R \cdot C) \cdot v_B = (1 + i\omega \cdot R \cdot C) \cdot \dfrac{v_O}{A}$

Legge di kirchhoff al nodo v_A:

$$v_A = \frac{\frac{v_S}{R} + i\omega \cdot C \cdot v_O}{\frac{1}{R} + \frac{i\omega \cdot C}{1 + i\omega \cdot R \cdot C} + i\omega \cdot C} = \frac{v_S + i\omega \cdot R \cdot C \cdot v_O}{1 + \frac{i\omega \cdot R \cdot C}{1 + i\omega \cdot R \cdot C} + i\omega \cdot R \cdot C}$$

$$(1 + i\omega \cdot R \cdot C) \cdot \frac{v_O}{A} = \frac{v_S + i\omega \cdot R \cdot C \cdot v_O}{1 + \frac{i\omega \cdot C \cdot R}{1 + i\omega \cdot R \cdot C} + i\omega \cdot R \cdot C}$$

$$\left[1 + i\omega \cdot 3R \cdot C + (i\omega)^2 \cdot R^2 \cdot C^2\right] \cdot \frac{v_O}{A} = v_S + i\omega \cdot R \cdot C \cdot v_O$$

$$\left[1 + i\omega \cdot (3 - A)R \cdot C + (i\omega)^2 \cdot R^2 \cdot C^2\right] \cdot \frac{v_O}{A} = v_S$$

$$v_O = \frac{A}{1 + i\omega \cdot (3 - A)R \cdot C + (i\omega)^2 \cdot R^2 \cdot C^2} \cdot v_S$$

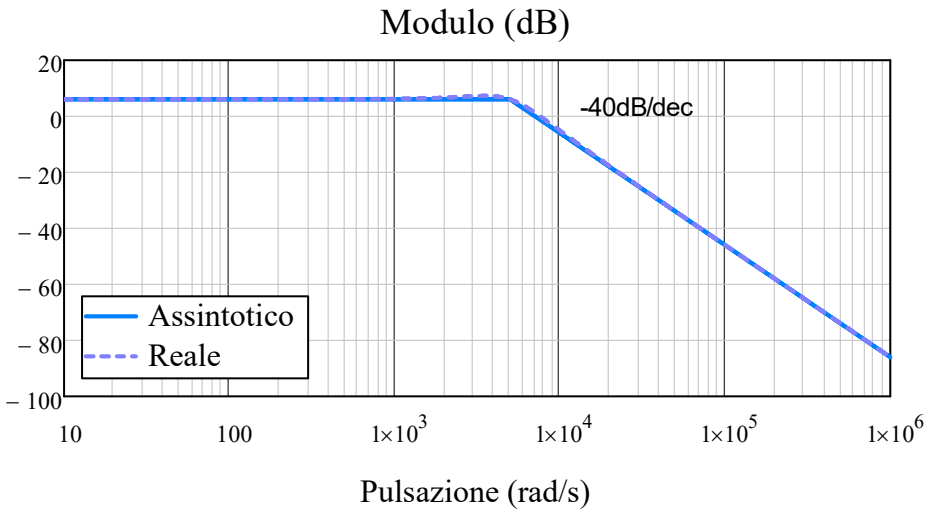
Poniamo: $\omega_P = \dfrac{1}{R \cdot C} \qquad \delta = \dfrac{1}{2} \cdot (3 - A) = 0.5$

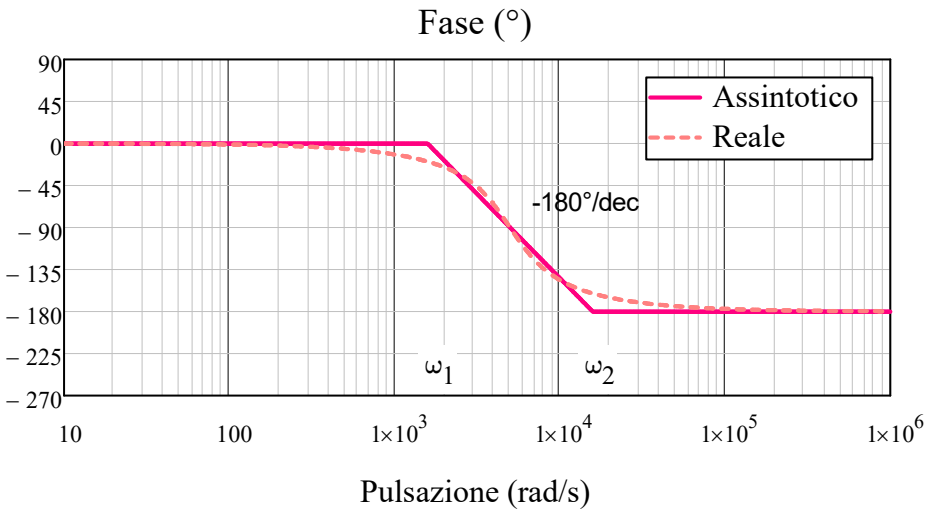
$$\boxed{W(s) = \frac{A}{1 + 2\delta \cdot \frac{s}{\omega_P} + \left(\frac{s}{\omega_P}\right)^2}}$$

2) calcolare C in modo tale che $\omega_P = 5000\,s^{-1}$

$$C = \frac{1}{R \cdot \omega_P} = 20 \cdot nF$$

3) Diagramma di Bode



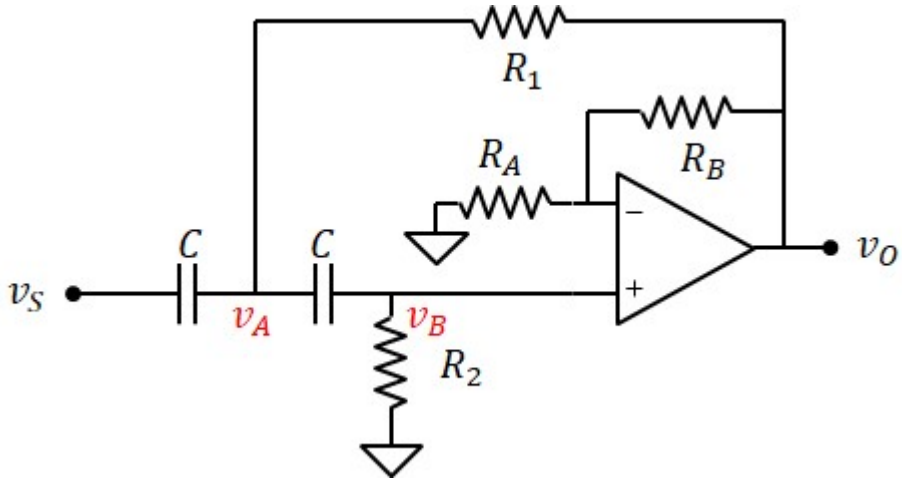


$$\omega_1 = \omega_P \cdot 10^{-\delta} = 1.58 \times 10^3 \cdot s^{-1}$$

$$\omega_2 = \omega_P \cdot 10^{\delta} = 1.58 \times 10^4 \cdot s^{-1}$$

Esercizio 35

DATI:
$R_A = 1k\Omega$, $R_B = 7k\Omega$
$C = 1\mu F$,
$R_1 = 40k\Omega$, $R_2 = 10k\Omega$



1) Funzione di trasferimento

Tensione di uscita: $v_O = v_B \cdot A \qquad A = 1 + \frac{R_B}{R_A} = 8$

Tensione v_B (partitore di tensione): $v_B = v_A \cdot \frac{R_2}{R_2 + \frac{1}{i\omega \cdot C}} = v_A \cdot \frac{i\omega \cdot R_2 \cdot C}{1 + i\omega \cdot R_2 \cdot C}$

tensione v_A $\quad v_A = v_B \cdot \left(\frac{1 + i\omega \cdot R_2 \cdot C}{i\omega \cdot R_2 \cdot C}\right) = \frac{1 + i\omega \cdot R_2 \cdot C}{i\omega \cdot R_2 \cdot C} \cdot \frac{v_O}{A}$

Legge di kirchhoff al nodo v_A: $\quad v_A = \frac{\frac{v_O}{R_1} + i\omega \cdot C \cdot v_S}{\frac{1}{R_1} + i\omega \cdot C + \frac{1}{Z_2}} = \frac{v_O + i\omega \cdot R_1 \cdot C \cdot v_S}{1 + i\omega \cdot R_1 \cdot C + \frac{i\omega \cdot C \cdot R_1}{1 + i\omega \cdot R_2 \cdot C}} \qquad Z_2 = \frac{1 + i\omega \cdot R_2 \cdot C}{i\omega \cdot C}$

$$\frac{1 + i\omega \cdot R_2 \cdot C}{i\omega \cdot R_2 \cdot C} \cdot \frac{v_O}{A} = \frac{v_O + i\omega \cdot R_1 \cdot C \cdot v_S}{1 + i\omega \cdot R_1 \cdot C + \frac{i\omega \cdot C \cdot R_1}{1 + i\omega \cdot R_2 \cdot C}}$$

$$\left[1 + i\omega \cdot R_2 \cdot C + i\omega \cdot R_1 \cdot C + R_1 \cdot C \cdot R_2 \cdot C \cdot (i\omega)^2 + i\omega \cdot C \cdot R_1\right] \cdot \frac{v_O}{A \cdot (i\omega \cdot R_2 \cdot C)} = v_O + i\omega \cdot R_1 \cdot C \cdot v_S$$

$$\left[1 + i\omega \cdot \left[R_2 \cdot (1 - A) + 2R_1\right] \cdot C + R_1 \cdot R_2 \cdot C^2 \cdot (i\omega)^2\right] \cdot \frac{v_O}{A \cdot (i\omega \cdot R_2 \cdot C)} = i\omega \cdot R_1 \cdot C \cdot v_S$$

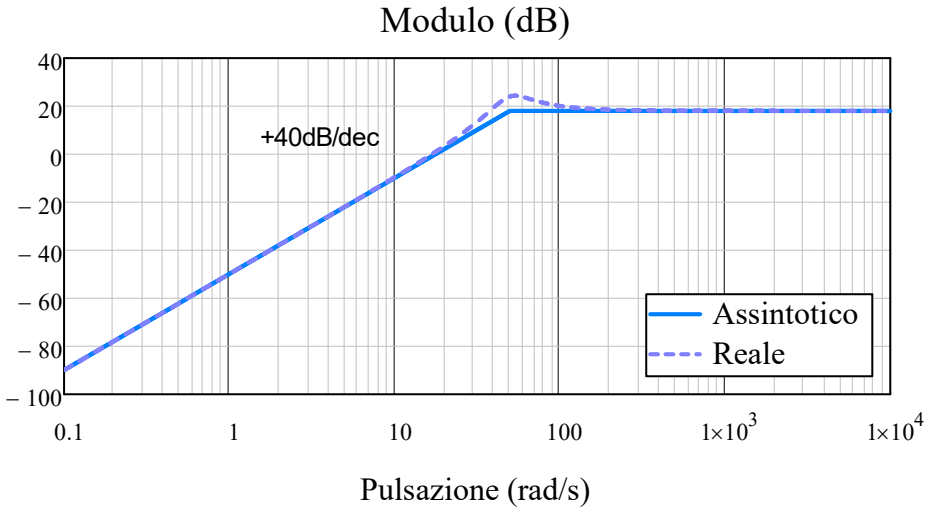
$$v_O = \frac{A \cdot (i\omega)^2 \cdot R_1 \cdot R_2 \cdot C^2}{1 + i\omega \cdot \left[R_2 \cdot (1 - A) + 2R_1\right] \cdot C + R_1 \cdot R_2 \cdot C^2 \cdot (i\omega)^2} \cdot v_S$$

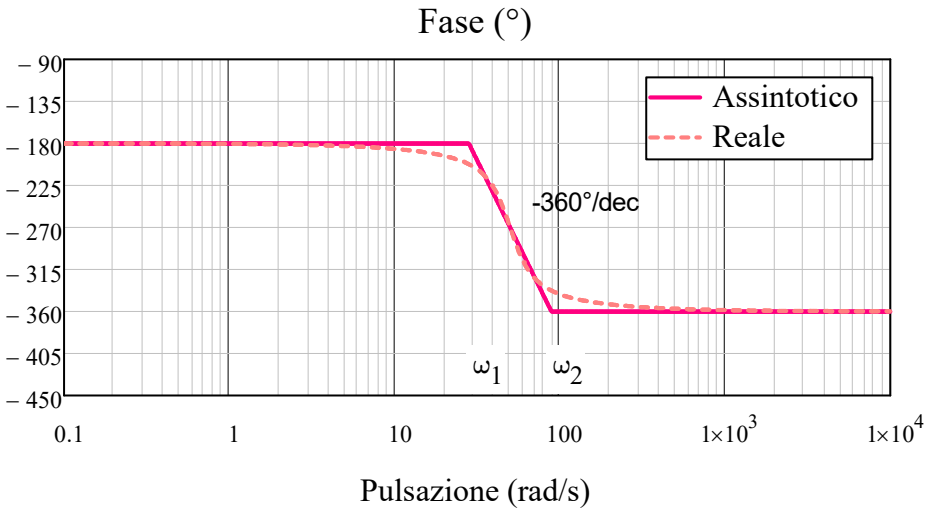
Poniamo: $\omega_P = \frac{1}{C \cdot \sqrt{R_1 \cdot R_2}} = 50 \cdot s^{-1}$

$$\delta = \frac{1}{2} \cdot \omega_P \cdot C \cdot \left[R_2 \cdot (1 - A) + 2R_1\right] = 0.25$$

$$\boxed{W(s) = \frac{A \cdot \left(\frac{s}{\omega_P}\right)^2}{1 + 2\delta \cdot \frac{s}{\omega_P} + \left(\frac{s}{\omega_P}\right)^2}}$$

2) Diagramma di Bode





$$\omega_1 = \omega_P \cdot 10^{-\delta} = 28.12 \cdot s^{-1}$$

$$\omega_2 = \omega_P \cdot 10^{\delta} = 88.91 \cdot s^{-1}$$

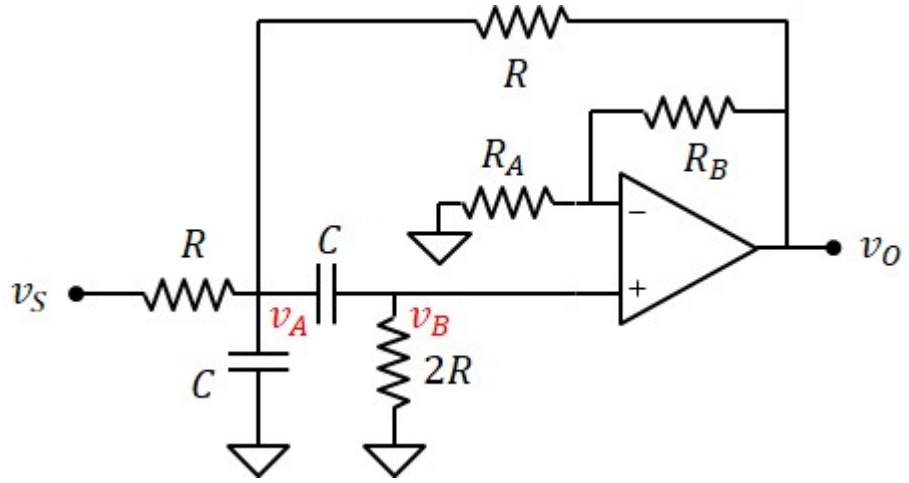
Esercizio 36

DATI:

$R_A = 10k\Omega$, $R_B = 14k\Omega$

$C = 200nF$,

$R = 5k\Omega$



1) Funzione di trasferimento

Tensione di uscita: $v_O = A \cdot V_B \qquad A = 1 + \frac{R_B}{R_A} = 2.4$

Tensione v_B (partitore di tensione): $v_B = v_A \cdot \frac{2R}{2R + \frac{1}{i\omega \cdot C}} = v_A \cdot \frac{i\omega \cdot 2R \cdot C}{1 + i\omega \cdot 2R \cdot C}$

tensione v_A $\quad v_A = \frac{1 + i\omega \cdot 2R \cdot C}{i\omega \cdot 2R \cdot C} \cdot v_B = \frac{1 + i\omega \cdot 2R \cdot C}{i\omega \cdot 2R \cdot C} \cdot \frac{v_O}{A}$

Legge di kirchhoff al nodo v_A:

$$v_A = \frac{\frac{v_S}{R} + \frac{v_O}{R}}{\frac{1}{R} + \frac{1}{R} + i\omega \cdot C + \frac{1}{Z}} = \frac{v_S + v_O}{2 + i\omega \cdot R \cdot C + \frac{i\omega \cdot R \cdot C}{1 + i\omega \cdot 2R \cdot C}} \qquad Z = \frac{1 + i\omega \cdot 2R \cdot C}{i\omega \cdot C}$$

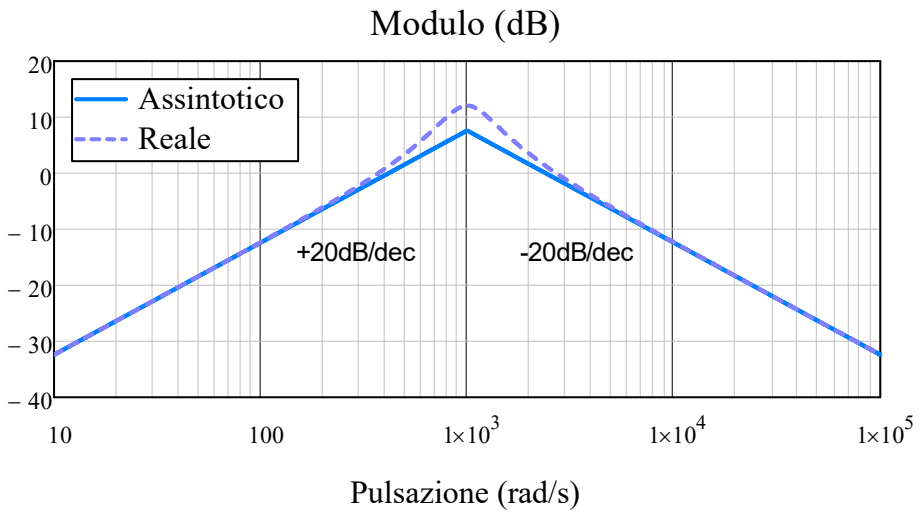
$$\left[1 + i\omega \cdot (3 - A) \cdot R \cdot C + (i\omega)^2 \cdot R^2 \cdot C^2\right] \cdot \frac{v_O}{A \cdot (i\omega \cdot R \cdot C)} = v_S$$

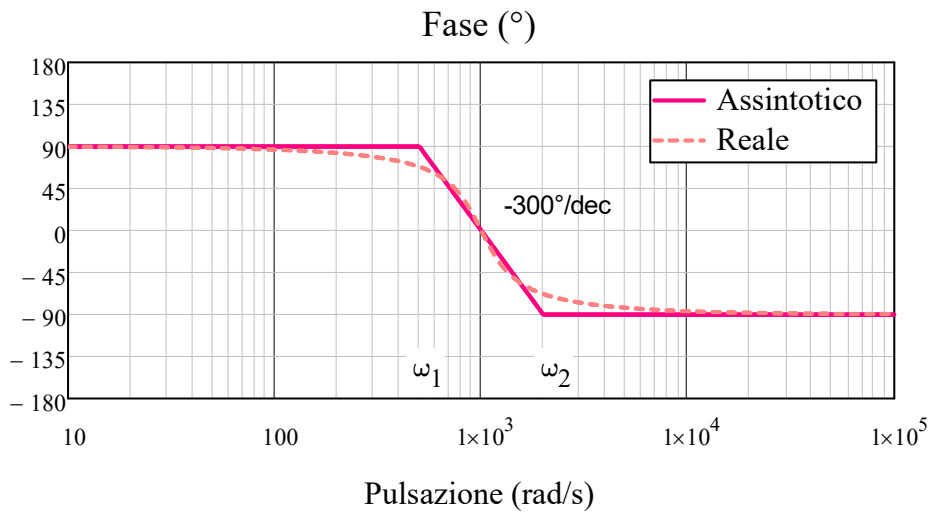
$$v_O = A \cdot \frac{(i\omega \cdot R \cdot C)}{\left[1 + i\omega \cdot (3 - A) \cdot R \cdot C + (i\omega)^2 \cdot R^2 \cdot C^2\right]} \cdot v_S$$

Poniamo: $\omega_P = \frac{1}{R \cdot C} = 1 \times 10^3 \cdot s^{-1} \qquad \delta = \frac{1}{2} \cdot (3 - A) = 0.3$

$$\boxed{W(s) = \frac{A \cdot \frac{s}{\omega_P}}{1 + 2 \cdot \delta \frac{s}{\omega_P} + \left(\frac{s}{\omega_P}\right)^2}}$$

2) Diagramma di Bode





$$\omega_1 = \omega_P \cdot 10^{-\delta} = 501.19 \cdot s^{-1}$$

$$\omega_2 = \omega_P \cdot 10^{\delta} = 2 \times 10^3 \cdot s^{-1}$$

3) Massimo reale del modulo di W

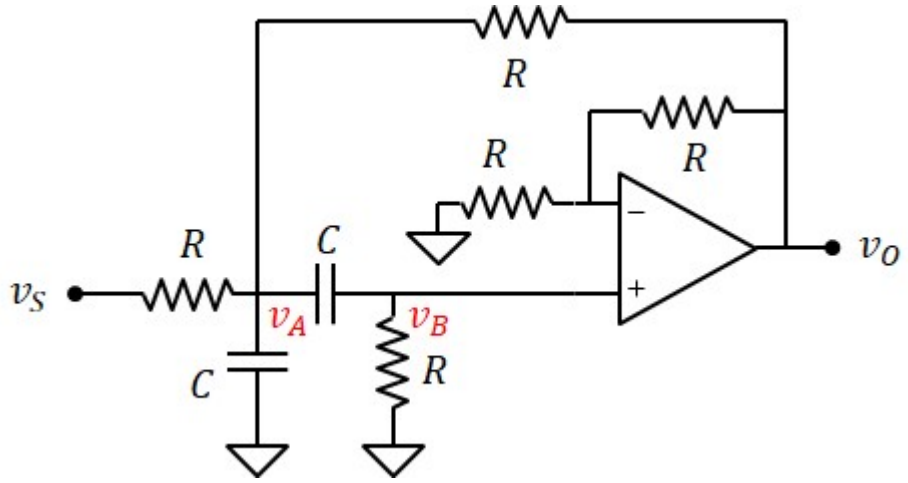
$$W(\omega_P) = \frac{A \cdot i}{1 + 2i \cdot \delta + (i)^2} = \frac{A}{2 \cdot \delta}$$

$$\boxed{W_{max} = \frac{A}{2\delta} = 4}$$

$$\boxed{W_{dB} = 20 \cdot \log\left(\frac{A}{2\delta}\right) = 12}$$

Esercizio 37

DATI:
$R = 10k\Omega, C = 20nF,$



1) Funzione di trasferimento

Tensione di uscita: $v_O = 2 \cdot v_B$

Tensione v_B (partitore di tensione): $v_B = v_A \cdot \dfrac{R}{R + \dfrac{1}{i\omega \cdot C}} = v_A \cdot \dfrac{i\omega \cdot R \cdot C}{1 + i\omega \cdot R \cdot C}$

tensione v_A $\quad v_A = \dfrac{1 + i\omega \cdot R \cdot C}{i\omega \cdot R \cdot C} \cdot v_B = \dfrac{1 + i\omega \cdot R \cdot C}{i\omega \cdot R \cdot C} \cdot \dfrac{v_O}{2}$

Legge di kirchhoff al nodo v_A:

$$v_A = \frac{\dfrac{v_S}{R} + \dfrac{v_O}{R}}{\dfrac{1}{R} + \dfrac{1}{R} + i\omega \cdot C + \dfrac{1}{Z}} = \frac{v_S + v_O}{2 + i\omega \cdot R \cdot C + \dfrac{i\omega \cdot R \cdot C}{1 + i\omega \cdot R \cdot C}} \qquad Z = \frac{1 + i\omega \cdot R \cdot C}{i\omega \cdot C}$$

$$\frac{1 + i\omega \cdot R \cdot C}{i\omega \cdot R \cdot C} \cdot \frac{v_O}{2} = \frac{v_S + v_O}{2 + i\omega \cdot R \cdot C + \dfrac{i\omega \cdot R \cdot C}{1 + i\omega \cdot R \cdot C}}$$

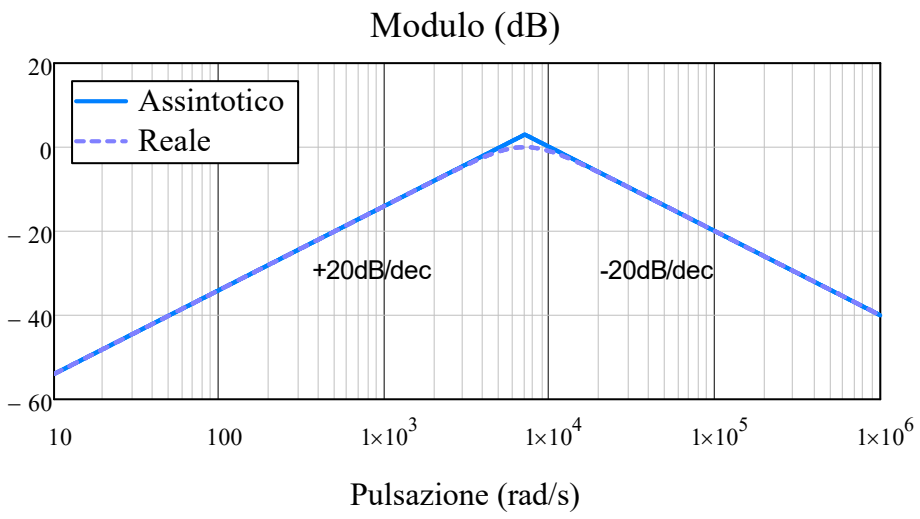
$$\left[2 + 4i\omega \cdot R \cdot C + (i\omega)^2 \cdot R^2 \cdot C^2\right] \cdot \frac{1}{i\omega \cdot R \cdot C} \cdot \frac{v_O}{2} = v_S + v_O$$

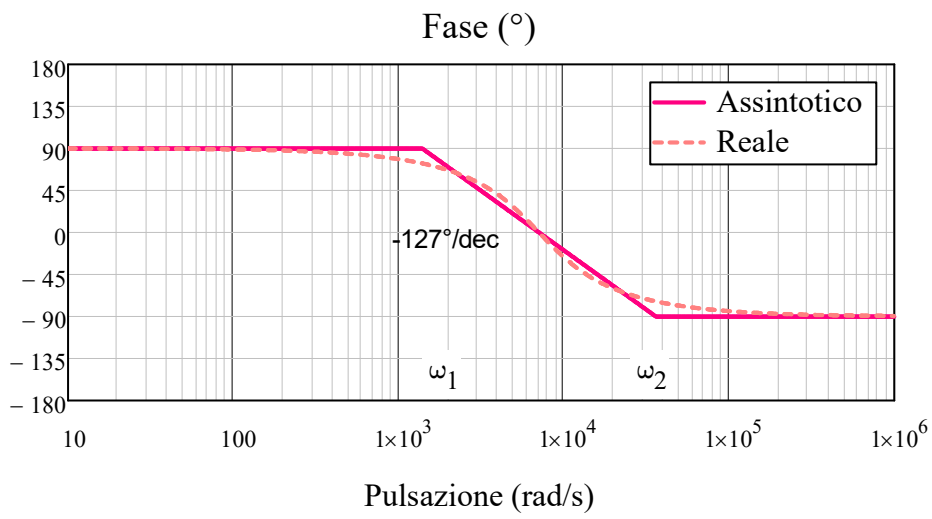
$$v_O = \frac{i\omega \cdot R \cdot C}{\left[1 + i\omega \cdot R \cdot C + (i\omega)^2 \cdot \dfrac{R^2 \cdot C^2}{2}\right]} \cdot v_S$$

Poniamo: $\omega_P = \dfrac{\sqrt{2}}{R \cdot C} = 7.071 \times 10^3 \cdot s^{-1} \qquad \delta = \dfrac{\sqrt{2}}{2} = 0.707 \qquad A = \sqrt{2}$

$$\boxed{W(s) = \frac{A \cdot \dfrac{s}{\omega_P}}{1 + 2 \cdot \delta \dfrac{s}{\omega_P} + \left(\dfrac{s}{\omega_P}\right)^2}}$$

2) Diagramma di Bode





$$\omega_1 = \omega_P \cdot 10^{-\delta} = 1.39 \times 10^3 \cdot s^{-1}$$

$$\omega_2 = \omega_P \cdot 10^{\delta} = 3.6 \times 10^4 \cdot s^{-1}$$

3) Massimo reale del modulo di W

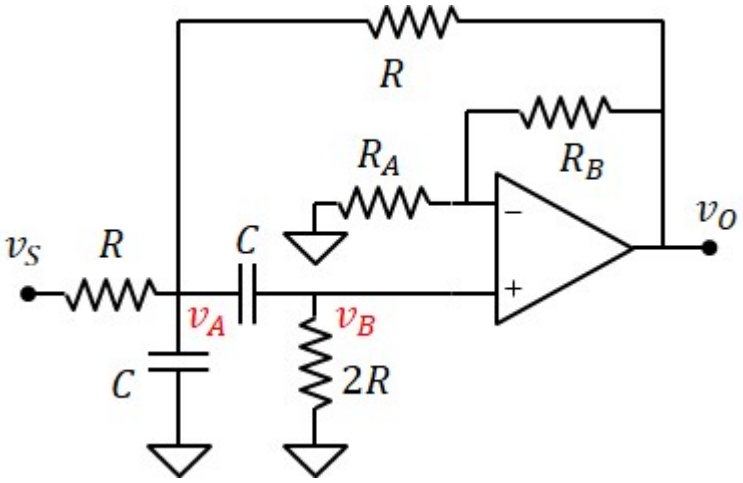
$$W(\omega_P) = \frac{A \cdot i}{1 + 2i \cdot \delta + (i)^2} = \frac{A}{2 \cdot \delta}$$

$$\boxed{W_{max} = \frac{A}{2\delta} = 1}$$

$$\boxed{W_{dB} = 20 \cdot \log\left(\frac{A}{2\delta}\right) = 0}$$

Esercizio 38

DATI:
$R = 5k\Omega, C = 100nF,$
$R_A = 10k\Omega$



1) Funzione di trasferimento

Tensione di uscita: $v_O = A \cdot v_B \qquad A = 1 + \frac{R_B}{R_A}$

Tensione v_B (partitore di tensione): $v_B = v_A \cdot \frac{2R}{2R + \frac{1}{i\omega \cdot C}} = v_A \cdot \frac{i\omega \cdot 2R \cdot C}{1 + i\omega \cdot 2R \cdot C}$

tensione v_A $\quad v_A = \frac{1 + i\omega \cdot 2R \cdot C}{i\omega \cdot 2R \cdot C} \cdot v_B = \frac{1 + i\omega \cdot 2R \cdot C}{i\omega \cdot 2R \cdot C} \cdot \frac{v_O}{A}$

Legge di kirchhoff al nodo v_A:

$$v_A = \frac{\frac{v_S}{R} + \frac{v_O}{R}}{\frac{1}{R} + \frac{1}{R} + i\omega \cdot C + \frac{1}{Z}} = \frac{v_S + v_O}{2 + i\omega \cdot R \cdot C + \frac{i\omega \cdot R \cdot C}{1 + i\omega \cdot 2R \cdot C}} \qquad Z = \frac{1 + i\omega \cdot 2R \cdot C}{i\omega \cdot C}$$

$$\frac{1 + i\omega \cdot 2R \cdot C}{i\omega \cdot 2R \cdot C} \cdot \frac{v_O}{A} = \frac{v_S + v_O}{2 + i\omega \cdot R \cdot C + \frac{i\omega \cdot R \cdot C}{1 + i\omega \cdot 2R \cdot C}}$$

$$\left[2 + 6i\omega \cdot R \cdot C + (i\omega)^2 \cdot 2R^2 \cdot C^2\right] \cdot \frac{1}{i\omega \cdot 2R \cdot C} \cdot \frac{v_O}{A} = v_S + v_O$$

$$v_O = \frac{A \cdot i\omega \cdot R \cdot C}{\left[1 + (3 - A)i\omega \cdot R \cdot C + (i\omega)^2 \cdot R^2 \cdot C^2\right]} \cdot v_S$$

Poniamo: $\omega_P = \frac{1}{R \cdot C} = 2 \times 10^3 \cdot s^{-1} \qquad \delta = \frac{1}{2} \cdot (3 - A)$

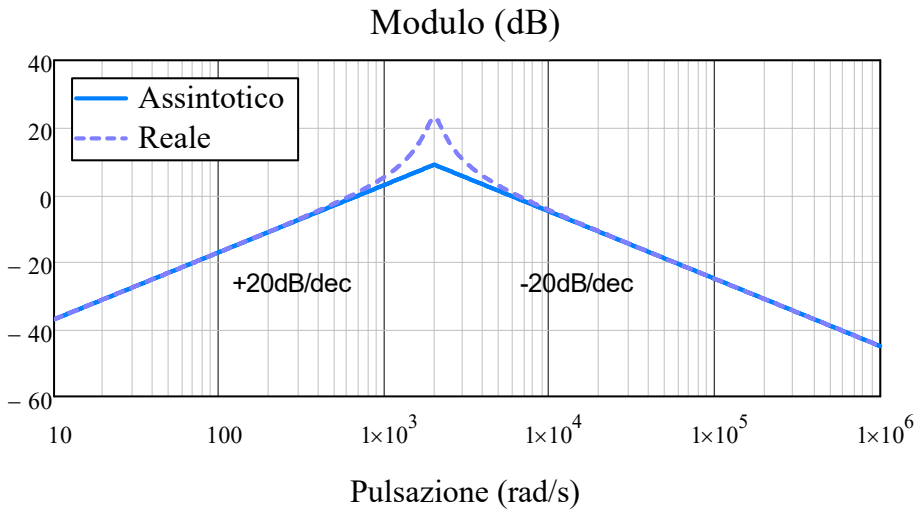
$$\boxed{W(s) = \frac{A \cdot \frac{s}{\omega_P}}{1 + 2 \cdot \delta \frac{s}{\omega_P} + \left(\frac{s}{\omega_P}\right)^2}}$$

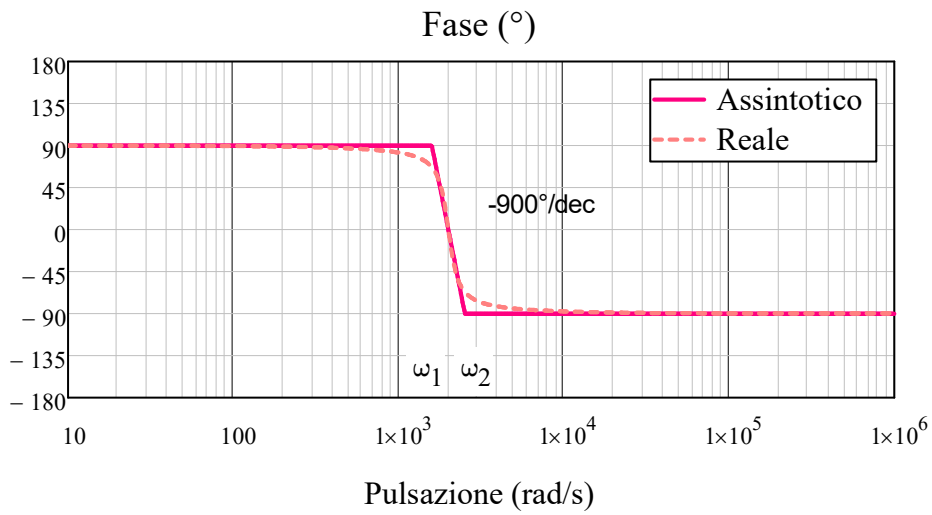
2) Valore di R_B per ottenere una pendenza del diagramma di bode della fase $S = \frac{900°}{dec}$

$$S = \frac{90°}{\delta} \qquad \delta = \frac{90°}{S} = 0.1$$

$$\delta = \frac{1}{2}\cdot(3 - A) \qquad A = 3 - 2\delta = 2.8 \qquad \boxed{R_B = R_A\cdot(A - 1) = 18\cdot k\Omega}$$

3) Diagramma di Bode





$$\omega_1 = \omega_P\cdot 10^{-\delta} = 1.59 \times 10^3\cdot s^{-1}$$

$$\omega_2 = \omega_P\cdot 10^{\delta} = 2.52 \times 10^3\cdot s^{-1}$$

4) Massimo reale del modulo di W

$$W(\omega_P) = \frac{A\cdot i}{1 + 2i\cdot\delta + (i)^2} = \frac{A}{2\cdot\delta} \qquad \boxed{W_{max} = \frac{A}{2\delta} = 14} \qquad \boxed{W_{dB} = 20\cdot\log\left(\frac{A}{2\delta}\right) = 22.9}$$

Esercizio 39

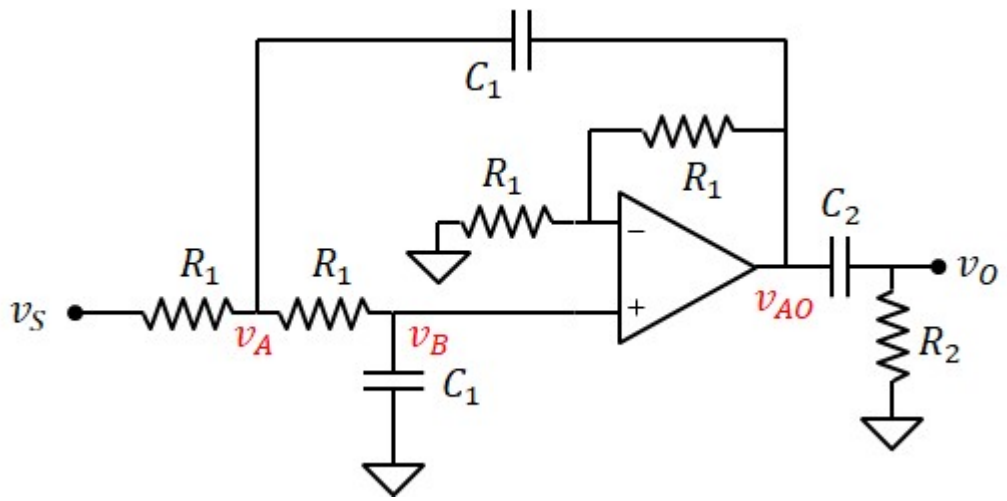
DATI:

$R_1 = 25k\Omega$,

$C_1 = 4nF$,

$R_2 = 100k\Omega$,

$C_2 = 100nF$



1) Funzione di trasferimento

Tensione di uscita: $v_{AO} = A \cdot v_B \qquad A = \left(1 + \frac{R_1}{R_1}\right) = 2$

Tensione v_B (partitore di tensione): $v_B = v_A \cdot \dfrac{\frac{1}{i\omega \cdot C_1}}{R_1 + \frac{1}{i\omega \cdot C_1}} = v_A \cdot \dfrac{1}{1 + i\omega \cdot R_1 \cdot C_1}$

tensione v_A $\qquad v_A = \left(1 + i\omega \cdot R_1 \cdot C_1\right) \cdot v_B = \left(1 + i\omega \cdot R_1 \cdot C_1\right) \cdot \dfrac{v_{AO}}{A}$

Legge di kirchhoff al nodo v_A:

$$v_A = \frac{\frac{v_S}{R_1} + i\omega \cdot C_1 \cdot v_{AO}}{\frac{1}{R_1} + \frac{i\omega \cdot C_1}{1 + i\omega \cdot R_1 \cdot C_1} + i\omega \cdot C_1} = \frac{v_S + i\omega \cdot R_1 \cdot C_1 \cdot v_{AO}}{1 + \frac{i\omega \cdot R_1 \cdot C_1}{1 + i\omega \cdot R_1 \cdot C_1} + i\omega \cdot R \cdot C}$$

$$\left(1 + i\omega \cdot R_1 \cdot C_1\right) \cdot \frac{v_{AO}}{A} = \frac{v_S + i\omega \cdot R_1 \cdot C_1 \cdot v_{AO}}{1 + \frac{i\omega \cdot C_1 \cdot R_1}{1 + i\omega \cdot R_1 \cdot C_1} + i\omega \cdot R_1 \cdot C_1}$$

$$\left[1 + i\omega \cdot 3R_1 \cdot C_1 + (i\omega)^2 \cdot R_1^{\,2} \cdot C_1^{\,2}\right] \cdot \frac{v_{AO}}{A} = v_S + i\omega \cdot R_1 \cdot C_1 \cdot v_{AO}$$

$$v_{AO} = \frac{A}{1 + i\omega \cdot (3 - A) R_1 \cdot C_1 + (i\omega)^2 \cdot R_1^{\,2} \cdot C_1^{\,2}} \cdot v_S$$

Poniamo: $\omega_{P_1} = \dfrac{1}{R_1 \cdot C_1} = 1 \times 10^4 \cdot s^{-1} \qquad \delta = \dfrac{1}{2} \cdot (3 - A) = 0.5$

$$v_{AO} = \frac{A}{1 + 2\delta \cdot \frac{s}{\omega_{P_1}} + \left(\frac{s}{\omega_{P_1}}\right)^2} \cdot v_S$$

Partitore di tensione:

$$v_O = \frac{i\omega \cdot R_2 \cdot C_2}{1 + i\omega \cdot R_2 \cdot C_2} \cdot v_{AO} = \frac{i\omega \cdot R_2 \cdot C_2}{1 + i\omega \cdot R_2 \cdot C_2} \cdot \frac{A}{1 + 2\delta \cdot \frac{s}{\omega_{P_1}} + \left(\frac{s}{\omega_{P_1}}\right)^2} \cdot v_S$$

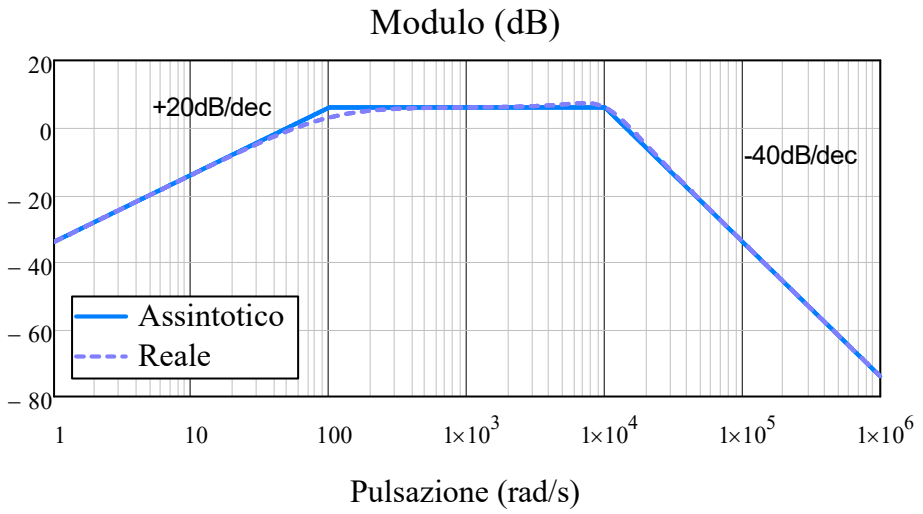
Poniamo: $\omega_{P_2} = \frac{1}{R_2 \cdot C_2} = 100 \cdot s^{-1}$

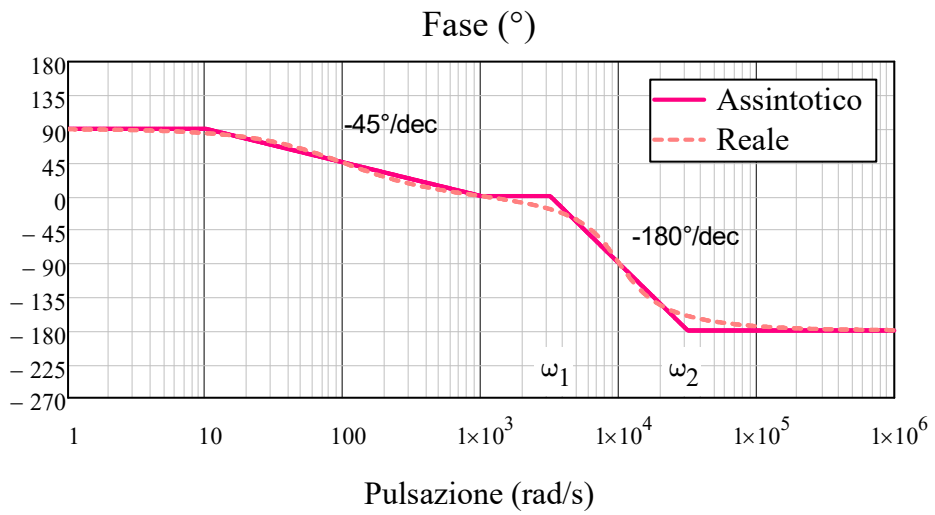
$$W(s) = \frac{A \cdot \frac{s}{\omega_{P_2}}}{\left(1 + \frac{s}{\omega_{P_2}}\right) \cdot \left[1 + 2\delta \cdot \frac{s}{\omega_{P_1}} + \left(\frac{s}{\omega_{P_1}}\right)^2\right]}$$

$\omega_{P_1} = 1 \times 10^4 \cdot s^{-1}$ $\delta = 0.5$

$\omega_{P_2} = 100 \cdot s^{-1}$ $A = 2$

2) Diagramma di Bode





$$\omega_1 = \omega_P \cdot 10^{-\delta} = \begin{pmatrix} 0 \\ 3.16 \times 10^3 \end{pmatrix} \cdot s^{-}$$

$$\omega_2 = \omega_P \cdot 10^{\delta} = \begin{pmatrix} 0 \\ 3.16 \times 10^4 \\ 316.23 \end{pmatrix} \cdot s^{-1}$$